TM Forum Specification

Performance

TMF628

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Direct inquiries to the TM Forum office:

181 New Road, Suite 304
Parsippany, NJ 07054, USA
Tel No. +1 862 227 1648
TM Forum Web Page: www.tmforum.org

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Introduction

The following document is the user guide of the REST API for Performance management. It includes the model definition as well as all available operations.

The following document is the specification of the REST API for Performance Management. It includes the model definition as well as all available operations. Possible actions include:

- Creating, updating and retrieving Measurement Collection Jobs
- Creating, updating and deleting Performance Indicator Specifications and Performance Indicator Group Specifications
- Retrieving Data Access Endpoints (which are created during Measurement Collection Jobs creation or update)
- Retrieving Tracking Records (which are created during Measurement Collection Jobs creation or update)
- Retrieving Performance Measurements (observed values of Performance Indicators)

The Measurement Collection is intended to be used with a recurring scheduled interval and continual collection but it also supports a one-off collection. The different mechanisms for the measurements collection are realized by using file transfer or by specifying a data access endpoint or by using the individual retrieval of the 'PerformanceMeasurement' object (https endpoints(GET)).

Events currently supported include:

- Create
- executionStateChange
- delete
- attributeValueChange
- filesReady
- filesPreparationError

The PerformanceIndicatorSpecification is intended to be used with a defining indicators for measuring performance. PerformanceIndicatorGroupSpecification is intented to group the performance indicators.

The Performance Measurements can be retrieved individually or collectively and result from the operations performed during a Measurement Collection Job. Said differently, Performance Measurements are generated/created by a "MeasurementCollectionJob" process. Events currently support include:

- create
- collectionReady

Using a Data Access Endpoint:

- The data access endpoint specifies where the peformance measurements can be

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accessed once taken. The configuration of the data access could vary. For example, this could be a publish-subcribe subscription endpoint or an endpoint that returns the data directly or an endpoint that follows a query pattern using a POST to iterate through the data using the url return in the POST response.

- The measurement collection job should contain the data access enpoint that its related/collected performance measurements can be accessed from.
- When the measurement collection job has taken/collected the measurements, the PerformanceMeasurementCreate event will be emitted for individual measurements

Using file transfer:

- The measurement collection job can be created with a fileTransferData attribute specified.
- When the measurement collection job has taken the measurements, the MeasurementCollectionJobFilesReady event will be emitted and the file can then be retrieved.

Using filtering in MeasurementCollectionJob and DataAccessEndpoint:

- The Measurement Collection Job contains an optional filtering definition attribute to further limit the data being collected.
- The DataAccessEndpoint, and Measurement Collection Job contain an optional 'filtering definition' provided by the CSP. In the payload response of the POST (“Create”), that 'filtering definition' attribute will defines the filtering attributes with their respective usage(range filtering, regex filtering,…).

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Sample Use Cases

Reader will find example of use cases using Usage API in “Open Digital Business Scenarios and Use Cases” document.

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Support of polymorphism and extension patterns

Support of polymorphic collections and types and schema based extension is provided by means of a list of generic meta-attributes that we describe below. Polymorphism in collections occurs when entities inherit from base entities, for instance a BillingAccount and SettlementAccount inheriting properties from the abstract Account entity.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines, Part 2 (TMF630).

The @type attribute provides a way to represent the actual class type of an entity. For example, within a list of Account instances some may be instances of BillingAccount where other could be instances of SettlementAccount. The @type gives this information. All resources and sub-resources of this API have a @type attributes that can be provided when this is useful.

The @referredType can be used within reference entities (like for instance an AccountRef object) to explicitly denote the actual entity type of the referred class. Notice that in reference entities the @type, when used, denotes the class type of the reference itself, such as BillingAccountRef or SettlementAccountRef, and not the class type of the referred object. However since reference classes are rarely sub-classed, @type is generally not useful in reference objects.

The @schemaLocation property can be used in resources to allow specifying user-defined properties of an Entity or to specify the expected characteristics of an entity.

The @baseType attribute gives a way to provide explicitly the base of class of a given resource that has been extended.

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RESOURCE MODEL

Managed Entity and Task Resource Models

MeasurementCollectionJob resource

Used to control the periodic collection of performance indicators, implemented as a sub-entity of the PM Measurement Job.

Resource model

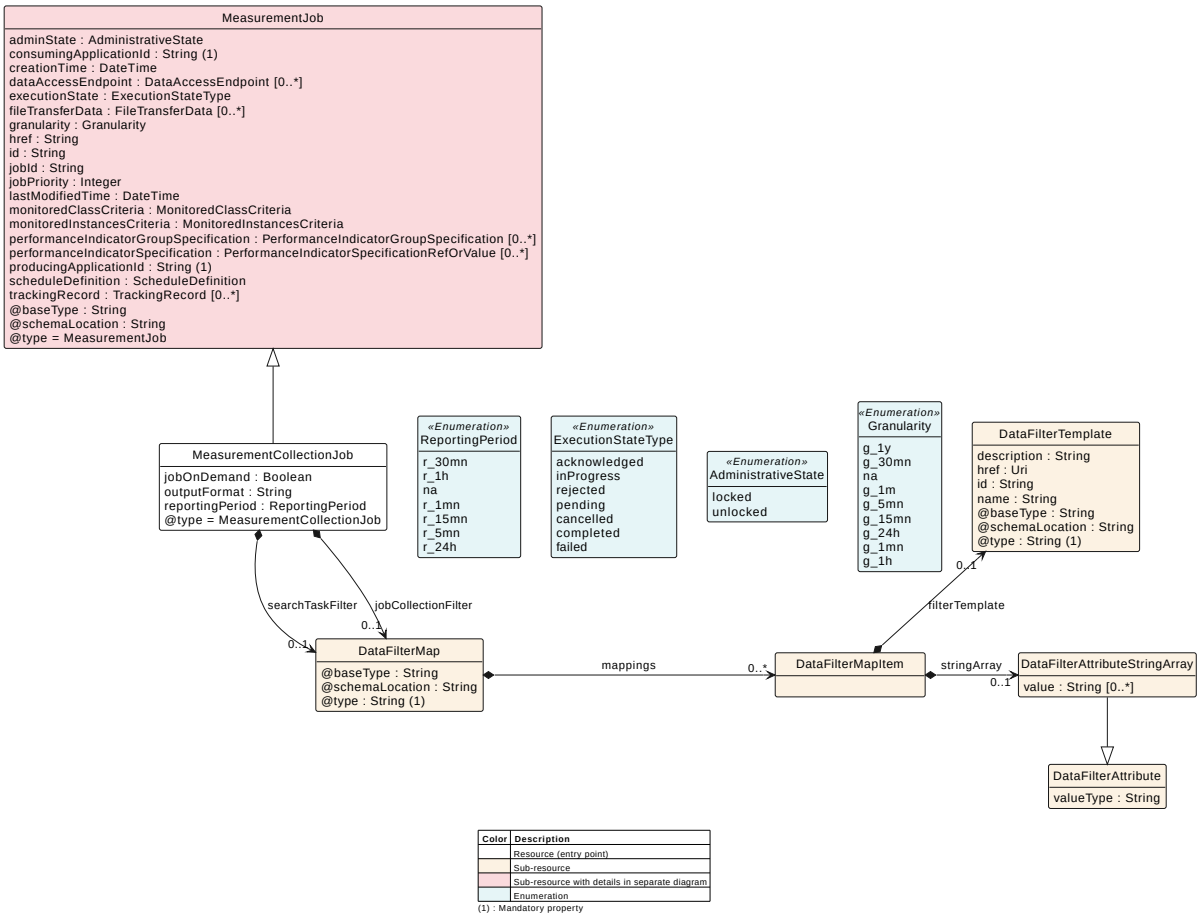


Figure 1 - MeasurementCollectionJob

[.] | .//

Figure 2 - MeasurementCollectionJob

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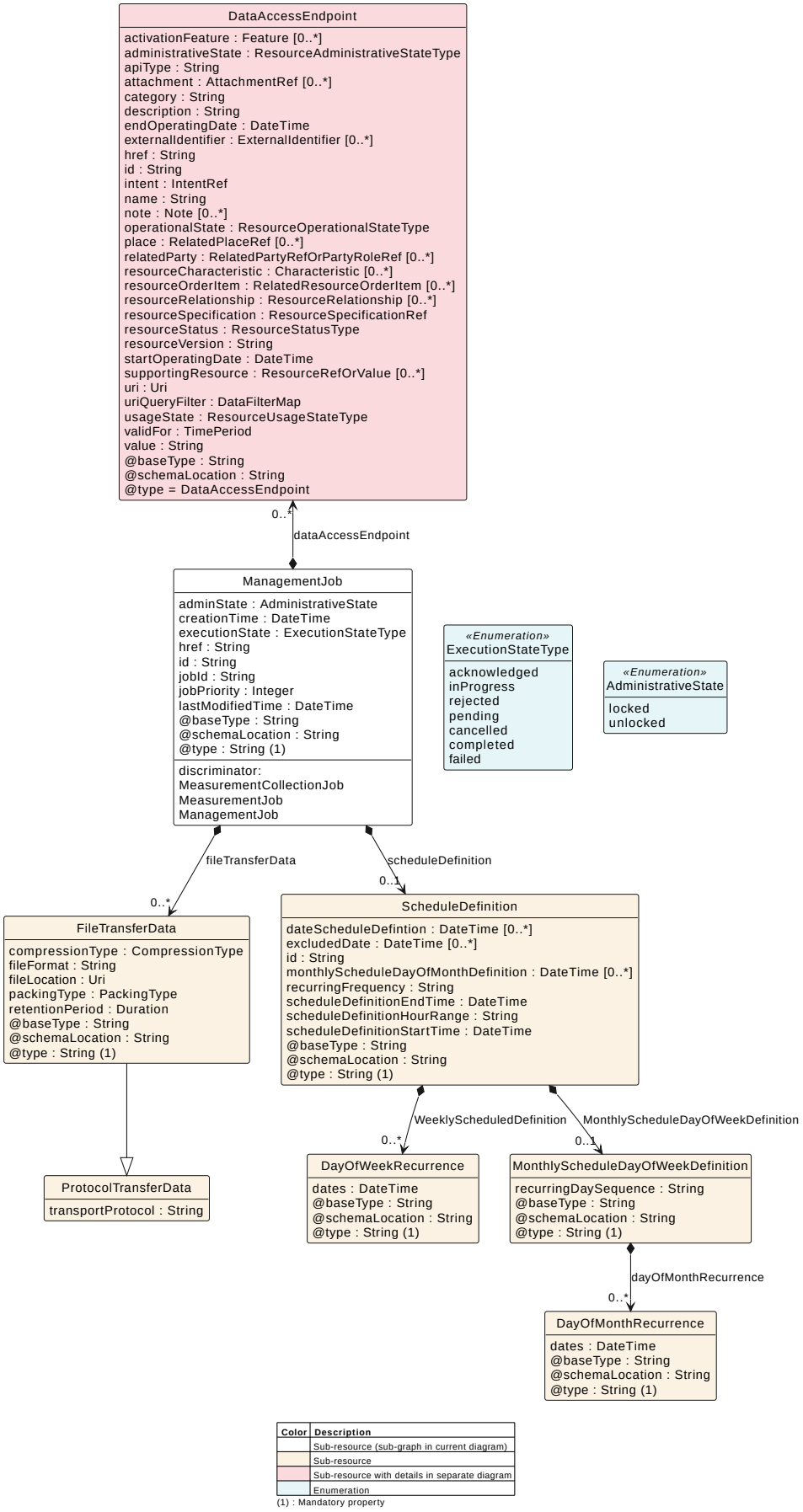


Figure 3 - ManagementJob

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[.] | .//

Figure 4 - ManagementJob

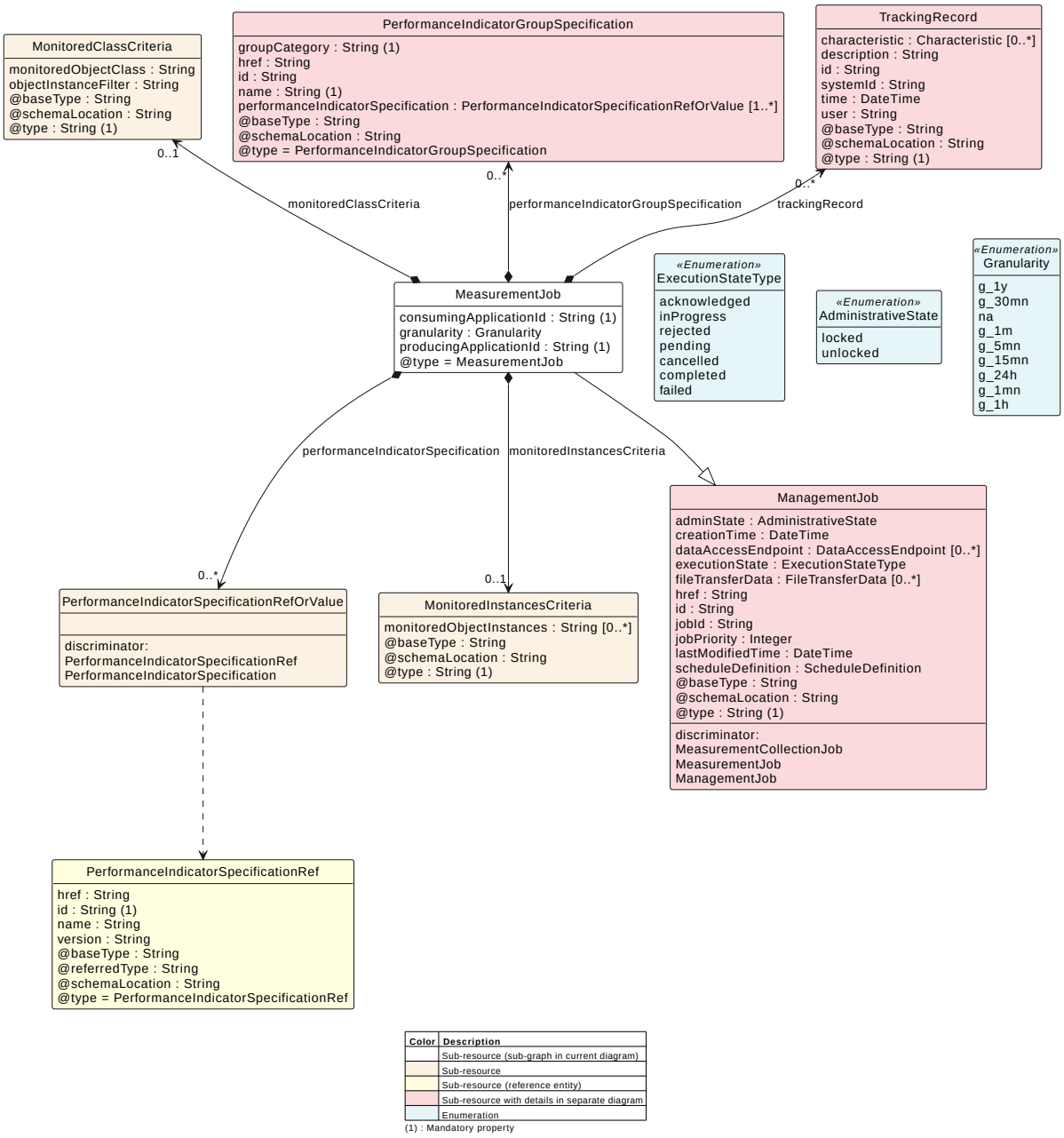


Figure 5 - MeasurementJob

[.] | .//

Figure 6 - MeasurementJob

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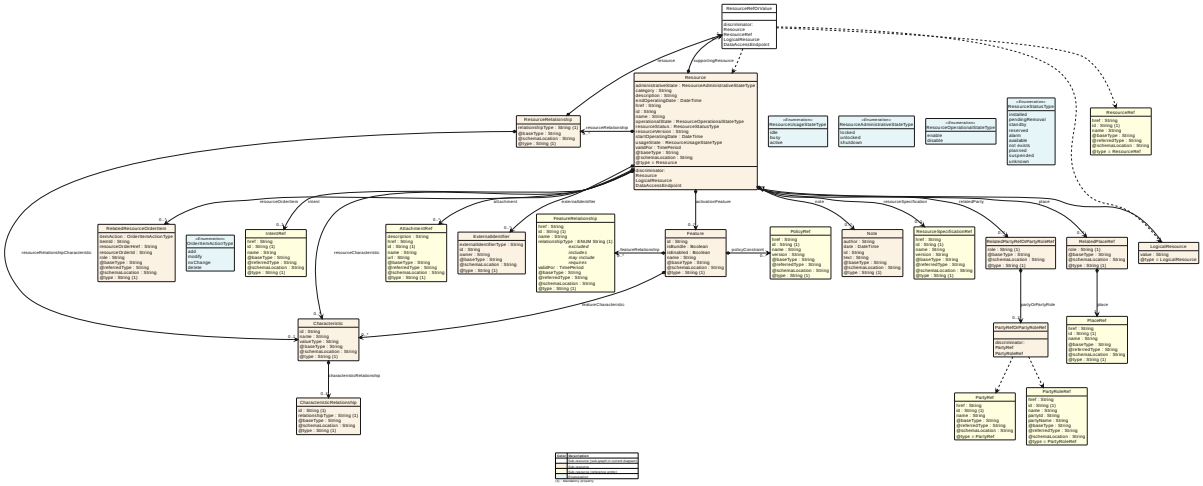


Figure 7 - ResourceRefOrValue

[.] | .//

Figure 8 - ResourceRefOrValue

Field descriptions

MeasurementCollectionJob fields

adminState	An AdministrativeState. This is enumeration for Administrative state. ENUMERATED with values: * locked * unlocked
consumingApplicationId	A String. The identifier of the application that consumes performance indicators.
creationTime	A DateTime. The measurement job creation time.
dataAccessEndpoint	A DataAccessEndpoint. This is the endpoint exposed by the IoT Device to authorized users.
executionState	An ExecutionStateType. Possible values for the state of the execution. ENUMERATED with values: * acknowledged * rejected * pending * inProgress * cancelled * completed * failed
fileTransferData	A FileTransferData.

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granularity	A Granularity. Sampling rate of the collection or production of performance indicators. ENUMERATED with values: * g_1mn * g_5mn * g_15mn * g_30mn * g_1h * g_24h * g_1m * g_1y * na
href	A String. Hyperlink reference.
id	A String. Unique identifier.
jobCollectionFilter	A DataFilterMap. Entity to Describe a given filter attributes/fields set using a mapping by type/template.
jobId	A String. The ID of the management job.
jobOnDemand	A Boolean. True if the job is a single job to be executed immediately in which case the reportingPeriod and scheduleDefinition would not be applicable, false otherwise.
jobPriority	An Integer. The priority of the management job. The way the management application will use the JobPriority to schedule job execution is application specific and outside the scope. Integer, limited to a range of 1 to 10.
lastModifiedTime	A DateTime. The last time that a measurement job was modified.
monitoredClassCriteria	A MonitoredClassCriteria. Specifies a monitored object class (a string) in conjunction with a filter object.
monitoredInstancesCriteria	A MonitoredInstancesCriteria. List of monitored instances.
outputFormat	A String.
performanceIndicatorGroupSpecification	A PerformanceIndicatorGroupSpecification.
performanceIndicatorSpecification	A PerformanceIndicatorSpecificationRefOrValue. A product to be created defined by value or existing defined by reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the product entity and not the RelatedProductRefOrValue class itself.
producingApplicationId	A String. The identifier of the application that produces performance indicators.
reportingPeriod	A ReportingPeriod. Possible values for the reporting period. ENUMERATED with values: * r_1mn * r_5mn * r_15mn * r_30mn * r_1h * r_24h * na
scheduleDefinition	A ScheduleDefinition. The schedule definition for running jobs.

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searchTaskFilter	A DataFilterMap. Entity to Describe a given filter attributes/fields set using a mapping by type/template.
trackingRecord	A TrackingRecord. Tracking records allow the tracking of modifications on the problem. The tracking records should not be embedded in the problem to allow retrieving the problem without the tracking records.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

DataFilterMap sub-resource fields

mappings	A DataFilterMapItem. Definition of a field used for filtering template with the associated attributes.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

DataFilterAttributeStringArray sub-resource fields

value	A String. An array of field names.
valueType	A String. Data type of the value of the attribute.

DataFilterTemplate sub-resource fields

description	A String. Description of this DataFilterTemplate.
href	A Uri. Hyperlink reference to this DataFilterTemplate.
id	A String. Unique identifier of this DataFilterTemplate.
name	A String. Name given to this DataFilterTemplate.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

MeasurementJob sub-resource fields

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adminState	An AdministrativeState. This is enumeration for Administrative state. ENUMERATED with values: * locked * unlocked
consumingApplicationId	A String. The identifier of the application that consumes performance indicators.
creationTime	A DateTime. The measurement job creation time.
dataAccessEndpoint	A DataAccessEndpoint. This is the endpoint exposed by the IoT Device to authorized users.
executionState	An ExecutionStateType. Possible values for the state of the execution. ENUMERATED with values: * acknowledged * rejected * pending * inProgress * cancelled * completed * failed
fileTransferData	A FileTransferData.
granularity	A Granularity. Sampling rate of the collection or production of performance indicators. ENUMERATED with values: * g_1mn * g_5mn * g_15mn * g_30mn * g_1h * g_24h * g_1m * g_1y * na
href	A String. Hyperlink reference.
id	A String. Unique identifier.
jobId	A String. The ID of the management job.
jobPriority	An Integer. The priority of the management job. The way the management application will use the JobPriority to schedule job execution is application specific and outside the scope. Integer, limited to a range of 1 to 10.
lastModifiedTime	A DateTime. The last time that a measurement job was modified.
monitoredClassCriteria	A MonitoredClassCriteria. Specifies a monitored object class (a string) in conjunction with a filter object.
monitoredInstancesCriteria	A MonitoredInstancesCriteria. List of monitored instances.
performanceIndicatorGroupSpecification	A PerformanceIndicatorGroupSpecification.

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performanceIndicatorSpecification	A PerformanceIndicatorSpecificationRefOrValue. A product to be created defined by value or existing defined by reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the product entity and not the RelatedProductRefOrValue class itself.
producingApplicationId	A String. The identifier of the application that produces performance indicators.
scheduleDefinition	A ScheduleDefinition. The schedule definition for running jobs.
trackingRecord	A TrackingRecord. Tracking records allow the tracking of modifications on the problem. The tracking records should not be embedded in the problem to allow retrieving the problem without the tracking records.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	MeasurementJob can be instanciated as * MeasurementCollectionJob
jobCollectionFilter	This property is present in subclasses
jobOnDemand	This property is present in subclasses
outputFormat	This property is present in subclasses
reportingPeriod	This property is present in subclasses
searchTaskFilter	This property is present in subclasses

ManagementJob sub-resource fields

adminState	An AdministrativeState. This is enumeration for Administrative state. ENUMERATED with values: * locked * unlocked
creationTime	A DateTime. The measurement job creation time.
dataAccessEndpoint	A DataAccessEndpoint. This is the endpoint exposed by the IoT Device to authorized users.
executionState	An ExecutionStateType. Possible values for the state of the execution. ENUMERATED with values: * acknowledged * rejected * pending * inProgress * cancelled * completed * failed
fileTransferData	A FileTransferData.
href	A String. Hyperlink reference.

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id	A String. Unique identifier.
jobId	A String. The ID of the management job.
jobPriority	An Integer. The priority of the management job. The way the management application will use the JobPriority to schedule job execution is application specific and outside the scope. Integer, limited to a range of 1 to 10.
lastModifiedTime	A DateTime. The last time that a measurement job was modified.
scheduleDefinition	A ScheduleDefinition. The schedule definition for running jobs.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	ManagementJob can be instanciated as * MeasurementJob
consumingApplicationId	This property is present in subclasses
granularity	This property is present in subclasses
monitoredClassCriteria	This property is present in subclasses
monitoredInstancesCriteria	This property is present in subclasses
performanceIndicatorGroupSpecification	This property is present in subclasses
performanceIndicatorSpecification	This property is present in subclasses
producingApplicationId	This property is present in subclasses
trackingRecord	This property is present in subclasses

Json representation sample(s)

We provide below a JSON representation as example of the 'MeasurementCollectionJob' resource object.


```
{
  "id": "4000",
  "href": "https://host:port/tmf-
api/performanceManagement/v5/MeasurementCollectionJob/4000",
  "consumingApplicationId": "4000",
  "creationTime": "2020-06-23T14:12:01.078Z",
  "jobId": "400",
  "jobPriority": 3,
  "jobOnDemand": false,
  "lastModifiedTime": "2020-06-23T14:12:01.079Z",
  "outputFormat": "string",
  "producingApplicationId": "NT_101",
  "adminState": "locked",
  "executionState": "acknowledged",
  "fileTransferData": [
    {
      "fileFormat": "csv",
```


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```
      "fileLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
      "transportProtocol": "ftp",
      "compressionType": {
        "compressionEnumType": "no_compression"
      },
      "packingType": {
        "packingEnumType": "NO_PACKING"
      },
      "retentionPeriod": {
        "amount": 1,
        "units": "month",
        "@baseType": "RetentionPeriod",
        "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/RetentionPeriod",
        "@type": "RetentionPeriod"
      },
      "@baseType": "FileTransferData",
      "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/FileTransferData",
      "@type": "FileTransferData"
    }
  ],
  "granularity": "g_1mn",
  "monitoredClassCriteria": {
    "monitoredObjectClass": "MeasurementCollectionJob",
    "objectInstanceFilter": "outputFormat=string",
    "@baseType": "MonitoredClassCriteria",
    "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/MonitoredClassCriteria",
    "@type": "MonitoredClassCriteria"
  },
  "monitoredInstancesCriteria": {
    "monitoredObjectInstances": ["MeasurementCollectionJob_Instance1"],
    "@baseType": "MonitoredInstancesCriteria",
    "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/MonitoredInstancesCriteria",
    "@type": "MonitoredInstancesCriteria"
  },
  "performanceIndicatorGroupSpecification": [
    {
      "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSGrouppecification/600",
      "id": "600",
      "name": "My critical performance indicators",
      "performanceIndicatorSpecification": [
        {
          "id": "5000",
          "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5000",
          "name": "CPU speed",
          "version": "1.0",
          "@baseType": "PerformanceIndicatorSpecification",
          "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
          "@type": "PerformanceIndicatorSpecification"
        }
      ],
      "@baseType": "PerformanceIndicatorGroupSpecification",
      "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorGroupSpecificati
on.schema.json",
```


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```
      "@type": "PerformanceIndicatorGroupSpecification"
    }
  ],
  "performanceIndicatorSpecification": [
    {
      "id": "200",
      "href": "https://host:port/tmf-
api/performanceManagement/v5/PerformanceIndicatorSpecification/200",
      "derivationAlgorithm": "na",
      "derivationMethod": "average",
      "description": "The average CPU speed",
      "indicatorCategory": "CPU",
      "indicatorUnit": "GHz",
      "name": "CPU Speed",
      "perspective": "aggregation over a collection of systems",
      "collectionType": "cumulative",
      "indicatorType": "float",
      "performanceIndicatorSpecRelationship": [
        {
          "id": "321",
          "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecRelationship/321",
          "relationshipType": "reliesOn",
          "validFor": {
            "endDateTime": "1985-04-12T23:20:50.52Z",
            "startDateTime": "1985-04-12T23:20:50.52Z"
          },
          "@baseType": "PerformanceIndicatorSpecRelationship",
          "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecRelationship
.schema.json",
          "@type": "PerformanceIndicatorSpecRelationship"
        }
      ],
      "validFor": {
        "endDateTime": "1985-04-12T23:20:50.52Z",
        "startDateTime": "1985-04-12T23:20:50.52Z"
      },
      "@baseType": "PerformanceIndicatorSpecification",
      "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
      "@type": "PerformanceIndicatorSpecification"
    }
  ],
  "reportingPeriod": "r_1mn",
  "scheduleDefinition": [{
    "id": "1",
    "recurringFrequency": "weekly",
    "scheduleDefinitionEndTime": "2020-06-23T14:12:01.080Z",
    "scheduleDefinitionHourRange": "2",
    "scheduleDefinitionStartTime": "2020-06-23T14:12:01.080Z",
    "WeeklyScheduledDefinition": [
      {
        "dates": "2020-06-23T14:12:01.080Z",
        "@baseType": "WeeklyScheduledDefinition",
        "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/WeeklyScheduledDefinition.schema.jso
n",
        "@type": "WeeklyScheduledDefinition"
      }
    ],
    "@baseType": "ScheduleDefinition",
```


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```
    "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/ScheduleDefinition.schema.json",
    "@type": "ScheduleDefinition"
  }],
  "trackingRecord": [
  {
    "id": "TR_101",
    "description": "my tracking records",
    "systemId": "NT_101",
    "time": "2020-06-23T17:55:51.671Z",
    "user": "NP1",
    "@baseType": "TrackingRecord",
    "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/TrackingRecord",
    "@type": "TrackingRecord"
  }
  ],
  "@baseType": "MeasurementCollectionJob",
  "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/MeasurementCollectionJob",
  "@type": "MeasurementCollectionJob"
}
```


We provide below a JSON representation as example of the 'MeasurementCollectionJob' resource object.


```
{
  "id": "4000",
  "href": "https://host:port/tmf-
api/performanceManagement/v5/MeasurementCollectionJob/4000",
  "consumingApplicationId": "4000",
  "creationTime": "2020-06-23T14:12:01.078Z",
  "jobId": "400",
  "jobPriority": 3,
  "jobOnDemand": false,
  "lastModifiedTime": "2020-06-23T14:12:01.079Z",
  "outputFormat": "string",
  "producingApplicationId": "NT_101",
  "adminState": "locked",
  "executionState": "acknowledged",
  "dataAccessEndpoint": [
    {
        "id": "123-xxx",
        "href": "https://host:port/tmf-
api/performanceManagement/v5/dataAccessEndpoint/123-xxx",
        "apiType": "REST",
        "category": "topic",
        "description": "Measurement collection job 123-xxx subscriber topic",
        "endOperatingDate": "2022-05-13T10:13:42.971Z",
        "name": "123-xxx subscriber topic",
        "resourceVersion": "1.0",
        "startOperatingDate": "2022-05-13T10:13:42.971Z",
        "uri": "https://pubsub.googleapis.com/v1/",
        "value": "myLogicalResourceValueIp"
    }
  ],
  "fileTransferData": [
    {
      "fileFormat": "csv",
      "fileLocation": "https://host:port/tmf-
```


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```
api/performanceManagement/v5/fileTransferLocation",
      "transportProtocol": "ftp",
      "compressionType": {
        "compressionEnumType": "no_compression"
      },
      "packingType": {
        "packingEnumType": "NO_PACKING"
      },
      "retentionPeriod": {
        "amount": 1,
        "units": "month",
        "@baseType": "RetentionPeriod",
        "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/RetentionPeriod",
        "@type": "RetentionPeriod"
      },
      "@baseType": "FileTransferData",
      "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/FileTransferData",
      "@type": "FileTransferData"
    }
  ],
  "granularity": "g_1mn",
  "monitoredClassCriteria": {
    "monitoredObjectClass": "MeasurementCollectionJob",
    "objectInstanceFilter": "outputFormat=string",
    "@baseType": "MonitoredClassCriteria",
    "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/MonitoredClassCriteria",
    "@type": "MonitoredClassCriteria"
  },
  "monitoredInstancesCriteria": {
    "monitoredObjectInstances": ["MeasurementCollectionJob_Instance1"],
    "@baseType": "MonitoredInstancesCriteria",
    "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/MonitoredInstancesCriteria",
    "@type": "MonitoredInstancesCriteria"
  },
  "performanceIndicatorGroupSpecification": [
    {
      "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSGrouppecification/600",
      "id": "600",
      "name": "My critical performance indicators",
      "performanceIndicatorSpecification": [
        {
          "id": "5000",
          "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5000",
          "name": "CPU speed",
          "version": "1.0",
          "@baseType": "PerformanceIndicatorSpecification",
          "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
          "@type": "PerformanceIndicatorSpecification"
        }
      ],
      "@baseType": "PerformanceIndicatorGroupSpecification",
      "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorGroupSpecificati
on.schema.json",
      "@type": "PerformanceIndicatorGroupSpecification"
```


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```
    }
  ],
  "performanceIndicatorSpecification": [
    {
      "id": "200",
      "href": "https://host:port/tmf-
api/performanceManagement/v5/PerformanceIndicatorSpecification/200",
      "derivationAlgorithm": "na",
      "derivationMethod": "average",
      "description": "The average CPU speed",
      "indicatorCategory": "CPU",
      "indicatorUnit": "GHz",
      "name": "CPU Speed",
      "perspective": "aggregation over a collection of systems",
      "collectionType": "cumulative",
      "indicatorType": "float",
      "performanceIndicatorSpecRelationship": [
        {
          "id": "321",
          "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecRelationship/321",
          "relationshipType": "reliesOn",
          "validFor": {
            "endDateTime": "1985-04-12T23:20:50.52Z",
            "startDateTime": "1985-04-12T23:20:50.52Z"
          },
          "@baseType": "PerformanceIndicatorSpecRelationship",
          "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecRelationship
.schema.json",
          "@type": "PerformanceIndicatorSpecRelationship"
        }
      ],
      "validFor": {
        "endDateTime": "1985-04-12T23:20:50.52Z",
        "startDateTime": "1985-04-12T23:20:50.52Z"
      },
      "@baseType": "PerformanceIndicatorSpecification",
      "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
      "@type": "PerformanceIndicatorSpecification"
    }
  ],
  "reportingPeriod": "r_1mn",
  "scheduleDefinition": [{
    "id": "1",
    "recurringFrequency": "weekly",
    "scheduleDefinitionEndTime": "2020-06-23T14:12:01.080Z",
    "scheduleDefinitionHourRange": "2",
    "scheduleDefinitionStartTime": "2020-06-23T14:12:01.080Z",
    "WeeklyScheduledDefinition": [
      {
        "dates": "2020-06-23T14:12:01.080Z",
        "@baseType": "WeeklyScheduledDefinition",
        "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/WeeklyScheduledDefinition.schema.jso
n",
        "@type": "WeeklyScheduledDefinition"
      }
    ],
    "@baseType": "ScheduleDefinition",
    "@schemaLocation": "https://host:port/tmf-
```


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```
api/performanceManagement/v5/schemas/Tmf/common/ScheduleDefinition.schema.json",
    "@type": "ScheduleDefinition"
  }],
  "trackingRecord": [
   {
    "id": "TR_101",
    "description": "my tracking records",
    "systemId": "NT_101",
    "time": "2020-06-23T17:55:51.671Z",
    "user": "NP1",
    "@baseType": "TrackingRecord",
    "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/TrackingRecord",
    "@type": "TrackingRecord"
   }
  ],
  "jobCollectionFilter":{
    "@type" : "DataFilterMap",
    "mappings": [
        {
            "filterTemplate": {
                "name" : "myRangeFilterCollect0",
                "@type" : "DataFilterTemplateRange",
                "criteria":{
                    "lowerBound" : {
                        "boundValue": -10,
                        "isInclusive": true
                    },
                    "upperBound" : {
                        "boundValue": -10,
                        "isInclusive": true
                    }
                }
            },
            "stringArray": {
                "value": [
                "field01", "field02", "field03"
                 ]
            }
        },
        {
            "filterTemplate": {
                "name": "myRegExSampleCollect0",
                "@type": "DataFilterTemplateRegex",
                "criteria": "dallas|houston|venise|paris|new-york"
            },
            "stringArray": {
                "value":[
                    "field11", "field12", "field13"
                ]
            }
        },
        {
            "filterTemplate": {
                "name" : "myBooleanFilterCollect0",
                "criteria" : true,
                "@type" : "DataFilterTemplateBoolean"
            },
            "stringArray": {
                "value": [
                    "field1", "field2", "field3"
                ]
            }
```

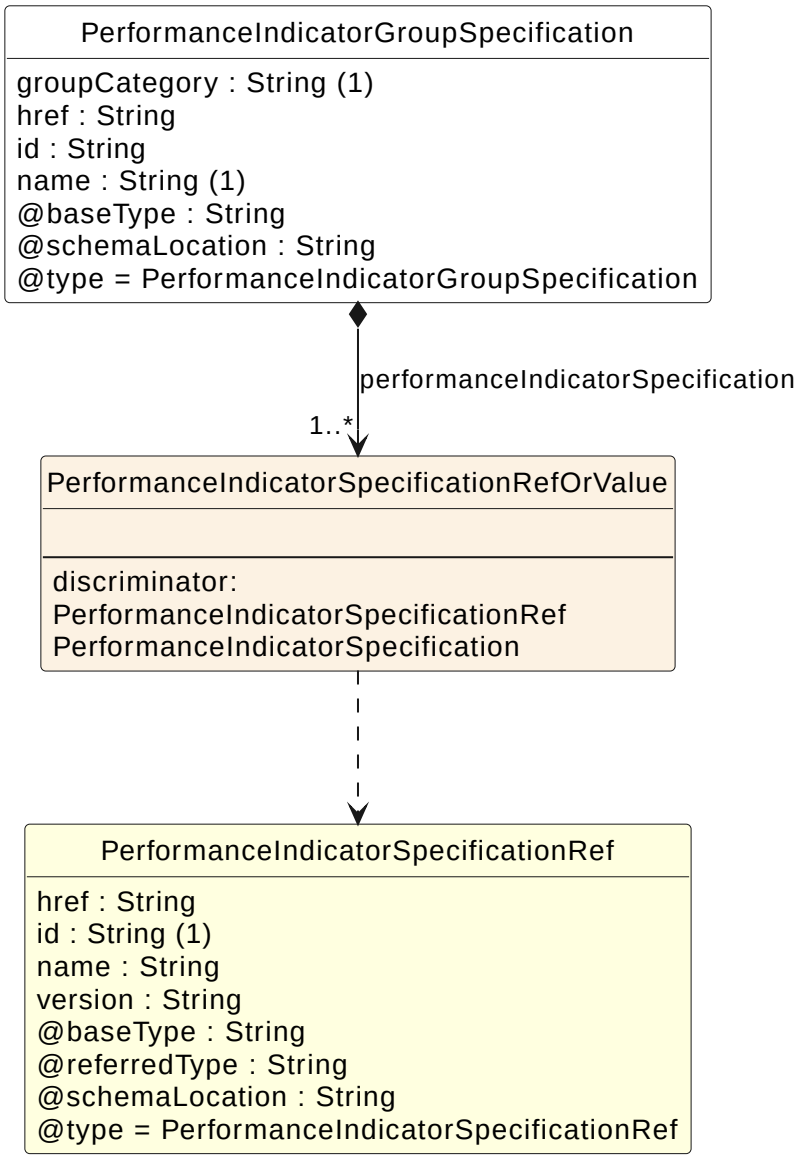
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```
        }
      ]
    },
    "searchTaskFilter":{
      "@type" : "DataFilterMap",
      "mappings": [
          {
              "filterTemplate": {
                  "name" : "myRangeFilterSearch0",
                  "@type" : "DataFilterTemplateRange",
                  "criteria":{
                      "lowerBound" : {
                          "boundValue": -10,
                          "isInclusive": true
                      },
                      "upperBound" : {
                          "boundValue": -10,
                          "isInclusive": true
                      }
                  }
              },
              "stringArray": {
                  "value": [
                  "field01", "field02", "field03"
                   ]
              }
          },
          {
              "filterTemplate": {
                  "name": "myRegExSampleSearch0",
                  "@type": "DataFilterTemplateRegex",
                  "criteria": "dallas|houston|venise|paris|new-york"
              },
              "stringArray": {
                  "value":[
                      "field11", "field12", "field13"
                  ]
              }
          },
          {
              "filterTemplate": {
                  "name" : "myBooleanFilterSearch0",
                  "criteria" : true,
                  "@type" : "DataFilterTemplateBoolean"
              },
              "stringArray": {
                  "value": [
                      "field1", "field2", "field3"
                  ]
              }
          }
      ]
    },
    "@baseType": "MeasurementCollectionJob",
    "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/MeasurementCollectionJob",
    "@type": "MeasurementCollectionJob"
}
```


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PerformanceIndicatorGroupSpecification resource

Resource model



Color	Description
	Resource (entry point)
	Sub-resource
	Sub-resource (reference entity)

(1) : Mandatory property

Figure 9 - PerformanceIndicatorGroupSpecification

[.] | .//

Figure 10 - PerformanceIndicatorGroupSpecification

Field descriptions

PerformanceIndicatorGroupSpecification fields

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groupCategory	A String. A grouping or set of PerformanceIndicatorGroupSpecifications that are classified together because of common characteristics, such as technology specific, service specific, or technology/service independent.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. A word, term, or phrase by which a PerformanceIndicatorGroupSpecification is tagged.
performanceIndicatorSpecification	A PerformanceIndicatorSpecificationRefOrValue. A product to be created defined by value or existing defined by reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the product entity and not the RelatedProductRefOrValue class itself.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PerformanceIndicatorSpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String. Version of the performance indicator specification.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PerformanceIndicatorSpecification sub-resource fields

collectionType	A CollectionType. This is enumeration for CollectionType state. ENUMERATED with values: * cumulative * delta * discrete_event * gauge * status_inspection
derivationAlgorithm	A String. A step-by-step procedure used to calculate the value of PerformanceIndicator.

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derivationMethod	A String. For simple calculations, the method used to calculate the value of a PerformanceIndicator, such as average, minimum, maximum, sum and so forth.
description	A String. A narrative that explains in detail what the PerformanceIndicatorSpecification is.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
indicatorCategory	A String. A grouping or set of PerformanceIndicatorSpecifications that are classified together because of common characteristics, such as technology specific, service specific, or technology/service independent.
indicatorType	An IndicatorType. This is enumeration for Indicator Type. ENUMERATED with values: * string * int * float * double
indicatorUnit	A String. The unit by which the indicator is measured. For example, seconds, KBs, rate per second, etc.
name	A String. A word, term, or phrase by which a PerformanceIndicatorSpecification is known and distinguished from other PerformanceIndicatorSpecifications.
performanceIndicatorSpecRelationship	A PerformanceIndicatorSpecRelationship. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.
perspective	A String. The point of view for the PerformanceIndicatorSpecification, such as a single user instance or aggregation.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PerformanceIndicatorSpecRelationship sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
relationshipType	A String.
role	A String. The association role for this service specification.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.

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@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'PerformanceIndicatorGroupSpecification' resource object.


```
{
  "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSGrouppecification/600",
  "id": "600",
  "name": "My critical performance indicators",
  "performanceIndicatorSpecification": [
    {
      "id": "5000",
      "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5000",
      "name": "CPU speed core 1",
      "version": "1.0",
      "@baseType": "PerformanceIndicatorSpecification",
      "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
      "@type": "PerformanceIndicatorSpecification"
    },
    {
      "id": "5001",
      "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5001",
      "name": "CPU speed core 2",
      "version": "1.0",
      "@baseType": "PerformanceIndicatorSpecification",
      "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
      "@type": "PerformanceIndicatorSpecification"
    },
    {
      "id": "5002",
      "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5002",
      "name": "CPU speed core 3",
      "version": "1.0",
      "@baseType": "PerformanceIndicatorSpecification",
      "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
      "@type": "PerformanceIndicatorSpecification"
    }
  ],
  "@baseType": "PerformanceIndicatorGroupSpecification",
  "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorGroupSpecificati
on.schema.json",
```


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```
    "@type": "PerformanceIndicatorGroupSpecification"
  }
```

PerformanceIndicatorSpecification resource

Holds the definition of the requested Performance indicator including: its name, value type, collection type, description and units.

Resource model

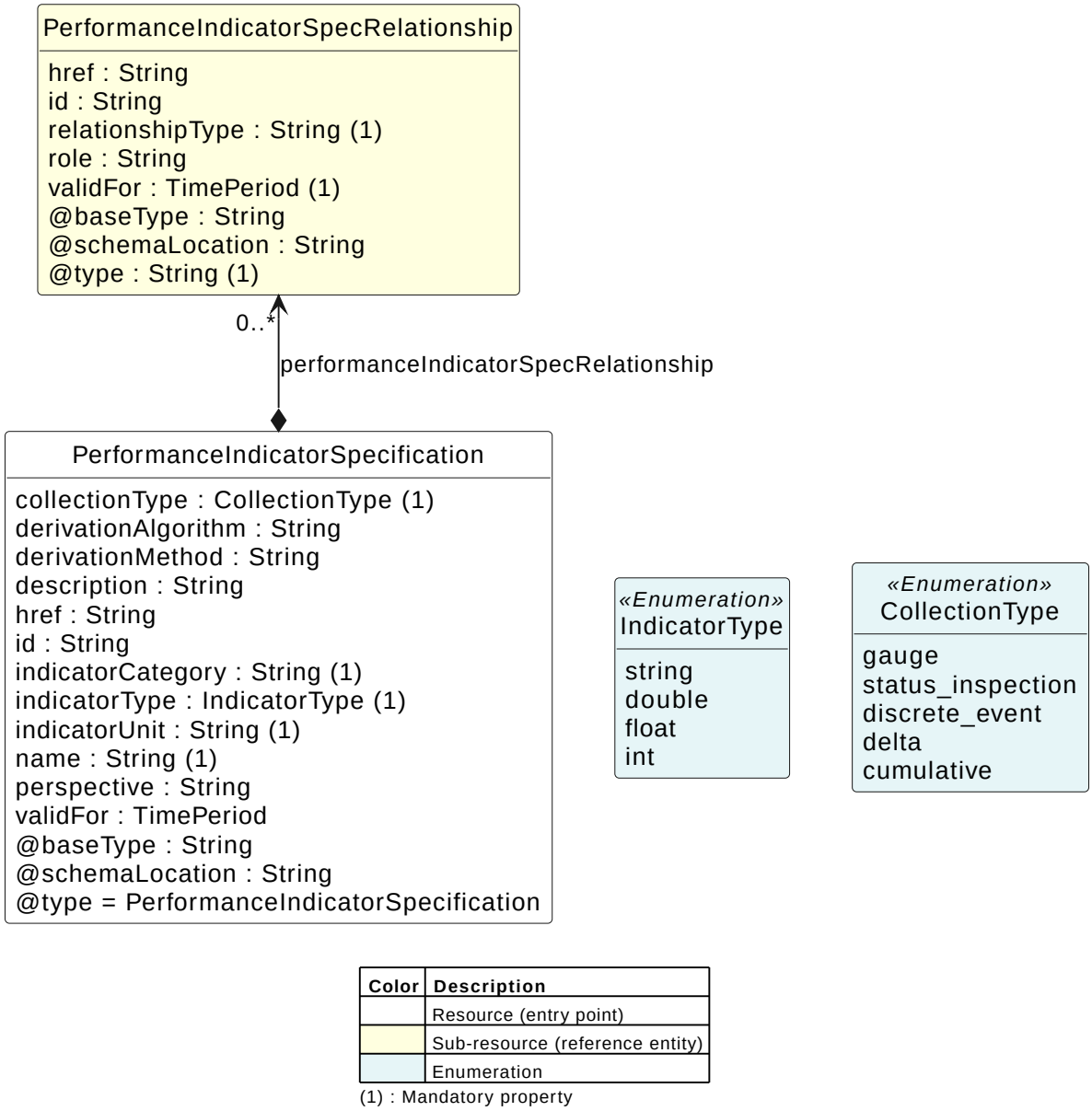


Figure 11 - PerformanceIndicatorSpecification

[.] | .//

Figure 12 - PerformanceIndicatorSpecification

Field descriptions

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PerformanceIndicatorSpecification fields

collectionType	A CollectionType. This is enumeration for CollectionType state. ENUMERATED with values: * cumulative * delta * discrete_event * gauge * status_inspection
derivationAlgorithm	A String. A step-by-step procedure used to calculate the value of PerformanceIndicator.
derivationMethod	A String. For simple calculations, the method used to calculate the value of a PerformanceIndicator, such as average, minimum, maximum, sum and so forth.
description	A String. A narrative that explains in detail what the PerformanceIndicatorSpecification is.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
indicatorCategory	A String. A grouping or set of PerformanceIndicatorSpecifications that are classified together because of common characteristics, such as technology specific, service specific, or technology/service independent.
indicatorType	An IndicatorType. This is enumeration for Indicator Type. ENUMERATED with values: * string * int * float * double
indicatorUnit	A String. The unit by which the indicator is measured. For example, seconds, KBs, rate per second, etc.
name	A String. A word, term, or phrase by which a PerformanceIndicatorSpecification is known and distinguished from other PerformanceIndicatorSpecifications.
performanceIndicatorSpecRelationship	A PerformanceIndicatorSpecRelationship. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.
perspective	A String. The point of view for the PerformanceIndicatorSpecification, such as a single user instance or aggregation.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

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PerformanceIndicatorSpecRelationship sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
relationshipType	A String.
role	A String. The association role for this service specification.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'PerformanceIndicatorSpecification' resource object.

```
{
    "id": "5000",
    "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5000",
    "name": "CPU speed",
    "description": "The average CPU speed",
    "perspective": "aggregation over a collection of systems",
    "indicatorCategory": "CPU",
    "indicatorType": "string",
    "derivationAlgorithm": "na",
    "derivationMethod": "average",
    "validFor": {
        "startDateTime": "2023-04-01T00:00:00.001Z",
        "endDateTime": "2023-04-30T23:59:59.999Z"
        },
    "collectionType": "cumulative",
    "indicatorUnit": "GHz",
    "performanceIndicatorSpecRelationship": [
        {
        "id": "string",
        "href": "string",
        "name": "string",
        "relationshipType": "string",
        "role": "string",
        "validFor": {
            "startDateTime": "2023-04-01T00:00:00.001Z",
            "endDateTime": "2023-04-30T23:59:59.999Z"
        },
        "@baseType": "PerformanceIndicatorSpecRelationship",
        "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecRelationship
.schema.json",
        "@type": "PerformanceIndicatorSpecRelationship"
        }
    ],
```

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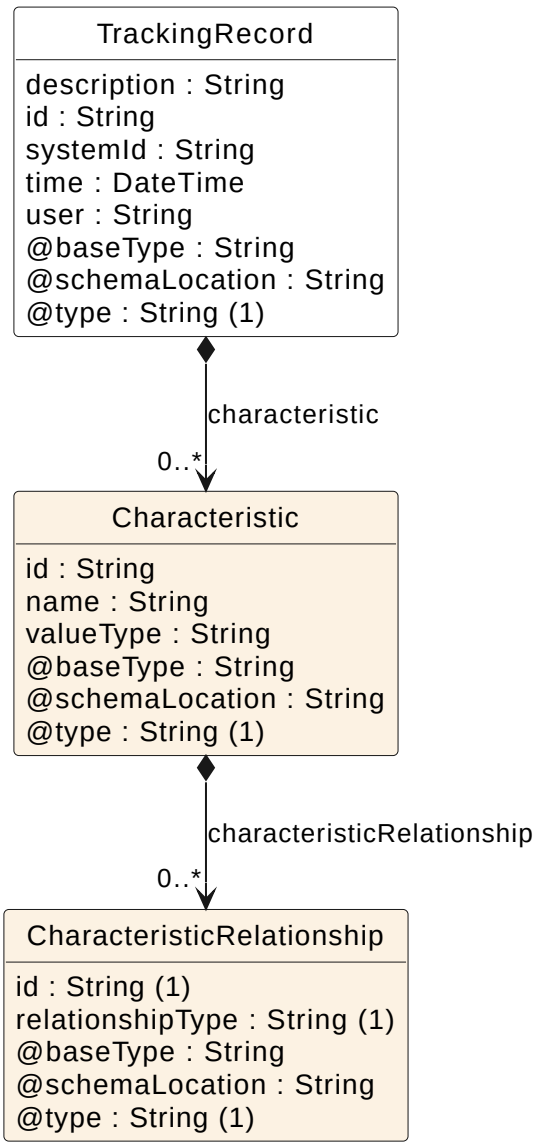
```
    "@baseType": "PerformanceIndicatorSpecification",
    "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
    "@type": "PerformanceIndicatorSpecification"
}
```

TrackingRecord resource

Tracking records allow the tracking of modifications on the problem. The tracking records should not be embedded in the problem to allow retrieving the problem without the tracking records.

Resource model

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Color	Description
	Resource (entry point)
	Sub-resource

(1) : Mandatory property

Figure 13 - TrackingRecord

[.] | .//

Figure 14 - TrackingRecord

Field descriptions

TrackingRecord fields

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characteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
description	A String. Describes the action being done, such as: ack, clear.
id	A String. Identifier of the TrackingRecord.
systemId	A String. Describes the system Id from which the action was done.
time	A DateTime. Describes the time at which the action was done.
user	A String. Describes the user doing the action.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Characteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CharacteristicRelationship sub-resource fields

id	A String. Unique identifier of the characteristic.
relationshipType	A String. The type of relationship.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

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Json representation sample(s)

We provide below a JSON representation as example of the 'TrackingRecord' resource object.


```
{
  "id": "1001",
  "description": "acknowledge ,clear",
  "systemId": "5001",
  "time": "2022-05-15T15:51:04.176Z",
  "user": "myUserName",
  "characteristic": [
    {
      "id": "001",
      "name": "myCharName",
      "valueType": "string",
      "characteristicRelationship": [
        {
          "id": "400",
          "href": "https://host:port/tmf-
api/performanceManagement/v5/characteristicRelationship/400",
          "relationshipType": "sub-part",
          "@baseType": "CharacteristicRelationship",
          "@schemaLocation":
"https://myschemas.com/CharacteristicRelationship.schema.json",
          "@type": "CharacteristicRelationship"
        }
      ],
      "value": "myCharactericticValue",
      "@baseType": "Characteristic",
      "@schemaLocation": "https://myschemas.com/StringCharacteristic.schema.json",
      "@type": "StringCharacteristic"
    }
  ],
  "@baseType": "TrackingRecord",
  "@schemaLocation": "https://myschemas.com/TrackingRecord.schema.json",
  "@type": "TrackingRecord"
}
```


PerformanceMeasurement resource

A measurement of a performance indicator or performance group indicator.

Resource model

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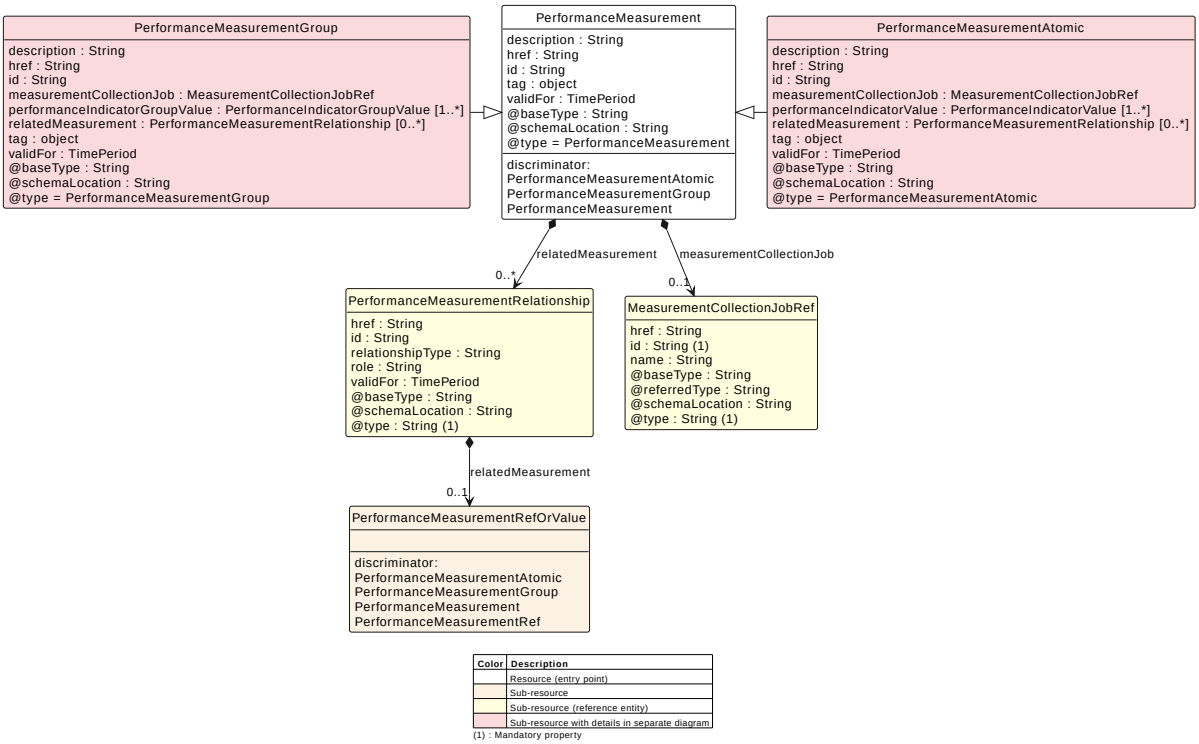


Figure 15 - PerformanceMeasurement

[.] | .//

Figure 16 - PerformanceMeasurement

Field descriptions

PerformanceMeasurement fields

description	A String. A free-text description of the performance measurement.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
measurementCollectionJob	A MeasurementCollectionJobRef. Reference to a MeasurementCollectionJob.
relatedMeasurement	A PerformanceMeasurementRelationship. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.
tag	An object. The optional tag object attached to this entire measurement.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String. When sub-classing, this defines the sub-class Extensible name.
	PerformanceMeasurement can be instanciated as * PerformanceMeasurementAtomic * PerformanceMeasurementGroup
performanceIndicatorGroupValue	This property is present in subclasses
performanceIndicatorValue	This property is present in subclasses

PerformanceMeasurementRelationship sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
relatedMeasurement	A PerformanceMeasurementRefOrValue. A PerformanceMeasurement to be created defined by value or existing defined by reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the product entity and not the RelatedProductRefOrValue class itself.
relationshipType	A String.
role	A String. The association role for this PerformanceMeasurement.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

MeasurementCollectionJobRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PerformanceMeasurementAtomic sub-resource fields

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description	A String. A free-text description of the performance measurement.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
measurementCollectionJob	A MeasurementCollectionJobRef. Reference to a MeasurementCollectionJob.
performanceIndicatorValue	A PerformanceIndicatorValue. A value of a performance indicator.
relatedMeasurement	A PerformanceMeasurementRelationship. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.
tag	An object. The optional tag object attached to this entire measurement.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PerformanceMeasurementGroup sub-resource fields

description	A String. A free-text description of the performance measurement.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
measurementCollectionJob	A MeasurementCollectionJobRef. Reference to a MeasurementCollectionJob.
performanceIndicatorGroupValue	A PerformanceIndicatorGroupValue. A value of a performance indicator.
relatedMeasurement	A PerformanceMeasurementRelationship. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.
tag	An object. The optional tag object attached to this entire measurement.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

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PerformanceMeasurementRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String. Version of the PerformanceMeasurement.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PerformanceIndicatorSpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String. Version of the performance indicator specification.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PerformanceIndicatorSpecification sub-resource fields

collectionType	A CollectionType. This is enumeration for CollectionType state. ENUMERATED with values: * cumulative * delta * discrete_event * gauge * status_inspection
derivationAlgorithm	A String. A step-by-step procedure used to calculate the value of PerformanceIndicator.
derivationMethod	A String. For simple calculations, the method used to calculate the value of a PerformanceIndicator, such as average, minimum, maximum, sum and so forth.
description	A String. A narrative that explains in detail what the PerformanceIndicatorSpecification is.

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href	A String. Hyperlink reference.
id	A String. Unique identifier.
indicatorCategory	A String. A grouping or set of PerformanceIndicatorSpecifications that are classified together because of common characteristics, such as technology specific, service specific, or technology/service independent.
indicatorType	An IndicatorType. This is enumeration for Indicator Type. ENUMERATED with values: * string * int * float * double
indicatorUnit	A String. The unit by which the indicator is measured. For example, seconds, KBs, rate per second, etc.
name	A String. A word, term, or phrase by which a PerformanceIndicatorSpecification is known and distinguished from other PerformanceIndicatorSpecifications.
performanceIndicatorSpecRelationship	A PerformanceIndicatorSpecRelationship. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.
perspective	A String. The point of view for the PerformanceIndicatorSpecification, such as a single user instance or aggregation.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PerformanceIndicatorSpecRelationship sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
relationshipType	A String.
role	A String. The association role for this service specification.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

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PerformanceIndicatorGroupSpecification sub-resource fields

groupCategory	A String. A grouping or set of PerformanceIndicatorGroupSpecifications that are classified together because of common characteristics, such as technology specific, service specific, or technology/service independent.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. A word, term, or phrase by which a PerformanceIndicatorGroupSpecification is tagged.
performanceIndicatorSpecification	A PerformanceIndicatorSpecificationRefOrValue. A product to be created defined by value or existing defined by reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the product entity and not the RelatedProductRefOrValue class itself.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PerformanceIndicatorGroupSpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String. Version of the performance indicator group specification.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'PerformanceMeasurement' resource object.

```
{
    "id": "1",
    "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceMeasurement/1",
    "description": "The first performance measurement in the database",
```

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```
    "measurementCollectionJob": {
      "id": "4000",
      "href": "https://host:port/tmf-
api/performanceManagement/v5/measurementCollectionJob/4000",
      "name": "Measurement Collection Job from system A",
      "@baseType": "MeasurementCollectionJobRef",
      "@schemaLocation": "https://myschemas/MeasurementCollectionJob.schema.json",
      "@type": "MeasurementCollectionJobRef",
      "@referredType": "MeasurementCollectionJob"
    },
    "performanceIndicatorValue": [
        {
          "performanceIndicatorSpecification":{
            "id": "1",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/1",
            "@baseType": "PerformanceIndicatorSpecificationRef",
            "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
            "@type": "PerformanceIndicatorSpecificationRef",
            "@referredType": "PerformanceIndicatorSpecification"
          },
          "observedValue": "2.1",
          "tag":{
            "imsi" : "1111000000",
            "zip" : "93165"
          }
        },
        {
          "performanceIndicatorSpecification":{
            "name": "CPU Speed Indicator Specification",
            "indicatorCategory": "SPEEDCategory",
            "indicatorName": "CPU Speed",
            "indicatorType": "Gauge",
            "indicatorUnit": "GHz",
            "@baseType": "PerformanceIndicatorSpecification",
            "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
            "@type": "PerformanceIndicatorSpecification"
          },
          "observedValue": "3.25"
        }
    ],
    "validFor": {
      "endDateTime": "1985-04-12T23:20:50.52Z",
      "startDateTime": "1985-04-12T23:20:50.52Z"
    },
    "tag":{
      "country" : "Belgium",
      "region" : "North"
    },
    "@baseType": "PerformanceMeasurement",
    "@schemaLocation":
"https://github.com/repo/PerformanceMeasurementAtomic.schema.json",
    "@type": "PerformanceMeasurementAtomic"
  }
```


We provide below a JSON representation as example of the 'PerformanceMeasurement' resource object.

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```
{
    "id": "1",
    "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceMeasurement/1",
    "description": "The first performance measurement in the database",
    "measurementCollectionJob": {
      "id": "4000",
      "href": "https://host:port/tmf-
api/performanceManagement/v5/measurementCollectionJob/4000",
      "name": "Measurement Collection Job from system A",
      "@baseType": "MeasurementCollectionJobRef",
      "@schemaLocation": "https://myschemas/MeasurementCollectionJob.schema.json",
      "@type": "MeasurementCollectionJobRef",
      "@referredType": "MeasurementCollectionJob"
    },
    "performanceIndicatorGroupValue": [{
          "performanceIndicatorGroupSpecification" : {
              "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSGrouppecification/600",
              "id": "600",
              "name": "My critical performance indicators",
              "groupCategory": "CPUGroup",
              "performanceIndicatorSpecification": [
                {
                  "id": "5000",
                  "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5000",
                  "name": "CPU speed",
                  "version": "1.0",
                  "@baseType": "PerformanceIndicatorSpecification",
                  "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
                  "@type": "PerformanceIndicatorSpecification"
                }
              ],
              "@baseType": "PerformanceIndicatorGroupSpecification",
              "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorGroupSpecificati
on.schema.json",
              "@type": "PerformanceIndicatorGroupSpecification"
            },
            "observedGroupValue": [
                  {
                   "performanceIndicatorSpecification":{
                      "id": "5000",
                      "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5000",
                      "@baseType": "PerformanceIndicatorSpecificationRef",
                      "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecificationRef.schema.json",
                      "@type": "PerformanceIndicatorSpecificationRef",
                      "@referredType": "PerformanceIndicatorSpecification"
                    },
                    "observedValue": "2.1",
                    "tag":{
                      "imsi" : "1111000000",
                      "zip" : "93165"
                    }
                  },
                  {
                    "performanceIndicatorSpecification":{
```


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```
                        "id": "5000",
                        "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5000",
                        "@baseType": "PerformanceIndicatorSpecificationRef",
                        "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecificationRef.schema.json",
                        "@type": "PerformanceIndicatorSpecificationRef",
                        "@referredType": "PerformanceIndicatorSpecification"
                      },
                      "observedValue": "2.2",
                      "tag":{
                        "imsi" : "1111000001",
                        "zip" : "93165"
                      }
                    }
              ]
      }],
    "validFor": {
      "endDateTime": "1985-04-12T23:20:50.52Z",
      "startDateTime": "1985-04-12T23:20:50.52Z"
    },
    "tag":{
      "country" : "Belgium",
      "region" : "North"
    },
    "@baseType": "PerformanceMeasurement",
    "@schemaLocation":
"https://github.com/repo/PerformanceMeasurementGroup.schema.json",
    "@type": "PerformanceMeasurementGroup"
  }
```


DataAccessEndpoint resource

This is the endpoint exposed by the IoT Device to authorized users.

Resource model

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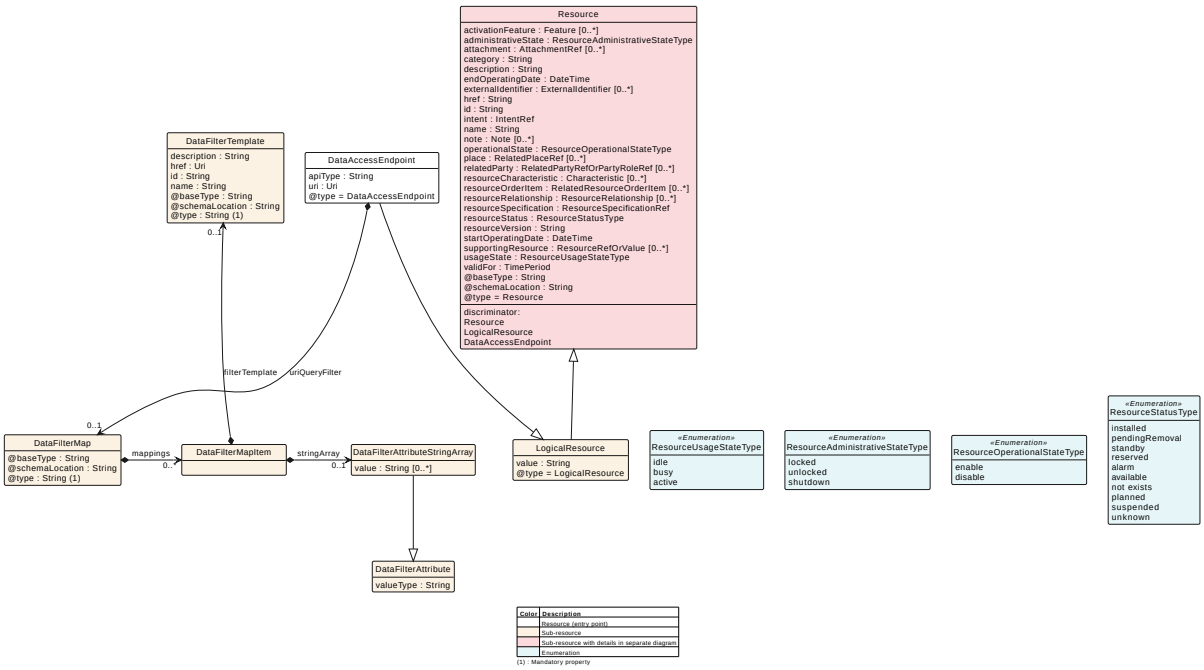


Figure 17 - DataAccessEndpoint

[.] | .//

Figure 18 - DataAccessEndpoint

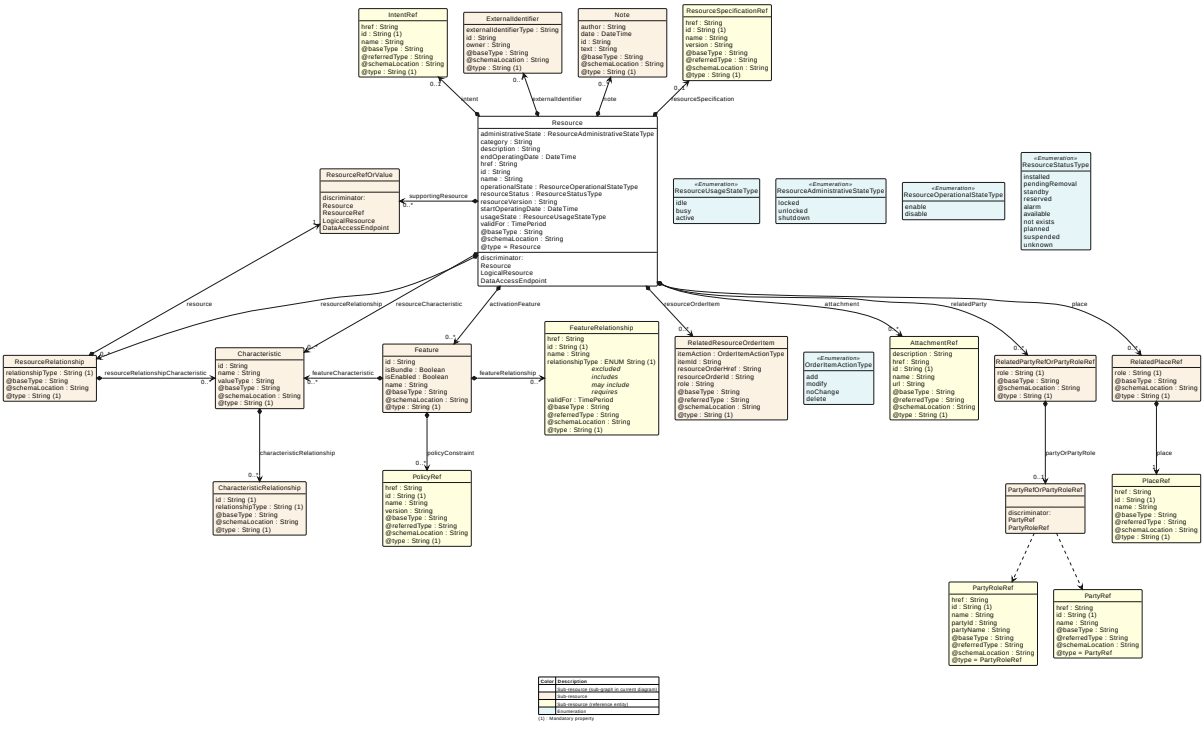


Figure 19 - Resource

[.] | .//

Figure 20 - Resource

Field descriptions

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DataAccessEndpoint fields

activationFeature	A Feature. Configuration feature.
administrativeState	A ResourceAdministrativeStateType. ResourceAdministrativeStateType enumerations; values defined by ITU X.731: 'locked': The resource is administratively prohibited from performing services for its users; 'shutdown': Use of the resource is administratively permitted to existing instances of use only. While the system remains in the shutting down state the manager may at any time cause the managed object to revert to the unlocked state; 'unlocked': The resource is administratively permitted to perform services for its users. This is independent of its inherent operability. ENUMERATED with values: * locked * unlocked * shutdown
apiType	A String.
attachment	An AttachmentRef. Attachment reference. An attachment complements the description of an element (for instance a product) through video, pictures.
category	A String. Category of the concrete resource. e.g Gold, Silver for MSISDN concrete resource.
description	A String. Free-text description of the resource.
endOperatingDate	A DateTime. A date time(DateTime). The date till the resource is operating.
externalIdentifier	An ExternalIdentifier. An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intent	An IntentRef. Intent reference, for when Intent is used by other entities.
name	A String. The name of the resource.
note	A Note. Extra information about a given entity.

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operationalState	A ResourceOperationalStateType. ResourceOperationalStateType enumerations; values defined by ITU X.731: 'disable': The resource is totally inoperable and unable to provide service to the user(s); 'enable': The resource is partially or fully operable and available for use. ENUMERATED with values: * enable * disable
place	A RelatedPlaceRef. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRef class itself.
relatedParty	A RelatedPartyRefOrPartyRoleRef. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.
resourceCharacteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
resourceOrderItem	A RelatedResourceOrderItem. RelatedResourceOrderItem (a ResourceOrder item) .The resource order item which triggered resource creation/change/termination.
resourceRelationship	A ResourceRelationship. Linked resources to the one instantiate, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (e.g. an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.
resourceSpecification	A ResourceSpecificationRef. Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.

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resourceStatus	A ResourceStatusType. ResourceStatusType enumerations. ENUMERATED with values: * alarm * available * installed * not exists * pendingRemoval * planned * reserved * standby * suspended * unknown
resourceVersion	A String. A field that identifies the specific version of an instance of a resource.
startOperatingDate	A DateTime. A date time(DateTime). The date from which the resource is operating.
supportingResource	A ResourceRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Resource entity and not the ResourceRefOrValue class itself.
uri	A Uri. URI for using the data access API.
uriQueryFilter	A DataFilterMap. Entity to Describe a given filter attributes/fields set using a mapping by type/template.
usageState	A ResourceUsageStateType. ResourceUsageStateType enumerations; values defined by ITU X.731: 'idle': The resource is not currently in use; 'active': The resource is in use, and has sufficient spare operating capacity to provide for additional users simultaneously; 'busy': The resource is in use, but it has no spare operating capacity to provide for additional users at this instant. ENUMERATED with values: * idle * active * busy
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A String. The value of the logical resource. E.g '0746712345' for MSISDN's.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

DataFilterMap sub-resource fields

mappings	A DataFilterMapItem. Definition of a field used for filtering template with the associated attributes.
@baseType	A String. When sub-classing, this defines the super-class.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

DataFilterAttributeStringArray sub-resource fields

value	A String. An array of field names.
valueType	A String. Data type of the value of the attribute.

DataFilterTemplate sub-resource fields

description	A String. Description of this DataFilterTemplate.
href	A Uri. Hyperlink reference to this DataFilterTemplate.
id	A String. Unique identifier of this DataFilterTemplate.
name	A String. Name given to this DataFilterTemplate.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Resource sub-resource fields

activationFeature	A Feature. Configuration feature.
administrativeState	A ResourceAdministrativeStateType. ResourceAdministrativeStateType enumerations; values defined by ITU X.731: 'locked': The resource is administratively prohibited from performing services for its users; 'shutdown': Use of the resource is administratively permitted to existing instances of use only. While the system remains in the shutting down state the manager may at any time cause the managed object to revert to the unlocked state; 'unlocked': The resource is administratively permitted to perform services for its users. This is independent of its inherent operability. ENUMERATED with values: * locked * unlocked * shutdown
attachment	An AttachmentRef. Attachment reference. An attachment complements the description of an element (for instance a product) through video, pictures.
category	A String. Category of the concrete resource. e.g Gold, Silver for MSISDN concrete resource.
description	A String. Free-text description of the resource.

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endOperatingDate	A DateTime. A date time(DateTime). The date till the resource is operating.
externalIdentifier	An ExternalIdentifier. An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intent	An IntentRef. Intent reference, for when Intent is used by other entities.
name	A String. The name of the resource.
note	A Note. Extra information about a given entity.
operationalState	A ResourceOperationalStateType. ResourceOperationalStateType enumerations; values defined by ITU X.731: 'disable': The resource is totally inoperable and unable to provide service to the user(s); 'enable': The resource is partially or fully operable and available for use. ENUMERATED with values: * enable * disable
place	A RelatedPlaceRef. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRef class itself.
relatedParty	A RelatedPartyRefOrPartyRoleRef. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.
resourceCharacteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
resourceOrderItem	A RelatedResourceOrderItem. RelatedResourceOrderItem (a ResourceOrder item) .The resource order item which triggered resource creation/change/termination.

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resourceRelationship	A ResourceRelationship. Linked resources to the one instantiate, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (e.g. an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.
resourceSpecification	A ResourceSpecificationRef. Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.
resourceStatus	A ResourceStatusType. ResourceStatusType enumerations. ENUMERATED with values: * alarm * available * installed * not exists * pendingRemoval * planned * reserved * standby * suspended * unknown
resourceVersion	A String. A field that identifies the specific version of an instance of a resource.
startOperatingDate	A DateTime. A date time(DateTime). The date from which the resource is operating.
supportingResource	A ResourceRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Resource entity and not the ResourceRefOrValue class itself.
usageState	A ResourceUsageStateType. ResourceUsageStateType enumerations; values defined by ITU X.731: 'idle': The resource is not currently in use; 'active': The resource is in use, and has sufficient spare operating capacity to provide for additional users simultaneously; 'busy': The resource is in use, but it has no spare operating capacity to provide for additional users at this instant. ENUMERATED with values: * idle * active * busy

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validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	Resource can be instanciated as * LogicalResource
value	This property is present in subclasses

LogicalResource sub-resource fields

activationFeature	A Feature. Configuration feature.
administrativeState	A ResourceAdministrativeStateType. ResourceAdministrativeStateType enumerations; values defined by ITU X.731: 'locked': The resource is administratively prohibited from performing services for its users; 'shutdown': Use of the resource is administratively permitted to existing instances of use only. While the system remains in the shutting down state the manager may at any time cause the managed object to revert to the unlocked state; 'unlocked': The resource is administratively permitted to perform services for its users. This is independent of its inherent operability. ENUMERATED with values: * locked * unlocked * shutdown
attachment	An AttachmentRef. Attachment reference. An attachment complements the description of an element (for instance a product) through video, pictures.
category	A String. Category of the concrete resource. e.g Gold, Silver for MSISDN concrete resource.
description	A String. Free-text description of the resource.
endOperatingDate	A DateTime. A date time(DateTime). The date till the resource is operating.
externalIdentifier	An ExternalIdentifier. An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list.
href	A String. Hyperlink reference.

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id	A String. Unique identifier.
intent	An IntentRef. Intent reference, for when Intent is used by other entities.
name	A String. The name of the resource.
note	A Note. Extra information about a given entity.
operationalState	A ResourceOperationalStateType. ResourceOperationalStateType enumerations; values defined by ITU X.731: 'disable': The resource is totally inoperable and unable to provide service to the user(s); 'enable': The resource is partially or fully operable and available for use. ENUMERATED with values: * enable * disable
place	A RelatedPlaceRef. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRef class itself.
relatedParty	A RelatedPartyRefOrPartyRoleRef. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.
resourceCharacteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
resourceOrderItem	A RelatedResourceOrderItem. RelatedResourceOrderItem (a ResourceOrder item) .The resource order item which triggered resource creation/change/termination.
resourceRelationship	A ResourceRelationship. Linked resources to the one instantiate, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (e.g. an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.

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resourceSpecification	A ResourceSpecificationRef. Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.
resourceStatus	A ResourceStatusType. ResourceStatusType enumerations. ENUMERATED with values: * alarm * available * installed * not exists * pendingRemoval * planned * reserved * standby * suspended * unknown
resourceVersion	A String. A field that identifies the specific version of an instance of a resource.
startOperatingDate	A DateTime. A date time(DateTime). The date from which the resource is operating.
supportingResource	A ResourceRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Resource entity and not the ResourceRefOrValue class itself.
usageState	A ResourceUsageStateType. ResourceUsageStateType enumerations; values defined by ITU X.731: 'idle': The resource is not currently in use; 'active': The resource is in use, and has sufficient spare operating capacity to provide for additional users simultaneously; 'busy': The resource is in use, but it has no spare operating capacity to provide for additional users at this instant. ENUMERATED with values: * idle * active * busy
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A String. The value of the logical resource. E.g '0746712345' for MSISDN's.
@baseType	A String. When sub-classing, this defines the super-class.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	LogicalResource can be instanciated as * DataAccessEndpoint
apiType	This property is present in subclasses
uri	This property is present in subclasses
uriQueryFilter	This property is present in subclasses

Json representation sample(s)

We provide below a JSON representation as example of the 'DataAccessEndpoint' resource object.

```
{
    "id": "123-xxx",
    "href": "https://host:port/tmf-
api/performanceManagement/v5/dataAccessEndpoint/123-xxx",
    "apiType": "REST",
    "category": "topic",
    "description": "Measurement collection job 123-xxx subscriber topic",
    "endOperatingDate": "2022-05-13T10:13:42.971Z",
    "name": "123-xxx subscriber topic",
    "resourceVersion": "1.0",
    "startOperatingDate": "2022-05-13T10:13:42.971Z",
    "uri": "https://pubsub.googleapis.com/v1/",
    "uriQueryFilter":{
      "mappings": [
          {
              "filterTemplate": {
                  "name" : "myRangeFilterSearch0",
                  "@type" : "DataFilterTemplateRange",
                  "criteria":{
                      "lowerBound" : {
                          "boundValue": -10,
                          "isInclusive": true
                      },
                      "upperBound" : {
                          "boundValue": -10,
                          "isInclusive": true
                      }
                  }
              },
              "stringArray": {
                  "value": [
                  "field01", "field02", "field03"
                   ]
              }
          },
          {
              "filterTemplate": {
                  "name": "myRegExSampleSearch0",
                  "@type": "DataFilterTemplateRegex",
                  "criteria": "dallas|houston|venise|paris|new-york"
              },
```

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```
                "stringArray": {
                    "value":[
                        "field11", "field12", "field13"
                    ]
                }
            },
            {
                "filterTemplate": {
                    "name" : "myBooleanFilterSearch0",
                    "criteria" : true,
                    "@type" : "DataFilterTemplateBoolean"
                },
                "stringArray": {
                    "value": [
                        "field1", "field2", "field3"
                    ]
                }
            }
        ]
    },
    "value": "myLogicalResourceValue",
    "administrativeState": "locked",
    "note": [
      {
        "id": "afa9-3d90aaa8da0f",
        "author": "Mr. Franklin",
        "date": "2022-13-05T08:00:00Z",
        "text": "For further details on how to create a subscription to this topic,
visit
https://cloud.google.com/pubsub/docs/reference/rest/v1/projects.subscriptions/create
",
        "@baseType": "Note",
        "@schemaLocation": "https://myschemas.com/Note.schema.json",
        "@type": "Note"
      }
    ],
    "operationalState": "enable",
    "resourceCharacteristic": [
      {
        "id": "2100",
        "name": "myCharName",
        "valueType": "string",
        "characteristicRelationship": [
          {
            "id": "21",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/characteristicRelationship/21",
            "relationshipType": "string",
            "@baseType": "CharacteristicRelationship",
            "@schemaLocation":
"https://myschemas.com/CharacteristicRelationship.schema.json",
            "@type": "CharacteristicRelationship"
          }
        ],
        "value": "myCharValue",
        "@baseType": "Characteristic",
        "@schemaLocation": "https://myschemas.com/StringCharacteristic.schema.json",
        "@type": "StringCharacteristic"
      }
    ],
    "resourceSpecification": {
      "id": "1",
      "href": "https://host:port/tmf-
```


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```
api/performanceManagement/v5/ResourceSpecification/1",
      "name": "mySpecName",
      "version": "1.0",
      "@baseType": "ResourceSpecification",
      "@schemaLocation": "https://mySchemas.com/ResourceSpecification.schema.json",
      "@type": "ResourceSpecification",
      "@referredType": "ResourceSpecification"
    },
    "resourceStatus": "standby",
    "usageState": "idle",
    "@baseType": "DataAccessEndpoint",
    "@schemaLocation": "https://myschemas.com/dataAccessEndpoint",
    "@type": "DataAccessEndpoint"
  }
```


PerformanceMeasurementAtomic resource

A measurement of a single performance indicator.

Resource model

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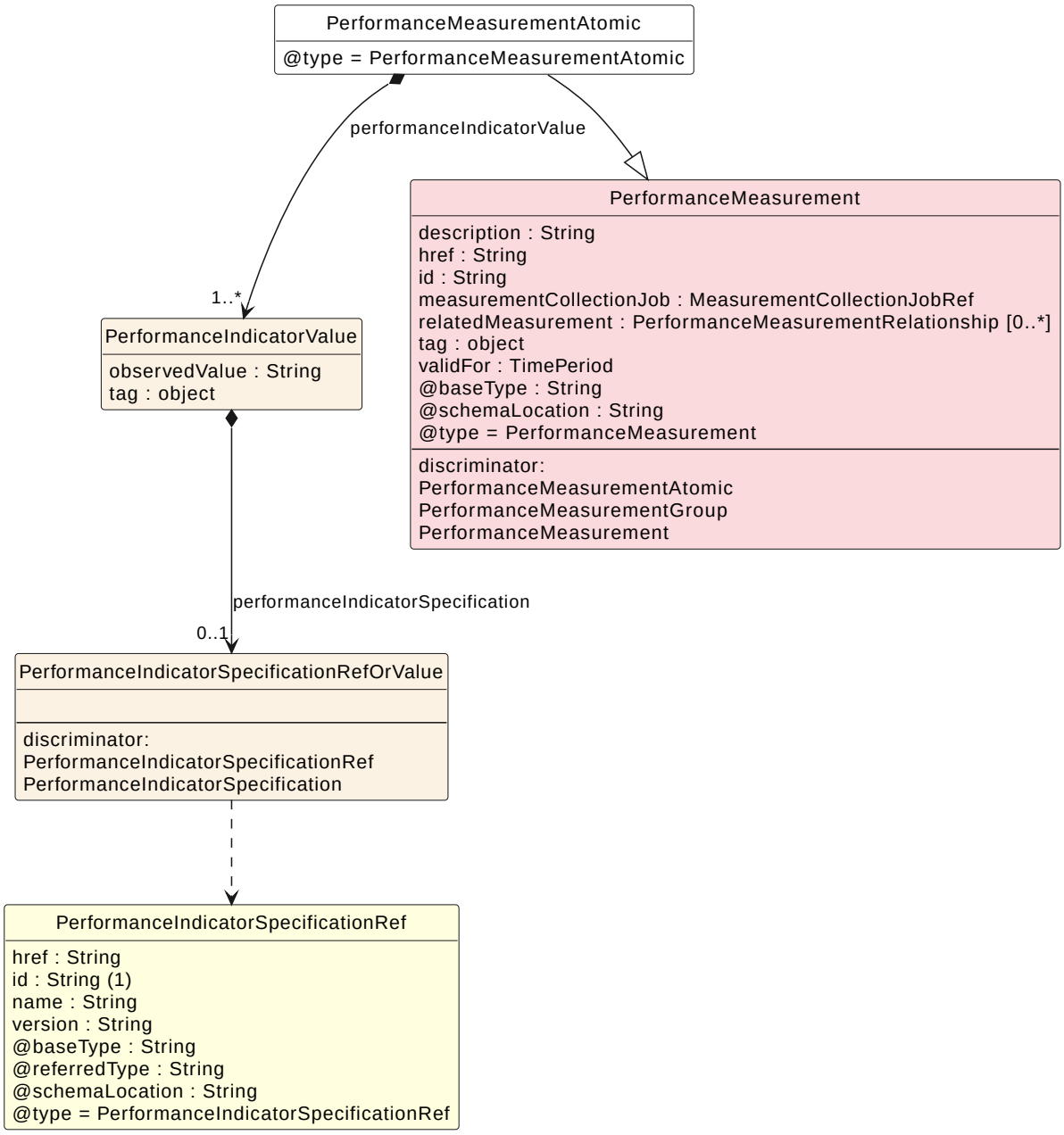


Figure 21 - PerformanceMeasurementAtomic

[.] | .//

Figure 22 - PerformanceMeasurementAtomic

Field descriptions

PerformanceMeasurementAtomic fields

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description	A String. A free-text description of the performance measurement.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
measurementCollectionJob	A MeasurementCollectionJobRef. Reference to a MeasurementCollectionJob.
performanceIndicatorValue	A PerformanceIndicatorValue. A value of a performance indicator.
relatedMeasurement	A PerformanceMeasurementRelationship. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.
tag	An object. The optional tag object attached to this entire measurement.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PerformanceIndicatorSpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String. Version of the performance indicator specification.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PerformanceIndicatorSpecification sub-resource fields

collectionType	A CollectionType. This is enumeration for CollectionType state. ENUMERATED with values: * cumulative * delta * discrete_event * gauge * status_inspection

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derivationAlgorithm	A String. A step-by-step procedure used to calculate the value of PerformanceIndicator.
derivationMethod	A String. For simple calculations, the method used to calculate the value of a PerformanceIndicator, such as average, minimum, maximum, sum and so forth.
description	A String. A narrative that explains in detail what the PerformanceIndicatorSpecification is.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
indicatorCategory	A String. A grouping or set of PerformanceIndicatorSpecifications that are classified together because of common characteristics, such as technology specific, service specific, or technology/service independent.
indicatorType	An IndicatorType. This is enumeration for Indicator Type. ENUMERATED with values: * string * int * float * double
indicatorUnit	A String. The unit by which the indicator is measured. For example, seconds, KBs, rate per second, etc.
name	A String. A word, term, or phrase by which a PerformanceIndicatorSpecification is known and distinguished from other PerformanceIndicatorSpecifications.
performanceIndicatorSpecRelationship	A PerformanceIndicatorSpecRelationship. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.
perspective	A String. The point of view for the PerformanceIndicatorSpecification, such as a single user instance or aggregation.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PerformanceIndicatorSpecRelationship sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
relationshipType	A String.
role	A String. The association role for this service specification.

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validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PerformanceMeasurementGroup sub-resource fields

description	A String. A free-text description of the performance measurement.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
measurementCollectionJob	A MeasurementCollectionJobRef. Reference to a MeasurementCollectionJob.
performanceIndicatorGroupValue	A PerformanceIndicatorGroupValue. A value of a performance indicator.
relatedMeasurement	A PerformanceMeasurementRelationship. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.
tag	An object. The optional tag object attached to this entire measurement.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PerformanceMeasurement sub-resource fields

description	A String. A free-text description of the performance measurement.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
measurementCollectionJob	A MeasurementCollectionJobRef. Reference to a MeasurementCollectionJob.
relatedMeasurement	A PerformanceMeasurementRelationship. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.

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tag	An object. The optional tag object attached to this entire measurement.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	PerformanceMeasurement can be instanciated as * PerformanceMeasurementAtomic * PerformanceMeasurementGroup
performanceIndicatorGroupValue	This property is present in subclasses
performanceIndicatorValue	This property is present in subclasses

Json representation sample(s)

We provide below a JSON representation as example of the 'PerformanceMeasurementAtomic' resource object.


```
{
  "@baseType": "PerformanceMeasurement",
  "validFor": {
    "startDateTime": "1985-04-12T23:20:50.52Z",
    "endDateTime": "1985-04-12T23:20:50.52Z"
  },
  "@type": "PerformanceMeasurementAtomic",
  "measurementCollectionJob": {
    "@referredType": "MeasurementCollectionJob",
    "@baseType": "MeasurementCollectionJobRef",
    "@type": "MeasurementCollectionJobRef",
    "name": "Measurement Collection Job from system A",
    "id": "4000",
    "href": "https://host:port/tmf-
api/performanceManagement/v5/measurementCollectionJob/4000",
    "@schemaLocation": "https://myschemas/MeasurementCollectionJob.schema.json"
  },
  "description": "The first performance measurement in the database",
  "performanceIndicatorValue": [
    {
      "observedValue": "2.1",
      "tag": {
        "zip": "93165",
        "imsi": "1111000000"
      },
      "performanceIndicatorSpecification": {
        "@referredType": "PerformanceIndicatorSpecification",
        "@baseType": "PerformanceIndicatorSpecificationRef",
        "@type": "PerformanceIndicatorSpecificationRef",
        "id": "1",
        "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/1",
        "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json"
```


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```
      }
    },
    {
      "observedValue": "3.25",
      "performanceIndicatorSpecification": {
        "indicatorName": "CPU Speed",
        "@baseType": "PerformanceIndicatorSpecification",
        "indicatorType": "Gauge",
        "@type": "PerformanceIndicatorSpecification",
        "name": "CPU Speed Indicator Specification",
        "indicatorCategory": "SPEEDCategory",
        "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
        "indicatorUnit": "GHz"
      }
    }
  ],
  "id": "1",
  "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceMeasurement/1",
  "tag": {
    "country": "Belgium",
    "region": "North"
  },
  "@schemaLocation":
"https://github.com/repo/PerformanceMeasurementAtomic.schema.json"
}
```


PerformanceMeasurementGroup resource

A measurement of a performance group indicator.

Resource model

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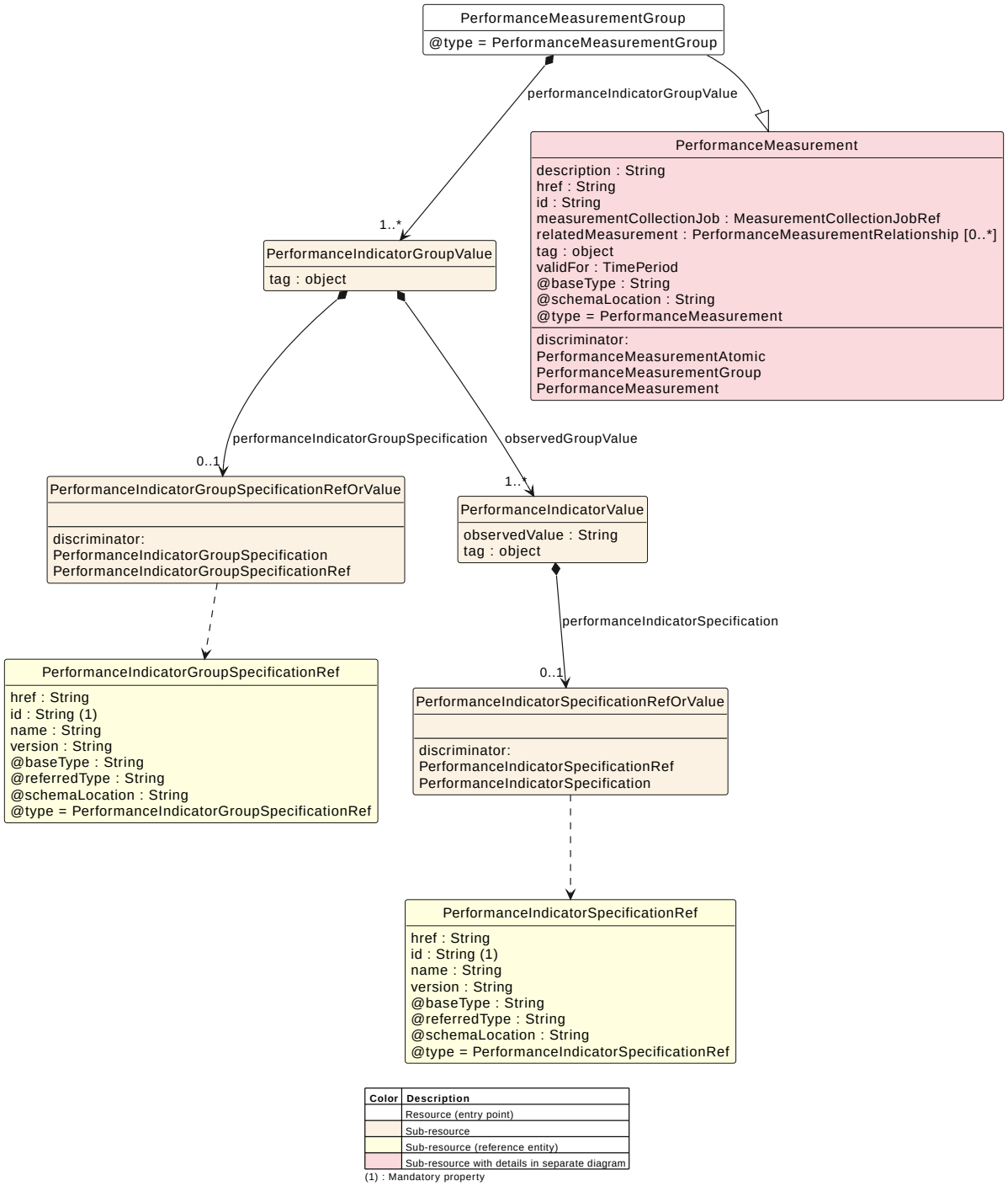


Figure 23 - PerformanceMeasurementGroup

[.] | .//

Figure 24 - PerformanceMeasurementGroup

Field descriptions

PerformanceMeasurementGroup fields

description	A String. A free-text description of the performance measurement.

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href	A String. Hyperlink reference.
id	A String. Unique identifier.
measurementCollectionJob	A MeasurementCollectionJobRef. Reference to a MeasurementCollectionJob.
performanceIndicatorGroupValue	A PerformanceIndicatorGroupValue. A value of a performance indicator.
relatedMeasurement	A PerformanceMeasurementRelationship. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.
tag	An object. The optional tag object attached to this entire measurement.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PerformanceIndicatorGroupSpecification sub-resource fields

groupCategory	A String. A grouping or set of PerformanceIndicatorGroupSpecifications that are classified together because of common characteristics, such as technology specific, service specific, or technology/service independent.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. A word, term, or phrase by which a PerformanceIndicatorGroupSpecification is tagged.
performanceIndicatorSpecification	A PerformanceIndicatorSpecificationRefOrValue. A product to be created defined by value or existing defined by reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the product entity and not the RelatedProductRefOrValue class itself.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PerformanceIndicatorGroupSpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.

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name	A String. Name of the referred entity.
version	A String. Version of the performance indicator group specification.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PerformanceIndicatorSpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String. Version of the performance indicator specification.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PerformanceIndicatorSpecification sub-resource fields

collectionType	A CollectionType. This is enumeration for CollectionType state. ENUMERATED with values: * cumulative * delta * discrete_event * gauge * status_inspection
derivationAlgorithm	A String. A step-by-step procedure used to calculate the value of PerformanceIndicator.
derivationMethod	A String. For simple calculations, the method used to calculate the value of a PerformanceIndicator, such as average, minimum, maximum, sum and so forth.
description	A String. A narrative that explains in detail what the PerformanceIndicatorSpecification is.
href	A String. Hyperlink reference.
id	A String. Unique identifier.

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indicatorCategory	A String. A grouping or set of PerformanceIndicatorSpecifications that are classified together because of common characteristics, such as technology specific, service specific, or technology/service independent.
indicatorType	An IndicatorType. This is enumeration for Indicator Type. ENUMERATED with values: * string * int * float * double
indicatorUnit	A String. The unit by which the indicator is measured. For example, seconds, KBs, rate per second, etc.
name	A String. A word, term, or phrase by which a PerformanceIndicatorSpecification is known and distinguished from other PerformanceIndicatorSpecifications.
performanceIndicatorSpecRelationship	A PerformanceIndicatorSpecRelationship. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.
perspective	A String. The point of view for the PerformanceIndicatorSpecification, such as a single user instance or aggregation.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PerformanceIndicatorSpecRelationship sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
relationshipType	A String.
role	A String. The association role for this service specification.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PerformanceMeasurementAtomic sub-resource fields

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description	A String. A free-text description of the performance measurement.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
measurementCollectionJob	A MeasurementCollectionJobRef. Reference to a MeasurementCollectionJob.
performanceIndicatorValue	A PerformanceIndicatorValue. A value of a performance indicator.
relatedMeasurement	A PerformanceMeasurementRelationship. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.
tag	An object. The optional tag object attached to this entire measurement.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PerformanceMeasurement sub-resource fields

description	A String. A free-text description of the performance measurement.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
measurementCollectionJob	A MeasurementCollectionJobRef. Reference to a MeasurementCollectionJob.
relatedMeasurement	A PerformanceMeasurementRelationship. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.
tag	An object. The optional tag object attached to this entire measurement.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

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	PerformanceMeasurement can be instanciated as * PerformanceMeasurementAtomic * PerformanceMeasurementGroup
performanceIndicatorGroupValue	This property is present in subclasses
performanceIndicatorValue	This property is present in subclasses

Json representation sample(s)

We provide below a JSON representation as example of the 'PerformanceMeasurementGroup' resource object.


```
{
  "@baseType": "PerformanceMeasurement",
  "validFor": {
    "startDateTime": "1985-04-12T23:20:50.52Z",
    "endDateTime": "1985-04-12T23:20:50.52Z"
  },
  "@type": "PerformanceMeasurementGroup",
  "measurementCollectionJob": {
    "@referredType": "MeasurementCollectionJob",
    "@baseType": "MeasurementCollectionJobRef",
    "@type": "MeasurementCollectionJobRef",
    "name": "Measurement Collection Job from system A",
    "id": "4000",
    "href": "https://host:port/tmf-
api/performanceManagement/v5/measurementCollectionJob/4000",
    "@schemaLocation": "https://myschemas/MeasurementCollectionJob.schema.json"
  },
  "description": "The first performance measurement in the database",
  "performanceIndicatorGroupValue": [{
    "observedGroupValue": [
      {
        "observedValue": "2.1",
        "tag": {
          "zip": "93165",
          "imsi": "1111000000"
        },
        "performanceIndicatorSpecification": {
          "@referredType": "PerformanceIndicatorSpecification",
          "@baseType": "PerformanceIndicatorSpecificationRef",
          "@type": "PerformanceIndicatorSpecificationRef",
          "id": "5000",
          "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5000",
          "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecificationRef.schema.json"
        }
      },
      {
        "observedValue": "2.2",
        "tag": {
          "zip": "93165",
          "imsi": "1111000001"
        },
        "performanceIndicatorSpecification": {
          "@referredType": "PerformanceIndicatorSpecification",
          "@baseType": "PerformanceIndicatorSpecificationRef",
          "@type": "PerformanceIndicatorSpecificationRef",
```


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```
          "id": "5000",
          "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5000",
          "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecificationRef.schema.json"
        }
      }
    ],
    "performanceIndicatorGroupSpecification": {
      "@baseType": "PerformanceIndicatorGroupSpecification",
      "groupCategory": "CPUGroup",
      "@type": "PerformanceIndicatorGroupSpecification",
      "name": "My critical performance indicators",
      "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSGrouppecification/600",
      "id": "600",
      "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorGroupSpecificati
on.schema.json",
      "performanceIndicatorSpecification": [{
        "@baseType": "PerformanceIndicatorSpecification",
        "@type": "PerformanceIndicatorSpecification",
        "name": "CPU speed",
        "id": "5000",
        "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5000",
        "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
        "version": "1.0"
      }]
    }
  }],
  "id": "1",
  "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceMeasurement/1",
  "tag": {
    "country": "Belgium",
    "region": "North"
  },
  "@schemaLocation":
"https://github.com/repo/PerformanceMeasurementGroup.schema.json"
}
```


Notification Resource Models

8 notifications are defined for this API.

Notifications related to MeasurementCollectionJob:

- Files Ready Event
- Files Preparation Error Event
- Create Event
- Delete Event
- Attribute Value Change Event
- Execution State Change Event

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Notifications related to PerformanceMeasurement:

- Collection Ready Event
- Create Event

The notification structure for all notifications in this API follow the pattern depicted by the figure below. A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType). This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the resource concerned by the notification using the resource name as access field ("resourceName" placeholder).

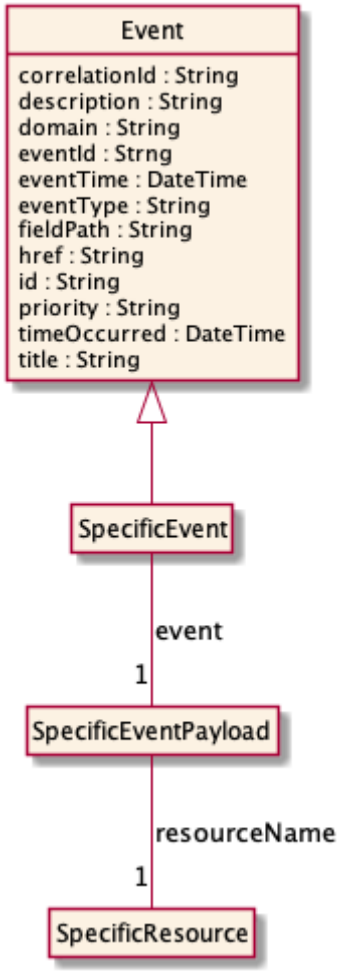


Figure 25 Notification Pattern

MeasurementCollectionJob

Files Ready Event

Message example for MeasurementCollectionJobFilesReadyEvent event

```
Content-Type: application/json
```

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```
{
  "correlationId": "fdd613a8-7396",
  "description": "MeasurementCollectionJobFilesReadyEvent illustration",
  "domain": "Commercial",
  "eventId": "456d-9572-d1d23be58dd4",
  "eventTime": "2022-08-25T12:18:25.017Z",
  "eventType": "MeasurementCollectionJobFilesReadyEvent",
  "priority": "4",
  "timeOcurred": "2022-08-25T12:18:16.825Z",
  "title": "MeasurementCollectionJobFilesReadyEvent",
  "event": {
      "measurementCollectionJob": {
          "id": "4000",
          "href": "https://host:port/tmf-
api/performanceManagement/v5/MeasurementCollectionJob/4000",
          "consumingApplicationId": "4000",
          "creationTime": "2020-06-23T14:12:01.078Z",
          "jobId": "400",
          "jobPriority": 3,
          "jobOnDemand": false,
          "lastModifiedTime": "2020-06-23T14:12:01.079Z",
          "outputFormat": "string",
          "producingApplicationId": "NT_101",
          "adminState": "locked",
          "executionState": "acknowledged",
          "fileTransferData": [
            {
              "fileFormat": "csv",
              "fileLocation": "https://host:port/tmf-
api/performanceManagement/v5/FileTransferData",
              "transportProtocol": "ftp",
              "compressionType": {
                "compressionEnumType": "no_compression"
              },
              "packingType": {
                "packingEnumType": "NO_PACKING"
              },
              "retentionPeriod": {
                "amount": 1,
                "units": "month",
                "@baseType": "RetentionPeriod",
                "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/RetentionPeriod",
                "@type": "RetentionPeriod"
              },
              "@baseType": "FileTransferData",
              "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
              "@type": "FileTransferData"
            }
          ],
          "granularity": "g_1mn",
          "monitoredClassCriteria": {
            "monitoredObjectClass": "string",
            "objectInstanceFilter": "string",
            "@baseType": "string",
            "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
            "@type": "string"
          },
          "monitoredInstancesCriteria": {
            "monitoredObjectInstances": ["string"],
            "@baseType": "string",
```


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```
            "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
            "@type": "string"
          },
          "performanceIndicatorGroupSpecification": [
            {
              "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSGrouppecification/600",
              "id": "600",
              "name": "My critical performance indicators",
              "performanceIndicatorSpecification": [
                {
                  "id": "5000",
                  "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5000",
                  "name": "CPU speed",
                  "version": "1.0",
                  "@baseType": "PerformanceIndicatorSpecification",
                  "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
                  "@type": "PerformanceIndicatorSpecification"
                }
              ],
              "@baseType": "PerformanceIndicatorGroupSpecification",
              "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorGroupSpecificati
on.schema.json",
              "@type": "PerformanceIndicatorGroupSpecification"
            }
          ],
          "performanceIndicatorSpecification": [
            {
              "id": "101",
              "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/101",
              "derivationAlgorithm": "string",
              "derivationMethod": "string",
              "description": "string",
              "indicatorCategory": "string",
              "indicatorUnit": "string",
              "name": "string",
              "perspective": "string",
              "collectionType": "cumulative",
              "indicatorType": "string",
              "performanceIndicatorSpecRelationship": [
                {
                  "id": "321",
                  "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecRelationship/321",
                  "relationshipType": "reliesOn",
                  "validFor": {
                    "endDateTime": "1985-04-12T23:20:50.52Z",
                    "startDateTime": "1985-04-12T23:20:50.52Z"
                  },
                  "@baseType": "PerformanceIndicatorSpecRelationship",
                  "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecRelationship
.schema.json",
                  "@type": "PerformanceIndicatorSpecRelationship"
                }
              ],
              "validFor": {
```


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```
                "endDateTime": "1985-04-12T23:20:50.52Z",
                "startDateTime": "1985-04-12T23:20:50.52Z"
              },
              "@baseType": "PerformanceIndicatorSpecification",
              "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
              "@type": "PerformanceIndicatorSpecification"
            }
          ],
          "reportingPeriod": "r_1mn",
          "scheduleDefinition": [{
            "id": "1",
            "recurringFrequency": "weekly",
            "scheduleDefinitionEndTime": "2020-06-23T14:12:01.080Z",
            "scheduleDefinitionHourRange": "2",
            "scheduleDefinitionStartTime": "2020-06-23T14:12:01.080Z",
            "MonthlyScheduleDayOfWeekDefinition": {
              "recurringDaySequence": "string",
              "dayOfMonthRecurrence": [
                {
                  "dates": "2020-06-23T14:12:01.080Z",
                  "@baseType": "DayOfMonthRecurrence",
                  "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/DayOfMonthRecurrence.schema.json",
                  "@type": "DayOfMonthRecurrence"
                }
              ],
              "@baseType": "MonthlyScheduleDayOfWeekDefinition",
              "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/MonthlyScheduleDayOfWeekDefinition.s
chema.json",
              "@type": "MonthlyScheduleDayOfWeekDefinition"
            },
            "WeeklyScheduledDefinition": [
              {
                "dates": "2020-06-23T14:12:01.080Z",
                "@baseType": "WeeklyScheduledDefinition",
                "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/WeeklyScheduledDefinition.schema.jso
n",
                "@type": "WeeklyScheduledDefinition"
              }
            ],
            "@baseType": "ScheduleDefinition",
            "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/ScheduleDefinition.schema.json",
            "@type": "ScheduleDefinition"
          }],
          "trackingRecord": [
          {
            "id": "TR_101",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
            "description": "string",
            "systemId": "NT_101",
            "time": "2020-06-23T17:55:51.671Z",
            "user": "NP1",
            "@baseType": "TrackingRecord",
            "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
            "@type": "TrackingRecord"
          }
```


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```
          ],
          "@baseType": "MeasurementCollectionJob",
          "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
          "@type": "MeasurementCollectionJob"
      }
  },
  "reportingSystem": {
      "id": "139",
      "name": "APP-110",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
  },
  "source": {
      "id": "471",
      "name": "APP-100",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "MeasurementCollectionJobFilesReadyEvent"
}
```


Files Preparation Error Event

Message example for MeasurementCollectionJobFilesPreparationErrorEvent event


```
Content-Type: application/json

{
  "correlationId": "416928f1-523a",
  "description": "MeasurementCollectionJobFilesPreparationErrorEvent illustration",
  "domain": "Commercial",
  "eventId": "4cc1-80c0-ba3b3efe2c75",
  "eventTime": "2022-08-25T12:18:25.029Z",
  "eventType": "MeasurementCollectionJobFilesPreparationErrorEvent",
  "priority": "4",
  "timeOcurred": "2022-08-25T12:18:19.038Z",
  "title": "MeasurementCollectionJobFilesPreparationErrorEvent",
  "event": {
      "measurementCollectionJob": {
          "id": "4000",
          "href": "https://host:port/tmf-
api/performanceManagement/v5/MeasurementCollectionJob/4000",
          "consumingApplicationId": "4000",
          "creationTime": "2020-06-23T14:12:01.078Z",
          "jobId": "400",
          "jobPriority": 3,
          "jobOnDemand": false,
          "lastModifiedTime": "2020-06-23T14:12:01.079Z",
          "outputFormat": "string",
          "producingApplicationId": "NT_101",
          "adminState": "locked",
          "executionState": "acknowledged",
          "fileTransferData": [
            {
              "fileFormat": "csv",
              "fileLocation": "https://host:port/tmf-
api/performanceManagement/v5/FileTransferData",
              "transportProtocol": "ftp",
```


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```
              "compressionType": {
                "compressionEnumType": "no_compression"
              },
              "packingType": {
                "packingEnumType": "NO_PACKING"
              },
              "retentionPeriod": {
                "amount": 1,
                "units": "month",
                "@baseType": "RetentionPeriod",
                "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/RetentionPeriod",
                "@type": "RetentionPeriod"
              },
              "@baseType": "FileTransferData",
              "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
              "@type": "FileTransferData"
            }
          ],
          "granularity": "g_1mn",
          "monitoredClassCriteria": {
            "monitoredObjectClass": "string",
            "objectInstanceFilter": "string",
            "@baseType": "string",
            "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
            "@type": "string"
          },
          "monitoredInstancesCriteria": {
            "monitoredObjectInstances": ["string"],
            "@baseType": "string",
            "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
            "@type": "string"
          },
          "performanceIndicatorGroupSpecification": [
            {
              "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSGrouppecification/600",
              "id": "600",
              "name": "My critical performance indicators",
              "performanceIndicatorSpecification": [
                {
                  "id": "5000",
                  "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5000",
                  "name": "CPU speed",
                  "version": "1.0",
                  "@baseType": "PerformanceIndicatorSpecification",
                  "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
                  "@type": "PerformanceIndicatorSpecification"
                }
              ],
              "@baseType": "PerformanceIndicatorGroupSpecification",
              "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorGroupSpecificati
on.schema.json",
              "@type": "PerformanceIndicatorGroupSpecification"
            }
          ],
```


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```
          "performanceIndicatorSpecification": [
            {
              "id": "101",
              "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/101",
              "derivationAlgorithm": "string",
              "derivationMethod": "string",
              "description": "string",
              "indicatorCategory": "string",
              "indicatorUnit": "string",
              "name": "string",
              "perspective": "string",
              "collectionType": "cumulative",
              "indicatorType": "string",
              "performanceIndicatorSpecRelationship": [
                {
                  "id": "321",
                  "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecRelationship/321",
                  "relationshipType": "reliesOn",
                  "validFor": {
                    "endDateTime": "1985-04-12T23:20:50.52Z",
                    "startDateTime": "1985-04-12T23:20:50.52Z"
                  },
                  "@baseType": "PerformanceIndicatorSpecRelationship",
                  "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecRelationship
.schema.json",
                  "@type": "PerformanceIndicatorSpecRelationship"
                }
              ],
              "validFor": {
                "endDateTime": "1985-04-12T23:20:50.52Z",
                "startDateTime": "1985-04-12T23:20:50.52Z"
              },
              "@baseType": "PerformanceIndicatorSpecification",
              "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
              "@type": "PerformanceIndicatorSpecification"
            }
          ],
          "reportingPeriod": "r_1mn",
          "scheduleDefinition": [{
            "id": "1",
            "recurringFrequency": "weekly",
            "scheduleDefinitionEndTime": "2020-06-23T14:12:01.080Z",
            "scheduleDefinitionHourRange": "2",
            "scheduleDefinitionStartTime": "2020-06-23T14:12:01.080Z",
            "MonthlyScheduleDayOfWeekDefinition": {
              "recurringDaySequence": "string",
              "dayOfMonthRecurrence": [
                {
                  "dates": "2020-06-23T14:12:01.080Z",
                  "@baseType": "DayOfMonthRecurrence",
                  "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/DayOfMonthRecurrence.schema.json",
                  "@type": "DayOfMonthRecurrence"
                }
              ],
              "@baseType": "MonthlyScheduleDayOfWeekDefinition",
              "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/MonthlyScheduleDayOfWeekDefinition.s
```


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```
chema.json",
              "@type": "MonthlyScheduleDayOfWeekDefinition"
            },
            "WeeklyScheduledDefinition": [
              {
                "dates": "2020-06-23T14:12:01.080Z",
                "@baseType": "WeeklyScheduledDefinition",
                "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/WeeklyScheduledDefinition.schema.jso
n",
                "@type": "WeeklyScheduledDefinition"
              }
            ],
            "@baseType": "ScheduleDefinition",
            "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/ScheduleDefinition.schema.json",
            "@type": "ScheduleDefinition"
          }],
          "trackingRecord": [
          {
            "id": "TR_101",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
            "description": "string",
            "systemId": "NT_101",
            "time": "2020-06-23T17:55:51.671Z",
            "user": "NP1",
            "@baseType": "TrackingRecord",
            "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
            "@type": "TrackingRecord"
          }
          ],
          "@baseType": "MeasurementCollectionJob",
          "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
          "@type": "MeasurementCollectionJob"
      }
  },
  "reportingSystem": {
      "id": "139",
      "name": "APP-110",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
  },
  "source": {
      "id": "471",
      "name": "APP-100",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "MeasurementCollectionJobFilesPreparationErrorEvent"
}
```


Create Event

Message example for MeasurementCollectionJobCreateEvent event

```
Content-Type: application/json
```

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```
{
  "correlationId": "2bc480b9-2dc0",
  "description": "MeasurementCollectionJobCreateEvent illustration",
  "domain": "Commercial",
  "eventId": "48e1-9989-2d1fce36bceb",
  "eventTime": "2022-08-25T12:18:24.978Z",
  "eventType": "MeasurementCollectionJobCreateEvent",
  "priority": "5",
  "timeOcurred": "2022-08-25T12:18:19.818Z",
  "title": "MeasurementCollectionJobCreateEvent",
  "event": {
      "measurementCollectionJob": {
          "id": "4000",
          "href": "https://host:port/tmf-
api/performanceManagement/v5/MeasurementCollectionJob/4000",
          "@type": "MeasurementCollectionJobRef"
      }
  },
  "reportingSystem": {
      "id": "139",
      "name": "APP-110",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
  },
  "source": {
      "id": "471",
      "name": "APP-100",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "MeasurementCollectionJobCreateEvent"
}
```


Delete Event

Message example for MeasurementCollectionJobDeleteEvent event


```
Content-Type: application/json

{
  "correlationId": "be9ea486-ec11",
  "description": "MeasurementCollectionJobDeleteEvent illustration",
  "domain": "Commercial",
  "eventId": "4739-824e-059be590beaf",
  "eventTime": "2022-08-25T12:18:24.997Z",
  "eventType": "MeasurementCollectionJobDeleteEvent",
  "priority": "5",
  "timeOcurred": "2022-08-25T12:18:18.505Z",
  "title": "MeasurementCollectionJobDeleteEvent",
  "event": {
      "measurementCollectionJob": {
          "id": "4000",
          "href": "https://host:port/tmf-
api/performanceManagement/v5/MeasurementCollectionJob/4000",
          "consumingApplicationId": "4000",
          "creationTime": "2020-06-23T14:12:01.078Z",
          "jobId": "400",
          "jobPriority": 3,
```


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```
          "jobOnDemand": false,
          "lastModifiedTime": "2020-06-23T14:12:01.079Z",
          "outputFormat": "string",
          "producingApplicationId": "NT_101",
          "adminState": "locked",
          "executionState": "acknowledged",
          "fileTransferData": [
            {
              "fileFormat": "csv",
              "fileLocation": "https://host:port/tmf-
api/performanceManagement/v5/FileTransferData",
              "transportProtocol": "ftp",
              "compressionType": {
                "compressionEnumType": "no_compression"
              },
              "packingType": {
                "packingEnumType": "NO_PACKING"
              },
              "retentionPeriod": {
                "amount": 1,
                "units": "month",
                "@baseType": "RetentionPeriod",
                "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/RetentionPeriod",
                "@type": "RetentionPeriod"
              },
              "@baseType": "FileTransferData",
              "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
              "@type": "FileTransferData"
            }
          ],
          "granularity": "g_1mn",
          "monitoredClassCriteria": {
            "monitoredObjectClass": "string",
            "objectInstanceFilter": "string",
            "@baseType": "string",
            "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
            "@type": "string"
          },
          "monitoredInstancesCriteria": {
            "monitoredObjectInstances": ["string"],
            "@baseType": "string",
            "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
            "@type": "string"
          },
          "performanceIndicatorGroupSpecification": [
            {
              "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSGrouppecification/600",
              "id": "600",
              "name": "My critical performance indicators",
              "performanceIndicatorSpecification": [
                {
                  "id": "5000",
                  "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5000",
                  "name": "CPU speed",
                  "version": "1.0",
                  "@baseType": "PerformanceIndicatorSpecification",
                  "@schemaLocation": "https://host:port/tmf-
```


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```
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
                  "@type": "PerformanceIndicatorSpecification"
                }
              ],
              "@baseType": "PerformanceIndicatorGroupSpecification",
              "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorGroupSpecificati
on.schema.json",
              "@type": "PerformanceIndicatorGroupSpecification"
            }
          ],
          "performanceIndicatorSpecification": [
            {
              "id": "101",
              "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/101",
              "derivationAlgorithm": "string",
              "derivationMethod": "string",
              "description": "string",
              "indicatorCategory": "string",
              "indicatorUnit": "string",
              "name": "string",
              "perspective": "string",
              "collectionType": "cumulative",
              "indicatorType": "string",
              "performanceIndicatorSpecRelationship": [
                {
                  "id": "321",
                  "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecRelationship/321",
                  "relationshipType": "reliesOn",
                  "validFor": {
                    "endDateTime": "1985-04-12T23:20:50.52Z",
                    "startDateTime": "1985-04-12T23:20:50.52Z"
                  },
                  "@baseType": "PerformanceIndicatorSpecRelationship",
                  "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecRelationship
.schema.json",
                  "@type": "PerformanceIndicatorSpecRelationship"
                }
              ],
              "validFor": {
                "endDateTime": "1985-04-12T23:20:50.52Z",
                "startDateTime": "1985-04-12T23:20:50.52Z"
              },
              "@baseType": "PerformanceIndicatorSpecification",
              "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
              "@type": "PerformanceIndicatorSpecification"
            }
          ],
          "reportingPeriod": "r_1mn",
          "scheduleDefinition": [{
            "id": "1",
            "recurringFrequency": "weekly",
            "scheduleDefinitionEndTime": "2020-06-23T14:12:01.080Z",
            "scheduleDefinitionHourRange": "2",
            "scheduleDefinitionStartTime": "2020-06-23T14:12:01.080Z",
            "MonthlyScheduleDayOfWeekDefinition": {
              "recurringDaySequence": "string",
```


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```
              "dayOfMonthRecurrence": [
                {
                  "dates": "2020-06-23T14:12:01.080Z",
                  "@baseType": "DayOfMonthRecurrence",
                  "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/DayOfMonthRecurrence.schema.json",
                  "@type": "DayOfMonthRecurrence"
                }
              ],
              "@baseType": "MonthlyScheduleDayOfWeekDefinition",
              "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/MonthlyScheduleDayOfWeekDefinition.s
chema.json",
              "@type": "MonthlyScheduleDayOfWeekDefinition"
            },
            "WeeklyScheduledDefinition": [
              {
                "dates": "2020-06-23T14:12:01.080Z",
                "@baseType": "WeeklyScheduledDefinition",
                "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/WeeklyScheduledDefinition.schema.jso
n",
                "@type": "WeeklyScheduledDefinition"
              }
            ],
            "@baseType": "ScheduleDefinition",
            "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/ScheduleDefinition.schema.json",
            "@type": "ScheduleDefinition"
          }],
          "trackingRecord": [
          {
            "id": "TR_101",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
            "description": "string",
            "systemId": "NT_101",
            "time": "2020-06-23T17:55:51.671Z",
            "user": "NP1",
            "@baseType": "TrackingRecord",
            "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
            "@type": "TrackingRecord"
          }
          ],
          "@baseType": "MeasurementCollectionJob",
          "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
          "@type": "MeasurementCollectionJob"
      }
  },
  "reportingSystem": {
      "id": "139",
      "name": "APP-110",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
  },
  "source": {
      "id": "471",
      "name": "APP-100",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
  },
```


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```
  "@baseType": "Event",
  "@type": "MeasurementCollectionJobDeleteEvent"
}
```

Attribute Value Change Event

Message example for MeasurementCollectionJobAttributeValueChangeEvent event

```
Content-Type: application/json

{
  "correlationId": "88d6563b-fef5",
  "description": "MeasurementCollectionJobAttributeValueChangeEvent illustration",
  "domain": "Commercial",
  "eventId": "470e-b052-17d87228ce4b",
  "eventTime": "2022-08-25T12:18:25.005Z",
  "eventType": "MeasurementCollectionJobAttributeValueChangeEvent",
  "priority": "5",
  "timeOcurred": "2022-08-25T12:18:22.736Z",
  "title": "MeasurementCollectionJobAttributeValueChangeEvent",
  "event": {
      "measurementCollectionJob": {
          "id": "4000",
          "href": "https://host:port/tmf-
api/performanceManagement/v5/MeasurementCollectionJob/4000",
          "consumingApplicationId": "4000",
          "creationTime": "2020-06-23T14:12:01.078Z",
          "jobId": "400",
          "jobPriority": 3,
          "jobOnDemand": false,
          "lastModifiedTime": "2020-06-23T14:12:01.079Z",
          "outputFormat": "string",
          "producingApplicationId": "NT_101",
          "adminState": "locked",
          "executionState": "acknowledged",
          "fileTransferData": [
            {
              "fileFormat": "csv",
              "fileLocation": "https://host:port/tmf-
api/performanceManagement/v5/FileTransferData",
              "transportProtocol": "ftp",
              "compressionType": {
                "compressionEnumType": "no_compression"
              },
              "packingType": {
                "packingEnumType": "NO_PACKING"
              },
              "retentionPeriod": {
                "amount": 1,
                "units": "month",
                "@baseType": "RetentionPeriod",
                "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/RetentionPeriod",
                "@type": "RetentionPeriod"
              },
              "@baseType": "FileTransferData",
              "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
              "@type": "FileTransferData"
            }
```

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```
          ],
          "granularity": "g_1mn",
          "monitoredClassCriteria": {
            "monitoredObjectClass": "string",
            "objectInstanceFilter": "string",
            "@baseType": "string",
            "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
            "@type": "string"
          },
          "monitoredInstancesCriteria": {
            "monitoredObjectInstances": ["string"],
            "@baseType": "string",
            "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
            "@type": "string"
          },
          "performanceIndicatorGroupSpecification": [
            {
              "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSGrouppecification/600",
              "id": "600",
              "name": "My critical performance indicators",
              "performanceIndicatorSpecification": [
                {
                  "id": "5000",
                  "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5000",
                  "name": "CPU speed",
                  "version": "1.0",
                  "@baseType": "PerformanceIndicatorSpecification",
                  "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
                  "@type": "PerformanceIndicatorSpecification"
                }
              ],
              "@baseType": "PerformanceIndicatorGroupSpecification",
              "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorGroupSpecificati
on.schema.json",
              "@type": "PerformanceIndicatorGroupSpecification"
            }
          ],
          "performanceIndicatorSpecification": [
            {
              "id": "101",
              "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/101",
              "derivationAlgorithm": "string",
              "derivationMethod": "string",
              "description": "string",
              "indicatorCategory": "string",
              "indicatorUnit": "string",
              "name": "string",
              "perspective": "string",
              "collectionType": "cumulative",
              "indicatorType": "string",
              "performanceIndicatorSpecRelationship": [
                {
                  "id": "321",
                  "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecRelationship/321",
```


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```
                  "relationshipType": "reliesOn",
                  "validFor": {
                    "endDateTime": "1985-04-12T23:20:50.52Z",
                    "startDateTime": "1985-04-12T23:20:50.52Z"
                  },
                  "@baseType": "PerformanceIndicatorSpecRelationship",
                  "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecRelationship
.schema.json",
                  "@type": "PerformanceIndicatorSpecRelationship"
                }
              ],
              "validFor": {
                "endDateTime": "1985-04-12T23:20:50.52Z",
                "startDateTime": "1985-04-12T23:20:50.52Z"
              },
              "@baseType": "PerformanceIndicatorSpecification",
              "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
              "@type": "PerformanceIndicatorSpecification"
            }
          ],
          "reportingPeriod": "r_1mn",
          "scheduleDefinition": [{
            "id": "1",
            "recurringFrequency": "weekly",
            "scheduleDefinitionEndTime": "2020-06-23T14:12:01.080Z",
            "scheduleDefinitionHourRange": "2",
            "scheduleDefinitionStartTime": "2020-06-23T14:12:01.080Z",
            "MonthlyScheduleDayOfWeekDefinition": {
              "recurringDaySequence": "string",
              "dayOfMonthRecurrence": [
                {
                  "dates": "2020-06-23T14:12:01.080Z",
                  "@baseType": "DayOfMonthRecurrence",
                  "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/DayOfMonthRecurrence.schema.json",
                  "@type": "DayOfMonthRecurrence"
                }
              ],
              "@baseType": "MonthlyScheduleDayOfWeekDefinition",
              "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/MonthlyScheduleDayOfWeekDefinition.s
chema.json",
              "@type": "MonthlyScheduleDayOfWeekDefinition"
            },
            "WeeklyScheduledDefinition": [
              {
                "dates": "2020-06-23T14:12:01.080Z",
                "@baseType": "WeeklyScheduledDefinition",
                "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/WeeklyScheduledDefinition.schema.jso
n",
                "@type": "WeeklyScheduledDefinition"
              }
            ],
            "@baseType": "ScheduleDefinition",
            "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/ScheduleDefinition.schema.json",
            "@type": "ScheduleDefinition"
          }],
          "trackingRecord": [
```


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```
          {
            "id": "TR_101",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
            "description": "string",
            "systemId": "NT_101",
            "time": "2020-06-23T17:55:51.671Z",
            "user": "NP1",
            "@baseType": "TrackingRecord",
            "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
            "@type": "TrackingRecord"
          }
          ],
          "@baseType": "MeasurementCollectionJob",
          "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
          "@type": "MeasurementCollectionJob"
      }
  },
  "reportingSystem": {
      "id": "139",
      "name": "APP-110",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
  },
  "source": {
      "id": "471",
      "name": "APP-100",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "MeasurementCollectionJobAttributeValueChangeEvent"
}
```


Execution State Change Event

Message example for MeasurementCollectionJobExecutionStateChangeEvent event


```
Content-Type: application/json

{
  "correlationId": "e941b96a-b783",
  "description": "MeasurementCollectionJobExecutionStateChangeEvent illustration",
  "domain": "Commercial",
  "eventId": "40e9-a950-9da00d0b4a6e",
  "eventTime": "2022-08-25T12:18:24.988Z",
  "eventType": "MeasurementCollectionJobExecutionStateChangeEvent",
  "priority": "3",
  "timeOcurred": "2022-08-25T12:18:23.999Z",
  "title": "MeasurementCollectionJobExecutionStateChangeEvent",
  "event": {
      "measurementCollectionJob": {
          "id": "4000",
          "href": "https://host:port/tmf-
api/performanceManagement/v5/MeasurementCollectionJob/4000",
          "consumingApplicationId": "4000",
          "creationTime": "2020-06-23T14:12:01.078Z",
          "jobId": "400",
```


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```
          "jobPriority": 3,
          "jobOnDemand": false,
          "lastModifiedTime": "2020-06-23T14:12:01.079Z",
          "outputFormat": "string",
          "producingApplicationId": "NT_101",
          "adminState": "locked",
          "executionState": "acknowledged",
          "fileTransferData": [
            {
              "fileFormat": "csv",
              "fileLocation": "https://host:port/tmf-
api/performanceManagement/v5/FileTransferData",
              "transportProtocol": "ftp",
              "compressionType": {
                "compressionEnumType": "no_compression"
              },
              "packingType": {
                "packingEnumType": "NO_PACKING"
              },
              "retentionPeriod": {
                "amount": 1,
                "units": "month",
                "@baseType": "RetentionPeriod",
                "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/RetentionPeriod",
                "@type": "RetentionPeriod"
              },
              "@baseType": "FileTransferData",
              "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
              "@type": "FileTransferData"
            }
          ],
          "granularity": "g_1mn",
          "monitoredClassCriteria": {
            "monitoredObjectClass": "string",
            "objectInstanceFilter": "string",
            "@baseType": "string",
            "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
            "@type": "string"
          },
          "monitoredInstancesCriteria": {
            "monitoredObjectInstances": ["string"],
            "@baseType": "string",
            "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
            "@type": "string"
          },
          "performanceIndicatorGroupSpecification": [
            {
              "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSGrouppecification/600",
              "id": "600",
              "name": "My critical performance indicators",
              "performanceIndicatorSpecification": [
                {
                  "id": "5000",
                  "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5000",
                  "name": "CPU speed",
                  "version": "1.0",
                  "@baseType": "PerformanceIndicatorSpecification",
```


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```
                  "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
                  "@type": "PerformanceIndicatorSpecification"
                }
              ],
              "@baseType": "PerformanceIndicatorGroupSpecification",
              "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorGroupSpecificati
on.schema.json",
              "@type": "PerformanceIndicatorGroupSpecification"
            }
          ],
          "performanceIndicatorSpecification": [
            {
              "id": "101",
              "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/101",
              "derivationAlgorithm": "string",
              "derivationMethod": "string",
              "description": "string",
              "indicatorCategory": "string",
              "indicatorUnit": "string",
              "name": "string",
              "perspective": "string",
              "collectionType": "cumulative",
              "indicatorType": "string",
              "performanceIndicatorSpecRelationship": [
                {
                  "id": "321",
                  "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecRelationship/321",
                  "relationshipType": "reliesOn",
                  "validFor": {
                    "endDateTime": "1985-04-12T23:20:50.52Z",
                    "startDateTime": "1985-04-12T23:20:50.52Z"
                  },
                  "@baseType": "PerformanceIndicatorSpecRelationship",
                  "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecRelationship
.schema.json",
                  "@type": "PerformanceIndicatorSpecRelationship"
                }
              ],
              "validFor": {
                "endDateTime": "1985-04-12T23:20:50.52Z",
                "startDateTime": "1985-04-12T23:20:50.52Z"
              },
              "@baseType": "PerformanceIndicatorSpecification",
              "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
              "@type": "PerformanceIndicatorSpecification"
            }
          ],
          "reportingPeriod": "r_1mn",
          "scheduleDefinition": [{
            "id": "1",
            "recurringFrequency": "weekly",
            "scheduleDefinitionEndTime": "2020-06-23T14:12:01.080Z",
            "scheduleDefinitionHourRange": "2",
            "scheduleDefinitionStartTime": "2020-06-23T14:12:01.080Z",
            "MonthlyScheduleDayOfWeekDefinition": {
```


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```
              "recurringDaySequence": "string",
              "dayOfMonthRecurrence": [
                {
                  "dates": "2020-06-23T14:12:01.080Z",
                  "@baseType": "DayOfMonthRecurrence",
                  "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/DayOfMonthRecurrence.schema.json",
                  "@type": "DayOfMonthRecurrence"
                }
              ],
              "@baseType": "MonthlyScheduleDayOfWeekDefinition",
              "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/MonthlyScheduleDayOfWeekDefinition.s
chema.json",
              "@type": "MonthlyScheduleDayOfWeekDefinition"
            },
            "WeeklyScheduledDefinition": [
              {
                "dates": "2020-06-23T14:12:01.080Z",
                "@baseType": "WeeklyScheduledDefinition",
                "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/WeeklyScheduledDefinition.schema.jso
n",
                "@type": "WeeklyScheduledDefinition"
              }
            ],
            "@baseType": "ScheduleDefinition",
            "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/ScheduleDefinition.schema.json",
            "@type": "ScheduleDefinition"
          }],
          "trackingRecord": [
          {
            "id": "TR_101",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
            "description": "string",
            "systemId": "NT_101",
            "time": "2020-06-23T17:55:51.671Z",
            "user": "NP1",
            "@baseType": "TrackingRecord",
            "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
            "@type": "TrackingRecord"
          }
          ],
          "@baseType": "MeasurementCollectionJob",
          "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
          "@type": "MeasurementCollectionJob"
      }
  },
  "reportingSystem": {
      "id": "139",
      "name": "APP-110",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
  },
  "source": {
      "id": "471",
      "name": "APP-100",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
```


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```
  },
  "@baseType": "Event",
  "@type": "MeasurementCollectionJobExecutionStateChangeEvent"
}
```

PerformanceMeasurement

Collection Ready Event

Message example for PerformanceMeasurementCollectionReady event

```
Content-Type: application/json

{
  "correlationId": "2bc480b9-2dc0",
  "description": "PerformanceMeasurementCollectionReadyEvent illustration",
  "domain": "Commercial",
  "eventId": "48e1-9989-2d1fce36bceb",
  "eventTime": "2022-08-25T12:18:24.978Z",
  "eventType": "PerformanceMeasurementCollectionReadyEvent",
  "priority": "5",
  "timeOcurred": "2022-08-25T12:18:19.818Z",
  "title": "PerformanceMeasurementCollectionReadyEvent",
  "event": {
      "performanceMeasurement": {
        "id": "1",
        "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceMeasurement/1",
        "description": "The first performance measurement in the database",
        "measurementCollectionJob": {
          "id": "4000",
          "href": "https://host:port/tmf-
api/performanceManagement/v5/measurementCollectionJob/4000",
          "name": "Measurement Collection Job from system A",
          "@baseType": "MeasurementCollectionJobRef",
          "@schemaLocation":
"https://myschemas/MeasurementCollectionJob.schema.json",
          "@type": "MeasurementCollectionJobRef",
          "@referredType": "MeasurementCollectionJob"
        },
        "performanceIndicatorValue": [
            {
              "performanceIndicatorSpecification":{
                "id": "1",
                "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/1",
                "@baseType": "PerformanceIndicatorSpecificationRef",
                "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
                "@type": "PerformanceIndicatorSpecificationRef",
                "@referredType": "PerformanceIndicatorSpecification"
              },
              "observedValue": "2.1"
            },
            {
              "performanceIndicatorSpecification":{
                "id": "2",
                "name": "CPU Speed Indicator Specification",
                "indicatorCategory": "SPEEDCategory",
```

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```
                "indicatorName": "CPU Speed",
                "indicatorType": "float",
                "indicatorUnit": "GHz",
                "@baseType": "PerformanceIndicatorSpecification",
                "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
                "@type": "PerformanceIndicatorSpecification"
              },
              "observedValue": "3.25"
            }
        ],
        "validFor": {
          "endDateTime": "1985-04-12T23:20:50.52Z",
          "startDateTime": "1985-04-12T23:20:50.52Z"
        },
        "@baseType": "PerformanceMeasurement",
        "@schemaLocation":
"https://github.com/repo/PerformanceMeasurementAtomic.schema.json",
        "@type": "PerformanceMeasurementAtomic"
      }
  },
  "reportingSystem": {
      "id": "139",
      "name": "APP-110",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
  },
  "source": {
      "id": "471",
      "name": "APP-100",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "PerformanceMeasurementCollectionReadyEvent"
}
```

Create Event

Message example for PerformanceMeasurementCreateEvent event

```
Content-Type: application/json

{
  "correlationId": "2bc480b9-2dc0",
  "description": "PerformanceMeasurementCreateEvent illustration",
  "domain": "Commercial",
  "eventId": "48e1-9989-2d1fce36bceb",
  "eventTime": "2022-08-25T12:18:24.978Z",
  "eventType": "PerformanceMeasurementCreateEvent",
  "priority": "5",
  "timeOcurred": "2022-08-25T12:18:19.818Z",
  "title": "PerformanceMeasurementCreateEvent",
  "event": {
      "performanceMeasurement": {
        "id": "1",
        "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceMeasurement/1",
        "@type": "PerformanceMeasurementAtomic"
      }
```

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```
    },
    "reportingSystem": {
        "id": "139",
        "name": "APP-110",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "471",
        "name": "APP-100",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "PerformanceMeasurementCreateEvent"
}
```


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API OPERATIONS

Remember the following Uniform Contract:

Operation on Entities	Uniform API Operation	Description
Query Entities	GET MeasurementCollectionJob GET PerformanceIndicatorGroupSpecification GET PerformanceIndicatorSpecification	GET must be used to retrieve a representation of a resource.
Create Entity	POST MeasurementCollectionJob POST PerformanceIndicatorGroupSpecification POST PerformanceIndicatorSpecification	POST must be used to create a new resource
Partial Update of an Entity	PATCH MeasurementCollectionJob PATCH PerformanceIndicatorGroupSpecification PATCH PerformanceIndicatorSpecification	PATCH must be used to partially update a resource
Remove an Entity	DELETE MeasurementCollectionJob DELETE PerformanceIndicatorGroupSpecification DELETE PerformanceIndicatorSpecification	DELETE must be used to remove a resource

Filtering and attribute selection rules are described in the TMF REST Design Guidelines.

Notifications are also described in a subsequent section.

Operations on MeasurementCollectionJob

List or find MeasurementCollectionJob objects

`GET /measurementCollectionJob?fields=…`

Description

This operation list MeasurementCollectionJob entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

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Here's an example of a request for retrieving Measurement Collection Job(s).

Request

```
GET /measurementCollectionJob?consumingApplicationId=4000
Content-Type: application/json
```

Response

```
200

[
  {
    "id": "4000",
    "href": "https://host:port/tmf-
api/performanceManagement/v5/MeasurementCollectionJob/4000",
    "consumingApplicationId": "4000",
    "creationTime": "2020-06-23T14:12:01.078Z",
    "jobId": "400",
    "jobPriority": 3,
    "jobOnDemand": false,
    "lastModifiedTime": "2020-06-23T14:12:01.079Z",
    "outputFormat": "string",
    "producingApplicationId": "NT_101",
    "adminState": "locked",
    "executionState": "acknowledged",
    "fileTransferData": [
      {
        "fileFormat": "csv",
        "fileLocation": "https://host:port/tmf-
api/performanceManagement/v5/FileTransferData",
        "transportProtocol": "ftp",
        "compressionType": {
          "compressionEnumType": "no_compression"
        },
        "packingType": {
          "packingEnumType": "NO_PACKING"
        },
        "retentionPeriod": {
          "amount": 1,
          "units": "month",
          "@baseType": "RetentionPeriod",
          "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/RetentionPeriod",
          "@type": "RetentionPeriod"
        },
        "@baseType": "FileTransferData",
        "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
        "@type": "FileTransferData"
      }
    ],
    "granularity": "g_1mn",
    "performanceIndicatorGroupSpecification": [
      {
        "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSGrouppecification/600",
        "id": "600",
        "name": "My critical performance indicators",
```

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```
        "groupCategory": "CPUGroup",
        "performanceIndicatorSpecification": [
          {
            "id": "5000",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5000",
            "name": "CPU speed",
            "version": "1.0",
            "@baseType": "PerformanceIndicatorSpecification",
            "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
            "@type": "PerformanceIndicatorSpecification"
          }
        ],
        "@baseType": "PerformanceIndicatorGroupSpecification",
        "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorGroupSpecificati
on.schema.json",
        "@type": "PerformanceIndicatorGroupSpecification"
      }
    ],
    "performanceIndicatorSpecification": [
      {
        "id": "101",
        "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/101",
        "derivationAlgorithm": "na",
        "derivationMethod": "average",
        "description": "The average CPU speed",
        "indicatorCategory": "CPU",
        "indicatorUnit": "GHz",
        "name": "CPU Speed",
        "perspective": "aggregation over a collection of systems",
        "collectionType": "cumulative",
        "indicatorType": "float",
        "performanceIndicatorSpecRelationship": [
          {
            "id": "321",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecRelationship/321",
            "relationshipType": "reliesOn",
            "validFor": {
              "endDateTime": "1985-04-12T23:20:50.52Z",
              "startDateTime": "1985-04-12T23:20:50.52Z"
            },
            "@baseType": "PerformanceIndicatorSpecRelationship",
            "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecRelationship
.schema.json",
            "@type": "PerformanceIndicatorSpecRelationship"
          }
        ],
        "validFor": {
          "endDateTime": "1985-04-12T23:20:50.52Z",
          "startDateTime": "1985-04-12T23:20:50.52Z"
        },
        "@baseType": "PerformanceIndicatorSpecification",
        "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
        "@type": "PerformanceIndicatorSpecification"
      }
```


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```
    ],
    "reportingPeriod": "r_1mn",
    "scheduleDefinition": [{
      "id": "1",
      "recurringFrequency": "weekly",
      "scheduleDefinitionEndTime": "2020-06-23T14:12:01.080Z",
      "scheduleDefinitionHourRange": "2",
      "scheduleDefinitionStartTime": "2020-06-23T14:12:01.080Z",
      "WeeklyScheduledDefinition": [
        {
          "dates": "2020-06-23T14:12:01.080Z",
          "@baseType": "WeeklyScheduledDefinition",
          "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/WeeklyScheduledDefinition.schema.jso
n",
          "@type": "WeeklyScheduledDefinition"
        }
      ],
      "@baseType": "ScheduleDefinition",
      "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/ScheduleDefinition.schema.json",
      "@type": "ScheduleDefinition"
    }],
    "trackingRecord": [
    {
      "id": "TR_101",
      "href": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
      "description": "NT_101 tracking",
      "systemId": "NT_101",
      "time": "2020-06-23T17:55:51.671Z",
      "user": "NP1",
      "@baseType": "TrackingRecord",
      "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
      "@type": "TrackingRecord"
    }
    ],
    "@baseType": "MeasurementCollectionJob",
    "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
    "@type": "MeasurementCollectionJob"
  },
  {
    "id": "4001",
    "href": "https://host:port/tmf-
api/performanceManagement/v5/MeasurementCollectionJob/4001",
    "consumingApplicationId": "4000",
    "creationTime": "2020-06-23T15:12:01.078Z",
    "jobId": "401",
    "jobPriority": 3,
    "jobOnDemand": false,
    "lastModifiedTime": "2020-06-23T15:12:01.079Z",
    "outputFormat": "string",
    "producingApplicationId": "NT_101",
    "adminState": "locked",
    "executionState": "acknowledged",
    "fileTransferData": [
      {
        "fileFormat": "csv",
        "fileLocation": "https://host:port/tmf-
api/performanceManagement/v5/FileTransferData",
        "transportProtocol": "ftp",
```


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```
        "compressionType": {
          "compressionEnumType": "no_compression"
        },
        "packingType": {
          "packingEnumType": "NO_PACKING"
        },
        "retentionPeriod": {
          "amount": 1,
          "units": "month",
          "@baseType": "RetentionPeriod",
          "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/RetentionPeriod",
          "@type": "RetentionPeriod"
        },
        "@baseType": "FileTransferData",
        "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
        "@type": "FileTransferData"
      }
    ],
    "granularity": "g_1mn",
    "performanceIndicatorGroupSpecification": [
      {
        "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSGrouppecification/600",
        "id": "600",
        "name": "My critical performance indicators",
        "groupCategory": "CPUGroup",
        "performanceIndicatorSpecification": [
          {
            "id": "5000",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5000",
            "name": "CPU speed",
            "version": "1.0",
            "@baseType": "PerformanceIndicatorSpecification",
            "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
            "@type": "PerformanceIndicatorSpecification"
          }
        ],
        "@baseType": "PerformanceIndicatorGroupSpecification",
        "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorGroupSpecificati
on.schema.json",
        "@type": "PerformanceIndicatorGroupSpecification"
      }
    ],
    "performanceIndicatorSpecification": [
      {
        "id": "101",
        "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/101",
        "derivationAlgorithm": "na",
        "derivationMethod": "average",
        "description": "The average CPU speed",
        "indicatorCategory": "CPU",
        "indicatorUnit": "GHz",
        "name": "CPU Speed",
        "perspective": "aggregation over a collection of systems",
        "collectionType": "cumulative",
        "indicatorType": "float",
```


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```
        "performanceIndicatorSpecRelationship": [
          {
            "id": "321",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecRelationship/321",
            "relationshipType": "reliesOn",
            "validFor": {
              "endDateTime": "1985-04-12T23:20:50.52Z",
              "startDateTime": "1985-04-12T23:20:50.52Z"
            },
            "@baseType": "PerformanceIndicatorSpecRelationship",
            "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecRelationship
.schema.json",
            "@type": "PerformanceIndicatorSpecRelationship"
          }
        ],
        "validFor": {
          "endDateTime": "1985-04-12T23:20:50.52Z",
          "startDateTime": "1985-04-12T23:20:50.52Z"
        },
        "@baseType": "PerformanceIndicatorSpecification",
        "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
        "@type": "PerformanceIndicatorSpecification"
      }
    ],
    "reportingPeriod": "r_1mn",
    "scheduleDefinition": [{
      "id": "2",
      "recurringFrequency": "weekly",
      "scheduleDefinitionEndTime": "2020-06-23T14:12:01.080Z",
      "scheduleDefinitionHourRange": "2",
      "scheduleDefinitionStartTime": "2020-06-23T14:12:01.080Z",
      "WeeklyScheduledDefinition": [
        {
          "dates": "2020-06-23T14:12:01.080Z",
          "@baseType": "WeeklyScheduledDefinition",
          "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/WeeklyScheduledDefinition.schema.jso
n",
          "@type": "WeeklyScheduledDefinition"
        }
      ],
      "@baseType": "ScheduleDefinition",
      "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/ScheduleDefinition.schema.json",
      "@type": "ScheduleDefinition"
    }],
    "trackingRecord": [
    {
      "id": "TR_102",
      "href": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
      "description": "NT_101 tracking",
      "systemId": "NT_101",
      "time": "2020-06-23T17:55:51.671Z",
      "user": "NP1",
      "@baseType": "TrackingRecord",
      "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
      "@type": "TrackingRecord"
```


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```
    }
    ],
    "@baseType": "MeasurementCollectionJob",
    "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
    "@type": "MeasurementCollectionJob"
  }
]
```

Retrieves a MeasurementCollectionJob by ID

`GET /measurementCollectionJob/{id}?fields=…`

Description

This operation retrieves a MeasurementCollectionJob entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving a specific Measurement Collection Job.

Request

```
GET /measurementCollectionJob/4000
Content-Type: application/json
```

Response

```
200

{
  "id": "4000",
  "href": "https://host:port/tmf-
api/performanceManagement/v5/MeasurementCollectionJob/4000",
  "consumingApplicationId": "4000",
  "creationTime": "2020-06-23T14:12:01.078Z",
  "jobId": "400",
  "jobPriority": 3,
  "jobOnDemand": false,
  "lastModifiedTime": "2020-06-23T14:12:01.079Z",
  "outputFormat": "string",
  "producingApplicationId": "NT_101",
  "adminState": "locked",
  "executionState": "acknowledged",
  "fileTransferData": [
    {
      "fileFormat": "csv",
      "fileLocation": "https://host:port/tmf-
api/performanceManagement/v5/FileTransferData",
      "transportProtocol": "ftp",
      "compressionType": {
        "compressionEnumType": "no_compression"
      },
```

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```
      "packingType": {
        "packingEnumType": "NO_PACKING"
      },
      "retentionPeriod": {
        "amount": 1,
        "units": "month",
        "@baseType": "RetentionPeriod",
        "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/RetentionPeriod",
        "@type": "RetentionPeriod"
      },
      "@baseType": "FileTransferData",
      "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
      "@type": "FileTransferData"
    }
  ],
  "granularity": "g_1mn",
  "performanceIndicatorGroupSpecification": [
    {
      "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSGrouppecification/600",
      "id": "600",
      "name": "My critical performance indicators",
      "groupCategory": "CPUGroup",
      "performanceIndicatorSpecification": [
        {
          "id": "5000",
          "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5000",
          "name": "CPU speed",
          "version": "1.0",
          "@baseType": "PerformanceIndicatorSpecification",
          "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
          "@type": "PerformanceIndicatorSpecification"
        }
      ],
      "@baseType": "PerformanceIndicatorGroupSpecification",
      "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorGroupSpecificati
on.schema.json",
      "@type": "PerformanceIndicatorGroupSpecification"
    }
  ],
  "performanceIndicatorSpecification": [
    {
      "id": "101",
      "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/101",
      "derivationAlgorithm": "na",
      "derivationMethod": "average",
      "description": "The average CPU speed",
      "indicatorCategory": "CPU",
      "indicatorUnit": "GHz",
      "name": "CPU Speed",
      "perspective": "aggregation over a collection of systems",
      "collectionType": "cumulative",
      "indicatorType": "float",
      "performanceIndicatorSpecRelationship": [
        {
          "id": "321",
```


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```
          "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecRelationship/321",
          "relationshipType": "reliesOn",
          "validFor": {
            "endDateTime": "1985-04-12T23:20:50.52Z",
            "startDateTime": "1985-04-12T23:20:50.52Z"
          },
          "@baseType": "PerformanceIndicatorSpecRelationship",
          "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecRelationship
.schema.json",
          "@type": "PerformanceIndicatorSpecRelationship"
        }
      ],
      "validFor": {
        "endDateTime": "1985-04-12T23:20:50.52Z",
        "startDateTime": "1985-04-12T23:20:50.52Z"
      },
      "@baseType": "PerformanceIndicatorSpecification",
      "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
      "@type": "PerformanceIndicatorSpecification"
    }
  ],
  "reportingPeriod": "r_1mn",
  "scheduleDefinition": [{
    "id": "1",
    "recurringFrequency": "weekly",
    "scheduleDefinitionEndTime": "2020-06-23T14:12:01.080Z",
    "scheduleDefinitionHourRange": "2",
    "scheduleDefinitionStartTime": "2020-06-23T14:12:01.080Z",
    "WeeklyScheduledDefinition": [
      {
        "dates": "2020-06-23T14:12:01.080Z",
        "@baseType": "WeeklyScheduledDefinition",
        "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/WeeklyScheduledDefinition.schema.jso
n",
        "@type": "WeeklyScheduledDefinition"
      }
    ],
    "@baseType": "ScheduleDefinition",
    "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/ScheduleDefinition.schema.json",
    "@type": "ScheduleDefinition"
  }],
  "trackingRecord": [
  {
    "id": "TR_101",
    "href": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
    "description": "string",
    "systemId": "NT_101",
    "time": "2020-06-23T17:55:51.671Z",
    "user": "NP1",
    "@baseType": "TrackingRecord",
    "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
    "@type": "TrackingRecord"
  }
  ],
  "@baseType": "MeasurementCollectionJob",
```


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```
  "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
  "@type": "MeasurementCollectionJob"
}
```

Creates a MeasurementCollectionJob

`POST /measurementCollectionJob?fields=…`

Description

This operation creates a MeasurementCollectionJob entity.

Mandatory Attributes

Mandatory Attributes	Rule
consumingApplicationId	
performanceIndicatorSpecification	
performanceIndicatorSpecification .@type	
performanceIndicatorSpecification .collectionType	
performanceIndicatorSpecification .indicatorCategory	
performanceIndicatorSpecification .indicatorType	
performanceIndicatorSpecification .indicatorUnit	
performanceIndicatorSpecification .name	
producingApplicationId	

Usage samples

Creation of a new MeasurementCollectionJob with POST operation

Request

```
POST /measurementCollectionJob
Content-Type: application/json

{
  "consumingApplicationId": "4000",
  "creationTime": "2020-06-23T14:12:01.078Z",
  "jobId": "400",
  "jobPriority": 3,
  "lastModifiedTime": "2020-06-23T14:12:01.079Z",
  "outputFormat": "string",
  "producingApplicationId": "NT_101",
```

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```
  "adminState": "locked",
  "executionState": "acknowledged",
  "fileTransferData": [
    {
      "fileFormat": "csv",
      "fileLocation": "https://host:port/tmf-
api/performanceManagement/v5/FileTransferData",
      "transportProtocol": "ftp",
      "compressionType": {
        "compressionEnumType": "no_compression"
      },
      "packingType": {
        "packingEnumType": "NO_PACKING"
      },
      "retentionPeriod": {
        "amount": 1,
        "units": "month",
        "@baseType": "RetentionPeriod",
        "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/RetentionPeriod",
        "@type": "RetentionPeriod"
      },
      "@baseType": "FileTransferData",
      "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
      "@type": "FileTransferData"
    }
  ],
  "granularity": "g_1mn",
  "performanceIndicatorGroupSpecification": [
    {
      "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorGroupSpecification/600",
      "id": "600",
      "name": "My critical performance indicators",
      "groupCategory": "CPUGroup",
      "performanceIndicatorSpecification": [
        {
          "id": "5000",
          "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5000",
          "@type": "PerformanceIndicatorSpecificationRef"
        }
      ],
      "@baseType": "PerformanceIndicatorGroupSpecification",
      "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorGroupSpecificati
on.schema.json",
      "@type": "PerformanceIndicatorGroupSpecification"
    }
  ],
  "performanceIndicatorSpecification": [
    {
      "id": "101",
      "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/101",
      "derivationAlgorithm": "na",
      "derivationMethod": "average",
      "description": "The average CPU speed",
      "indicatorCategory": "CPU",
      "indicatorUnit": "GHz",
      "name": "CPU Speed",
      "perspective": "aggregation over a collection of systems",
```


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```
      "collectionType": "cumulative",
      "indicatorType": "float",
      "performanceIndicatorSpecRelationship": [
        {
          "id": "321",
          "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecRelationship/321",
          "relationshipType": "reliesOn",
          "validFor": {
            "endDateTime": "1985-04-12T23:20:50.52Z",
            "startDateTime": "1985-04-12T23:20:50.52Z"
          },
          "@baseType": "PerformanceIndicatorSpecRelationship",
          "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecRelationship
.schema.json",
          "@type": "PerformanceIndicatorSpecRelationship"
        }
      ],
      "validFor": {
        "endDateTime": "1985-04-12T23:20:50.52Z",
        "startDateTime": "1985-04-12T23:20:50.52Z"
      },
      "@baseType": "PerformanceIndicatorSpecification",
      "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
      "@type": "PerformanceIndicatorSpecification"
    }
  ],
  "reportingPeriod": "r_1mn",
  "scheduleDefinition": [{
    "id": "1",
    "recurringFrequency": "weekly",
    "scheduleDefinitionEndTime": "2020-06-23T14:12:01.080Z",
    "scheduleDefinitionHourRange": "2",
    "scheduleDefinitionStartTime": "2020-06-23T14:12:01.080Z",
    "WeeklyScheduledDefinition": [
      {
        "dates": "2020-06-23T14:12:01.080Z",
        "@baseType": "WeeklyScheduledDefinition",
        "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/WeeklyScheduledDefinition.schema.jso
n",
        "@type": "WeeklyScheduledDefinition"
      }
    ],
    "@baseType": "ScheduleDefinition",
    "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/ScheduleDefinition.schema.json",
    "@type": "ScheduleDefinition"
  }],
  "trackingRecord": [
  {
    "id": "TR_101",
    "href": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
    "description": "NT_101 tracking",
    "systemId": "NT_101",
    "time": "2020-06-23T17:55:51.671Z",
    "user": "NP1",
    "@baseType": "TrackingRecord",
    "@schemaLocation": "https://host:port/tmf-
```


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```
api/performanceManagement/v5/fileTransferLocation",
    "@type": "TrackingRecord"
  }
  ],
  "@baseType": "MeasurementCollectionJob",
  "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
  "@type": "MeasurementCollectionJob"
}
```

Response

```
201

{
  "id": "4000",
  "href": "https://host:port/tmf-
api/performanceManagement/v5/MeasurementCollectionJob/4000",
  "consumingApplicationId": "4000",
  "creationTime": "2020-06-23T14:12:01.078Z",
  "jobId": "400",
  "jobPriority": 3,
  "jobOnDemand": false,
  "lastModifiedTime": "2020-06-23T14:12:01.079Z",
  "outputFormat": "string",
  "producingApplicationId": "NT_101",
  "adminState": "locked",
  "executionState": "acknowledged",
  "fileTransferData": [
    {
      "fileFormat": "csv",
      "fileLocation": "https://host:port/tmf-
api/performanceManagement/v5/FileTransferData",
      "transportProtocol": "ftp",
      "compressionType": {
        "compressionEnumType": "no_compression"
      },
      "packingType": {
        "packingEnumType": "NO_PACKING"
      },
      "retentionPeriod": {
        "amount": 1,
        "units": "month",
        "@baseType": "RetentionPeriod",
        "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/RetentionPeriod",
        "@type": "RetentionPeriod"
      },
      "@baseType": "FileTransferData",
      "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
      "@type": "FileTransferData"
    }
  ],
  "granularity": "g_1mn",
  "performanceIndicatorGroupSpecification": [
    {
      "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSGrouppecification/600",
      "id": "600",
```

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```
      "name": "My critical performance indicators",
      "groupCategory": "CPUGroup",
      "performanceIndicatorSpecification": [
        {
          "id": "5000",
          "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5000",
          "@type": "PerformanceIndicatorSpecificationRef"
        }
      ],
      "@baseType": "PerformanceIndicatorGroupSpecification",
      "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorGroupSpecificati
on.schema.json",
      "@type": "PerformanceIndicatorGroupSpecification"
    }
  ],
  "performanceIndicatorSpecification": [
    {
      "id": "101",
      "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/101",
      "derivationAlgorithm": "na",
      "derivationMethod": "average",
      "description": "The average CPU speed",
      "indicatorCategory": "CPU",
      "indicatorUnit": "GHz",
      "name": "CPU Speed",
      "perspective": "aggregation over a collection of systems",
      "collectionType": "cumulative",
      "indicatorType": "float",
      "performanceIndicatorSpecRelationship": [
        {
          "id": "321",
          "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecRelationship/321",
          "relationshipType": "reliesOn",
          "validFor": {
            "endDateTime": "1985-04-12T23:20:50.52Z",
            "startDateTime": "1985-04-12T23:20:50.52Z"
          },
          "@baseType": "PerformanceIndicatorSpecRelationship",
          "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecRelationship
.schema.json",
          "@type": "PerformanceIndicatorSpecRelationship"
        }
      ],
      "validFor": {
        "endDateTime": "1985-04-12T23:20:50.52Z",
        "startDateTime": "1985-04-12T23:20:50.52Z"
      },
      "@baseType": "PerformanceIndicatorSpecification",
      "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
      "@type": "PerformanceIndicatorSpecification"
    }
  ],
  "reportingPeriod": "r_1mn",
  "scheduleDefinition": [{
    "id": "1",
    "recurringFrequency": "weekly",
```


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```
    "scheduleDefinitionEndTime": "2020-06-23T14:12:01.080Z",
    "scheduleDefinitionHourRange": "2",
    "scheduleDefinitionStartTime": "2020-06-23T14:12:01.080Z",
    "WeeklyScheduledDefinition": [
      {
        "dates": "2020-06-23T14:12:01.080Z",
        "@baseType": "WeeklyScheduledDefinition",
        "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/WeeklyScheduledDefinition.schema.jso
n",
        "@type": "WeeklyScheduledDefinition"
      }
    ],
    "@baseType": "ScheduleDefinition",
    "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/ScheduleDefinition.schema.json",
    "@type": "ScheduleDefinition"
  }],
  "trackingRecord": [
  {
    "id": "TR_101",
    "href": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
    "description": "NT_101 tracking",
    "systemId": "NT_101",
    "time": "2020-06-23T17:55:51.671Z",
    "user": "NP1",
    "@baseType": "TrackingRecord",
    "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
    "@type": "TrackingRecord"
  }
  ],
  "@baseType": "MeasurementCollectionJob",
  "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/fileTransferLocation",
  "@type": "MeasurementCollectionJob"
}
```


Updates partially a MeasurementCollectionJob

`PATCH /measurementCollectionJob/{id}?fields=…`

Description

This operation allows partial updates of a MeasurementCollectionJob entity. Support of json/merge (https://tools.ietf.org/html/rfc7386) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optiona. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
adminState	

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Non Patchable Attributes	Rule
creationTime	
executionState	
href	
jobId	
jobPriority	
lastModifiedTime	
monitoredClassCriteria	
monitoredInstancesCriteria	

Patchable Attributes	Rule
consumingApplicationId	
dataAccessEndpoint	
fileTransferData	
granularity	
id	
jobCollectionFilter	
jobOnDemand	
outputFormat	
performanceIndicatorGroupSpecification	
performanceIndicatorSpecification	
producingApplicationId	
reportingPeriod	
scheduleDefinition	
searchTaskFilter	
trackingRecord	

Usage samples

Update a MeasurementCollectionJob using json

Request

```
PATCH /measurementCollectionJob/{id}
Content-Type: application/json

{
    "reportingPeriod": "r_1mn",
    "dataAccessEndpoint": [
        {
```

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```
            "id": "123-xxx",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/dataAccessEndpoint/123-xxx",
            "apiType": "REST",
            "category": "topic",
            "description": "Measurement collection job 123-xxx subscriber topic",
            "endOperatingDate": "2022-05-13T10:13:42.971Z",
            "name": "123-xxx subscriber topic",
            "resourceVersion": "1.0",
            "startOperatingDate": "2022-05-13T10:13:42.971Z",
            "uri": "https://pubsub.googleapis.com/v1/",
            "value": "myLogicalResourceValueIp",
            "@type": "DataAccessEndpoint"
        }
    ],
    "@type": "MeasurementCollectionJob"
}
```


Response


```
200

{
    "id": "4001",
    "href": "https://host:port/tmf-
api/performanceManagement/v5/MeasurementCollectionJob/4001",
    "creationTime": "2022-05-13T10:13:42.971Z",
    "jobId": "402",
    "jobPriority": 4,
    "jobOnDemand": false,
    "lastModifiedTime": "2022-05-13T10:13:42.971Z",
    "producingApplicationId": "NT_101",
    "consumingApplicationId": "4000",
    "outputFormat": "string",
    "adminState": "locked",
    "dataAccessEndpoint": [
        {
            "id": "123-xxx",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/dataAccessEndpoint/123-xxx",
            "apiType": "REST",
            "category": "topic",
            "description": "Measurement collection job 123-xxx subscriber topic",
            "endOperatingDate": "2022-05-13T10:13:42.971Z",
            "name": "123-xxx subscriber topic",
            "resourceVersion": "1.0",
            "startOperatingDate": "2022-05-13T10:13:42.971Z",
            "uri": "https://pubsub.googleapis.com/v1/",
            "value": "myLogicalResourceValueIp",
            "@type": "DataAccessEndpoint"
        }
    ],
    "executionState": "acknowledged",
    "performanceIndicatorSpecification": [
        {
            "id": "201",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/PerformanceIndicatorSpecification/201",
            "derivationAlgorithm": "na",
            "derivationMethod": "average",
```


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```
            "description": "The average CPU speed",
            "indicatorCategory": "CPU",
            "indicatorUnit": "GHz",
            "name": "CPU Speed",
            "perspective": "aggregation over a collection of systems",
            "collectionType": "cumulative",
            "indicatorType": "float",
            "validFor": {
                "endDateTime": "2022-04-12T23:20:50.52Z",
                "startDateTime": "2022-04-12T23:20:50.52Z"
            },
            "@baseType": "PerformanceIndicatorSpecification",
            "@schemaLocation":
"https://myschemas.com/PerformanceIndicatorSpecification.schema.json",
            "@type": "PerformanceIndicatorSpecification"
        }
    ],
    "reportingPeriod": "r_1mn",
    "scheduleDefinition": [
        {
            "id": "1",
            "recurringFrequency": "weekly",
            "scheduleDefinitionEndTime": "2022-05-13T10:13:42.971Z",
            "scheduleDefinitionHourRange": "08:00-12:00,16:00-19:00",
            "scheduleDefinitionStartTime": "2022-05-13T10:13:42.971Z",
            "weeklyScheduledDefinition": [
                {
                    "dates": "2022-05-13T10:13:42.971Z",
                    "@baseType": "DayOfWeekRecurrence",
                    "@schemaLocation":
"https://myschemas.com/DayOfWeekRecurrence.schema.json",
                    "@type": "DayOfWeekRecurrence"
                }
        ],
        "@baseType": "ScheduleDefinition",
        "@schemaLocation": "https://myschemas.com/ScheduleDefinition.schema.json",
        "@type": "ScheduleDefinition"
        }
    ],
    "jobCollectionFilter":{
        "@type" : "DataFilterMap",
        "mappings": [
            {
                "filterTemplate": {
                    "name" : "myRangeFilterCollect0",
                    "@type" : "DataFilterTemplateRange",
                    "criteria":{
                        "lowerBound" : {
                            "boundValue": -10,
                            "isInclusive": true
                        },
                        "upperBound" : {
                            "boundValue": -10,
                            "isInclusive": true
                        }
                    }
                },
                "stringArray": {
                    "value": [
                    "field01", "field02", "field03"
                     ]
                }
            },
```


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```
            {
                "filterTemplate": {
                    "name": "myRegExSampleCollect0",
                    "@type": "DataFilterTemplateRegex",
                    "criteria": "dallas|houston|venise|paris|new-york"
                },
                "stringArray": {
                    "value":[
                        "field11", "field12", "field13"
                    ]
                }
            },
            {
                "filterTemplate": {
                    "name" : "myBooleanFilterCollect0",
                    "criteria" : true,
                    "@type" : "DataFilterTemplateBoolean"
                },
                "stringArray": {
                    "value": [
                        "field1", "field2", "field3"
                    ]
                }
            }
        ]
    },
    "searchTaskFilter":{
        "@type" : "DataFilterMap",
        "mappings": [
            {
                "filterTemplate": {
                    "name" : "myRangeFilterSearch0",
                    "@type" : "DataFilterTemplateRange",
                    "criteria":{
                        "lowerBound" : {
                            "boundValue": -10,
                            "isInclusive": true
                        },
                        "upperBound" : {
                            "boundValue": -10,
                            "isInclusive": true
                        }
                    }
                },
                "stringArray": {
                    "value": [
                    "field01", "field02", "field03"
                     ]
                }
            },
            {
                "filterTemplate": {
                    "name": "myRegExSampleSearch0",
                    "@type": "DataFilterTemplateRegex",
                    "criteria": "dallas|houston|venise|paris|new-york"
                },
                "stringArray": {
                    "value":[
                        "field11", "field12", "field13"
                    ]
                }
            },
            {
```

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```
                "filterTemplate": {
                    "name" : "myBooleanFilterSearch0",
                    "criteria" : true,
                    "@type" : "DataFilterTemplateBoolean"
                },
                "stringArray": {
                    "value": [
                        "field1", "field2", "field3"
                    ]
                }
            }
        ]
    },
    "@baseType": "MeasurementCollectionJob",
    "@schemaLocation": "https://myschemas.com/MeasurementCollectionJob.schema.json",
    "@type": "MeasurementCollectionJob"
}
```

Update Without query parameters a MeasurementCollectionJob using json-patch+json

Request

```
PATCH /measurementCollectionJob/{id}
Content-Type: application/json-patch+json

[
    {   "op":"replace",
        "path": "/reportingPeriod",
        "value": "r_1mn"
    },
    {   "op":"replace",
        "path" : "/scheduleDefinition/0/recurringFrequency",
        "value" : "weekly"
    }
]
```

Response

```
200

{
    "id": "4001",
    "href": "https://host:port/tmf-
api/performanceManagement/v5/MeasurementCollectionJob/4001",
    "creationTime": "2022-05-13T10:13:42.971Z",
    "jobId": "402",
    "jobPriority": 4,
    "jobOnDemand": false,
    "lastModifiedTime": "2022-05-13T10:13:42.971Z",
    "producingApplicationId": "NT_101",
    "consumingApplicationId": "4000",
    "outputFormat": "string",
    "adminState": "locked",
    "dataAccessEndpoint": [
        {
            "id": "123-xxx",
            "href": "https://host:port/tmf-
```

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```
api/performanceManagement/v5/dataAccessEndpoint/123-xxx",
            "apiType": "REST",
            "category": "topic",
            "description": "Measurement collection job 123-xxx subscriber topic",
            "endOperatingDate": "2022-05-13T10:13:42.971Z",
            "name": "123-xxx subscriber topic",
            "resourceVersion": "1.0",
            "startOperatingDate": "2022-05-13T10:13:42.971Z",
            "uri": "https://pubsub.googleapis.com/v1/",
            "value": "myLogicalResourceValueIp",
            "@type": "DataAccessEndpoint"
        }
    ],
    "executionState": "acknowledged",
    "performanceIndicatorSpecification": [
        {
            "id": "201",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/PerformanceIndicatorSpecification/201",
            "derivationAlgorithm": "na",
            "derivationMethod": "average",
            "description": "The average CPU speed",
            "indicatorCategory": "CPU",
            "indicatorUnit": "GHz",
            "name": "CPU Speed",
            "perspective": "aggregation over a collection of systems",
            "collectionType": "cumulative",
            "indicatorType": "float",
            "validFor": {
                "endDateTime": "2022-04-12T23:20:50.52Z",
                "startDateTime": "2022-04-12T23:20:50.52Z"
            },
            "@baseType": "PerformanceIndicatorSpecification",
            "@schemaLocation":
"https://myschemas.com/PerformanceIndicatorSpecification.schema.json",
            "@type": "PerformanceIndicatorSpecification"
        }
    ],
    "reportingPeriod": "r_1mn",
    "scheduleDefinition": [
        {
            "id": "1",
            "recurringFrequency": "weekly",
            "scheduleDefinitionEndTime": "2022-05-13T10:13:42.971Z",
            "scheduleDefinitionHourRange": "08:00-12:00,16:00-19:00",
            "scheduleDefinitionStartTime": "2022-05-13T10:13:42.971Z",
            "weeklyScheduledDefinition": [
                {
                    "dates": "2022-05-13T10:13:42.971Z",
                    "@baseType": "DayOfWeekRecurrence",
                    "@schemaLocation":
"https://myschemas.com/DayOfWeekRecurrence.schema.json",
                    "@type": "DayOfWeekRecurrence"
                }
        ],
        "@baseType": "ScheduleDefinition",
        "@schemaLocation": "https://myschemas.com/ScheduleDefinition.schema.json",
        "@type": "ScheduleDefinition"
        }
    ],
    "jobCollectionFilter":{
        "@type" : "DataFilterMap",
        "mappings": [
```


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```
            {
                "filterTemplate": {
                    "name" : "myRangeFilterCollect0",
                    "@type" : "DataFilterTemplateRange",
                    "criteria":{
                        "lowerBound" : {
                            "boundValue": -10,
                            "isInclusive": true
                        },
                        "upperBound" : {
                            "boundValue": -10,
                            "isInclusive": true
                        }
                    }
                },
                "stringArray": {
                    "value": [
                    "field01", "field02", "field03"
                     ]
                }
            },
            {
                "filterTemplate": {
                    "name": "myRegExSampleCollect0",
                    "@type": "DataFilterTemplateRegex",
                    "criteria": "dallas|houston|venise|paris|new-york"
                },
                "stringArray": {
                    "value":[
                        "field11", "field12", "field13"
                    ]
                }
            },
            {
                "filterTemplate": {
                    "name" : "myBooleanFilterCollect0",
                    "criteria" : true,
                    "@type" : "DataFilterTemplateBoolean"
                },
                "stringArray": {
                    "value": [
                        "field1", "field2", "field3"
                    ]
                }
            }
        ]
    },
    "searchTaskFilter":{
        "@type" : "DataFilterMap",
        "mappings": [
            {
                "filterTemplate": {
                    "name" : "myRangeFilterSearch0",
                    "@type" : "DataFilterTemplateRange",
                    "criteria":{
                        "lowerBound" : {
                            "boundValue": -10,
                            "isInclusive": true
                        },
                        "upperBound" : {
                            "boundValue": -10,
                            "isInclusive": true
                        }
```

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```
                    }
                },
                "stringArray": {
                    "value": [
                    "field01", "field02", "field03"
                     ]
                }
            },
            {
                "filterTemplate": {
                    "name": "myRegExSampleSearch0",
                    "@type": "DataFilterTemplateRegex",
                    "criteria": "dallas|houston|venise|paris|new-york"
                },
                "stringArray": {
                    "value":[
                        "field11", "field12", "field13"
                    ]
                }
            },
            {
                "filterTemplate": {
                    "name" : "myBooleanFilterSearch0",
                    "criteria" : true,
                    "@type" : "DataFilterTemplateBoolean"
                },
                "stringArray": {
                    "value": [
                        "field1", "field2", "field3"
                    ]
                }
            }
        ]
    },
    "@baseType": "MeasurementCollectionJob",
    "@schemaLocation": "https://myschemas.com/MeasurementCollectionJob.schema.json",
    "@type": "MeasurementCollectionJob"
}
```

Update With query parameters a MeasurementCollectionJob using json-patch+json

Request

```
PATCH
/measurementCollectionJob/{id}?fields=id,href,lastModifiedTime,adminState,dataAccess
Endpoint,executionState,reportingPeriod,scheduleDefinition,@type
Content-Type: application/json-patch+json

[
    {   "op":"replace",
        "path": "/reportingPeriod",
        "value": "r_1mn"
    },
    {   "op":"replace",
        "path" : "/scheduleDefinition/0/recurringFrequency",
        "value" : "weekly"
    }
]
```

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Response

```
200

{
    "id": "4001",
    "href": "https://host:port/tmf-
api/performanceManagement/v5/MeasurementCollectionJob/4001",
    "lastModifiedTime": "2022-05-13T10:13:42.971Z",
    "adminState": "locked",
    "dataAccessEndpoint": [
        {
            "id": "123-xxx",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/dataAccessEndpoint/123-xxx",
            "apiType": "REST",
            "category": "topic",
            "description": "Measurement collection job 123-xxx subscriber topic",
            "endOperatingDate": "2022-05-13T10:13:42.971Z",
            "name": "123-xxx subscriber topic",
            "resourceVersion": "1.0",
            "startOperatingDate": "2022-05-13T10:13:42.971Z",
            "uri": "https://pubsub.googleapis.com/v1/",
            "value": "myLogicalResourceValueIp",
            "@type": "DataAccessEndpoint"
        }
    ],
    "executionState": "acknowledged",
    "reportingPeriod": "r_1mn",
    "scheduleDefinition": [
        {
            "id": "1",
            "recurringFrequency": "weekly",
            "scheduleDefinitionEndTime": "2022-05-13T10:13:42.971Z",
            "scheduleDefinitionHourRange": "08:00-12:00,16:00-19:00",
            "scheduleDefinitionStartTime": "2022-05-13T10:13:42.971Z",
            "weeklyScheduledDefinition": [
                {
                    "dates": "2022-05-13T10:13:42.971Z",
                    "@baseType": "DayOfWeekRecurrence",
                    "@schemaLocation":
"https://myschemas.com/DayOfWeekRecurrence.schema.json",
                    "@type": "DayOfWeekRecurrence"
                }
        ],
        "@baseType": "ScheduleDefinition",
        "@schemaLocation": "https://myschemas.com/ScheduleDefinition.schema.json",
        "@type": "ScheduleDefinition"
        }
    ],
    "@type": "MeasurementCollectionJob"
}
```

Update a MeasurementCollectionJob using json-patch-query+json

Request

```
PATCH /measurementCollectionJob/{id}
Content-Type: application/json-patch-query+json
```

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```
[
    {   "op":"replace",
        "path" : "scheduleDefinition[?(@.id='1')].recurringFrequency=weekly",
        "value" : "daily"
    }
]
```

Response

```
200

{
    "id": "4001",
    "href": "https://host:port/tmf-
api/performanceManagement/v5/MeasurementCollectionJob/4001",
    "creationTime": "2022-05-13T10:13:42.971Z",
    "jobId": "402",
    "jobPriority": 4,
    "jobOnDemand": false,
    "lastModifiedTime": "2022-05-13T10:13:42.971Z",
    "producingApplicationId": "NT_101",
    "consumingApplicationId": "4000",
    "outputFormat": "string",
    "adminState": "locked",
    "dataAccessEndpoint": [
        {
            "id": "123-xxx",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/dataAccessEndpoint/123-xxx",
            "apiType": "REST",
            "category": "topic",
            "description": "Measurement collection job 123-xxx subscriber topic",
            "endOperatingDate": "2022-05-13T10:13:42.971Z",
            "name": "123-xxx subscriber topic",
            "resourceVersion": "1.0",
            "startOperatingDate": "2022-05-13T10:13:42.971Z",
            "uri": "https://pubsub.googleapis.com/v1/",
            "value": "myLogicalResourceValueIp",
            "@type": "DataAccessEndpoint"
        }
    ],
    "executionState": "acknowledged",
    "performanceIndicatorSpecification": [
        {
            "id": "201",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/PerformanceIndicatorSpecification/201",
            "derivationAlgorithm": "na",
            "derivationMethod": "average",
            "description": "The average CPU speed",
            "indicatorCategory": "CPU",
            "indicatorUnit": "GHz",
            "name": "CPU Speed",
            "perspective": "aggregation over a collection of systems",
            "collectionType": "cumulative",
            "indicatorType": "float",
            "validFor": {
                "endDateTime": "2022-04-12T23:20:50.52Z",
                "startDateTime": "2022-04-12T23:20:50.52Z"
```

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```
            },
            "@baseType": "PerformanceIndicatorSpecification",
            "@schemaLocation":
"https://myschemas.com/PerformanceIndicatorSpecification.schema.json",
            "@type": "PerformanceIndicatorSpecification"
        }
    ],
    "reportingPeriod": "r_1mn",
    "scheduleDefinition": [
        {
            "id": "1",
            "recurringFrequency": "weekly",
            "scheduleDefinitionEndTime": "2022-05-13T10:13:42.971Z",
            "scheduleDefinitionHourRange": "08:00-12:00,16:00-19:00",
            "scheduleDefinitionStartTime": "2022-05-13T10:13:42.971Z",
            "weeklyScheduledDefinition": [
                {
                    "dates": "2022-05-13T10:13:42.971Z",
                    "@baseType": "DayOfWeekRecurrence",
                    "@schemaLocation":
"https://myschemas.com/DayOfWeekRecurrence.schema.json",
                    "@type": "DayOfWeekRecurrence"
                }
        ],
        "@baseType": "ScheduleDefinition",
        "@schemaLocation": "https://myschemas.com/ScheduleDefinition.schema.json",
        "@type": "ScheduleDefinition"
        }
    ],
    "jobCollectionFilter":{
        "@type" : "DataFilterMap",
        "mappings": [
            {
                "filterTemplate": {
                    "name" : "myRangeFilterCollect0",
                    "@type" : "DataFilterTemplateRange",
                    "criteria":{
                        "lowerBound" : {
                            "boundValue": -10,
                            "isInclusive": true
                        },
                        "upperBound" : {
                            "boundValue": -10,
                            "isInclusive": true
                        }
                    }
                },
                "stringArray": {
                    "value": [
                    "field01", "field02", "field03"
                     ]
                }
            },
            {
                "filterTemplate": {
                    "name": "myRegExSampleCollect0",
                    "@type": "DataFilterTemplateRegex",
                    "criteria": "dallas|houston|venise|paris|new-york"
                },
                "stringArray": {
                    "value":[
                        "field11", "field12", "field13"
                    ]
```


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```
                }
            },
            {
                "filterTemplate": {
                    "name" : "myBooleanFilterCollect0",
                    "criteria" : true,
                    "@type" : "DataFilterTemplateBoolean"
                },
                "stringArray": {
                    "value": [
                        "field1", "field2", "field3"
                    ]
                }
            }
        ]
    },
    "searchTaskFilter":{
        "@type" : "DataFilterMap",
        "mappings": [
            {
                "filterTemplate": {
                    "name" : "myRangeFilterSearch0",
                    "@type" : "DataFilterTemplateRange",
                    "criteria":{
                        "lowerBound" : {
                            "boundValue": -10,
                            "isInclusive": true
                        },
                        "upperBound" : {
                            "boundValue": -10,
                            "isInclusive": true
                        }
                    }
                },
                "stringArray": {
                    "value": [
                    "field01", "field02", "field03"
                     ]
                }
            },
            {
                "filterTemplate": {
                    "name": "myRegExSampleSearch0",
                    "@type": "DataFilterTemplateRegex",
                    "criteria": "dallas|houston|venise|paris|new-york"
                },
                "stringArray": {
                    "value":[
                        "field11", "field12", "field13"
                    ]
                }
            },
            {
                "filterTemplate": {
                    "name" : "myBooleanFilterSearch0",
                    "criteria" : true,
                    "@type" : "DataFilterTemplateBoolean"
                },
                "stringArray": {
                    "value": [
                        "field1", "field2", "field3"
                    ]
                }
```


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```
            }
        ]
    },
    "@baseType": "MeasurementCollectionJob",
    "@schemaLocation": "https://myschemas.com/MeasurementCollectionJob.schema.json",
    "@type": "MeasurementCollectionJob"
}
```

Update a MeasurementCollectionJob using merge-patch+json

Request

```
PATCH /measurementCollectionJob/{id}
Content-Type: application/merge-patch+json

{
    "reportingPeriod": "r_1mn",
    "dataAccessEndpoint": [
        {
            "id": "123-xxx",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/dataAccessEndpoint/123-xxx",
            "apiType": "REST",
            "category": "topic",
            "description": "Measurement collection job 123-xxx subscriber topic",
            "endOperatingDate": "2022-05-13T10:13:42.971Z",
            "name": "123-xxx subscriber topic",
            "resourceVersion": "1.0",
            "startOperatingDate": "2022-05-13T10:13:42.971Z",
            "uri": "https://pubsub.googleapis.com/v1/",
            "value": "myLogicalResourceValueIp",
            "@type": "DataAccessEndpoint"
        }
    ],
    "@type": "MeasurementCollectionJob"
}
```

Response

```
200

{
    "id": "4001",
    "href": "https://host:port/tmf-
api/performanceManagement/v5/MeasurementCollectionJob/4001",
    "creationTime": "2022-05-13T10:13:42.971Z",
    "jobId": "402",
    "jobPriority": 4,
    "jobOnDemand": false,
    "lastModifiedTime": "2022-05-13T10:13:42.971Z",
    "producingApplicationId": "NT_101",
    "consumingApplicationId": "4000",
    "outputFormat": "string",
    "adminState": "locked",
    "dataAccessEndpoint": [
        {
            "id": "123-xxx",
```

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```
            "href": "https://host:port/tmf-
api/performanceManagement/v5/dataAccessEndpoint/123-xxx",
            "apiType": "REST",
            "category": "topic",
            "description": "Measurement collection job 123-xxx subscriber topic",
            "endOperatingDate": "2022-05-13T10:13:42.971Z",
            "name": "123-xxx subscriber topic",
            "resourceVersion": "1.0",
            "startOperatingDate": "2022-05-13T10:13:42.971Z",
            "uri": "https://pubsub.googleapis.com/v1/",
            "value": "myLogicalResourceValueIp",
            "@type": "DataAccessEndpoint"
        }
    ],
    "executionState": "acknowledged",
    "performanceIndicatorSpecification": [
        {
            "id": "201",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/PerformanceIndicatorSpecification/201",
            "derivationAlgorithm": "na",
            "derivationMethod": "average",
            "description": "The average CPU speed",
            "indicatorCategory": "CPU",
            "indicatorUnit": "GHz",
            "name": "CPU Speed",
            "perspective": "aggregation over a collection of systems",
            "collectionType": "cumulative",
            "indicatorType": "float",
            "validFor": {
                "endDateTime": "2022-04-12T23:20:50.52Z",
                "startDateTime": "2022-04-12T23:20:50.52Z"
            },
            "@baseType": "PerformanceIndicatorSpecification",
            "@schemaLocation":
"https://myschemas.com/PerformanceIndicatorSpecification.schema.json",
            "@type": "PerformanceIndicatorSpecification"
        }
    ],
    "reportingPeriod": "r_1mn",
    "scheduleDefinition": [
        {
            "id": "1",
            "recurringFrequency": "weekly",
            "scheduleDefinitionEndTime": "2022-05-13T10:13:42.971Z",
            "scheduleDefinitionHourRange": "08:00-12:00,16:00-19:00",
            "scheduleDefinitionStartTime": "2022-05-13T10:13:42.971Z",
            "weeklyScheduledDefinition": [
                {
                    "dates": "2022-05-13T10:13:42.971Z",
                    "@baseType": "DayOfWeekRecurrence",
                    "@schemaLocation":
"https://myschemas.com/DayOfWeekRecurrence.schema.json",
                    "@type": "DayOfWeekRecurrence"
                }
        ],
        "@baseType": "ScheduleDefinition",
        "@schemaLocation": "https://myschemas.com/ScheduleDefinition.schema.json",
        "@type": "ScheduleDefinition"
        }
    ],
    "jobCollectionFilter":{
        "@type" : "DataFilterMap",
```


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```
        "mappings": [
            {
                "filterTemplate": {
                    "name" : "myRangeFilterCollect0",
                    "@type" : "DataFilterTemplateRange",
                    "criteria":{
                        "lowerBound" : {
                            "boundValue": -10,
                            "isInclusive": true
                        },
                        "upperBound" : {
                            "boundValue": -10,
                            "isInclusive": true
                        }
                    }
                },
                "stringArray": {
                    "value": [
                    "field01", "field02", "field03"
                     ]
                }
            },
            {
                "filterTemplate": {
                    "name": "myRegExSampleCollect0",
                    "@type": "DataFilterTemplateRegex",
                    "criteria": "dallas|houston|venise|paris|new-york"
                },
                "stringArray": {
                    "value":[
                        "field11", "field12", "field13"
                    ]
                }
            },
            {
                "filterTemplate": {
                    "name" : "myBooleanFilterCollect0",
                    "criteria" : true,
                    "@type" : "DataFilterTemplateBoolean"
                },
                "stringArray": {
                    "value": [
                        "field1", "field2", "field3"
                    ]
                }
            }
        ]
    },
    "searchTaskFilter":{
        "@type" : "DataFilterMap",
        "mappings": [
            {
                "filterTemplate": {
                    "name" : "myRangeFilterSearch0",
                    "@type" : "DataFilterTemplateRange",
                    "criteria":{
                        "lowerBound" : {
                            "boundValue": -10,
                            "isInclusive": true
                        },
                        "upperBound" : {
                            "boundValue": -10,
                            "isInclusive": true
```


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```
                        }
                    }
                },
                "stringArray": {
                    "value": [
                    "field01", "field02", "field03"
                     ]
                }
            },
            {
                "filterTemplate": {
                    "name": "myRegExSampleSearch0",
                    "@type": "DataFilterTemplateRegex",
                    "criteria": "dallas|houston|venise|paris|new-york"
                },
                "stringArray": {
                    "value":[
                        "field11", "field12", "field13"
                    ]
                }
            },
            {
                "filterTemplate": {
                    "name" : "myBooleanFilterSearch0",
                    "criteria" : true,
                    "@type" : "DataFilterTemplateBoolean"
                },
                "stringArray": {
                    "value": [
                        "field1", "field2", "field3"
                    ]
                }
            }
        ]
    },
    "@baseType": "MeasurementCollectionJob",
    "@schemaLocation": "https://myschemas.com/MeasurementCollectionJob.schema.json",
    "@type": "MeasurementCollectionJob"
}
```

Deletes a MeasurementCollectionJob

`DELETE /measurementCollectionJob/{id}`

Description

This operation deletes a MeasurementCollectionJob entity.

Usage samples

Here's an example of a request for deleting a Measurement Collection Job.

Request

```
DELETE /measurementCollectionJob/4000
Content-Type: application/json
```

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Response

```
204
```

Operations on PerformanceIndicatorGroupSpecification

List or find PerformanceIndicatorGroupSpecification objects

`GET /performanceIndicatorGroupSpecification?fields=…`

Description

This operation list PerformanceIndicatorGroupSpecification entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving multiple Performance Indicator Group Specifications.

Request

```
GET /performanceIndicatorGroupSpecification?groupCategory=CPUGroup
Content-Type: application/json
```

Response

```
200

[
  {
    "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSGrouppecification/600",
    "id": "600",
    "name": "CriticalPerformance",
    "groupCategory" : "CPUGroup",
    "performanceIndicatorSpecification": [
      {
        "id": "5000",
        "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5000",
        "name": "CPU speed",
        "version": "1.0",
        "indicatorCategory": "CPU",
        "indicatorType": "float",
        "collectionType": "cumulative",
        "indicatorUnit": "GHz",
        "@baseType": "PerformanceIndicatorSpecification",
        "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
        "@type": "PerformanceIndicatorSpecification"
```

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```
      }
    ],
    "@baseType": "PerformanceIndicatorGroupSpecification",
    "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorGroupSpecificati
on.schema.json",
    "@type": "PerformanceIndicatorGroupSpecification"
  },
  {
    "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSGrouppecification/601",
    "id": "601",
    "name": "CriticalPerformance",
    "groupCategory" : "CPUGroup",
    "performanceIndicatorSpecification": [
      {
        "id": "5001",
        "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5001",
        "name": "CPU speed",
        "version": "1.0",
        "indicatorCategory": "CPU",
        "indicatorType": "float",
        "collectionType": "cumulative",
        "indicatorUnit": "GHz",
        "@baseType": "PerformanceIndicatorSpecification",
        "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
        "@type": "PerformanceIndicatorSpecification"
      }
    ],
    "@baseType": "PerformanceIndicatorGroupSpecification",
    "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorGroupSpecificati
on.schema.json",
    "@type": "PerformanceIndicatorGroupSpecification"
  }
]
```


Retrieves a PerformanceIndicatorGroupSpecification by ID

`GET /performanceIndicatorGroupSpecification/{id}?fields=…`

Description

This operation retrieves a PerformanceIndicatorGroupSpecification entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving a specific Performance Indicator Group Specification.

Request

```
GET /performanceIndicatorGroupSpecification/5000
```

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```
Content-Type: application/json
```

Response

```
200

{
    "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSGrouppecification/600",
    "id": "600",
    "name": "My critical performance indicators",
    "performanceIndicatorSpecification": [
      {
        "id": "5001",
        "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5001",
        "name": "CPU speed core 1",
        "indicatorCategory": "CPU",
        "indicatorType": "float",
        "collectionType": "cumulative",
        "indicatorUnit": "GHz",
        "version": "1.0",
        "@baseType": "PerformanceIndicatorSpecification",
        "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
        "@type": "PerformanceIndicatorSpecification"
      },
      {
        "id": "5002",
        "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5002",
        "name": "CPU speed core 2",
        "indicatorCategory": "CPU",
        "indicatorType": "float",
        "collectionType": "cumulative",
        "indicatorUnit": "GHz",
        "version": "1.0",
        "@baseType": "PerformanceIndicatorSpecification",
        "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
        "@type": "PerformanceIndicatorSpecification"
      },
      {
        "id": "5003",
        "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5003",
        "name": "CPU speed core 3",
        "indicatorCategory": "CPU",
        "indicatorType": "float",
        "collectionType": "cumulative",
        "indicatorUnit": "GHz",
        "version": "1.0",
        "@baseType": "PerformanceIndicatorSpecification",
        "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
        "@type": "PerformanceIndicatorSpecification"
      }
```

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```
    ],
    "@baseType": "PerformanceIndicatorGroupSpecification",
    "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorGroupSpecificati
on.schema.json",
    "@type": "PerformanceIndicatorGroupSpecification"
  }
```

Creates a PerformanceIndicatorGroupSpecification

`POST /performanceIndicatorGroupSpecification?fields=…`

Description

This operation creates a PerformanceIndicatorGroupSpecification entity.

Mandatory Attributes

Mandatory Attributes	Rule
groupCategory	
name	
performanceIndicatorSpecification	
performanceIndicatorSpecification .@type	
performanceIndicatorSpecification .collectionType	
performanceIndicatorSpecification .indicatorCategory	
performanceIndicatorSpecification .indicatorType	
performanceIndicatorSpecification .indicatorUnit	
performanceIndicatorSpecification .name	

Usage samples

Here's an example of a request and response for creating a new PerformanceIndicatorGroupSpecification

Request

```
POST /performanceIndicatorGroupSpecification
Content-Type: application/json

{
    "name": "MyPerformanceIndicatorGroupSpecification",
    "groupCategory" : "CPUGroup",
    "performanceIndicatorSpecification": [
```

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```
        {
            "id": "5000",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5000",
            "name": "CPU speed",
            "indicatorCategory": "CPU",
            "indicatorType": "float",
            "collectionType": "cumulative",
            "indicatorUnit": "GHz",
            "@baseType": "PerformanceIndicatorSpecificationRef",
            "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
            "@type": "PerformanceIndicatorSpecificationRef",
            "@referredType": "PerformanceIndicatorSpecification"
        },
        {
            "id": "5001",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5001",
            "name": "Peak CPU speed",
            "indicatorCategory": "CPU",
            "indicatorType": "float",
            "collectionType": "cumulative",
            "indicatorUnit": "GHz",
            "@baseType": "PerformanceIndicatorSpecificationRef",
            "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
            "@type": "PerformanceIndicatorSpecificationRef",
            "@referredType": "PerformanceIndicatorSpecification"
        },
        {
            "id": "5002",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5002",
            "name": "Min CPU Speed",
            "indicatorCategory": "CPU",
            "indicatorType": "float",
            "collectionType": "cumulative",
            "indicatorUnit": "GHz",
            "version": "1.0",
            "@baseType": "PerformanceIndicatorSpecification",
            "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
            "@type": "PerformanceIndicatorSpecification"
          }
    ],
    "@baseType": "PerformanceIndicatorGroupSpecification",
    "@schemaLocation":
"https://myschemas/PerformanceIndicatorGroupSpecification.schema.json",
    "@type": "PerformanceIndicatorGroupSpecification"
}
```


Response


```
201

{
    "id": "100",
    "href": "https://host:port/tmf-
```


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```
api/performanceManagement/v5/performanceIndicatorGroupSpecification/100",
    "name": "MyPerformanceIndicatorGroupSpecification",
    "groupCategory" : "CPUGroup",
    "performanceIndicatorSpecification": [
        {
            "id": "5000",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5000",
            "name": "CPU speed",
            "indicatorCategory": "CPU",
            "indicatorType": "float",
            "collectionType": "cumulative",
            "indicatorUnit": "GHz",
            "@baseType": "PerformanceIndicatorSpecificationRef",
            "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
            "@type": "PerformanceIndicatorSpecificationRef",
            "@referredType": "PerformanceIndicatorSpecification"
        },
        {
            "id": "5001",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5001",
            "name": "Peak CPU speed",
            "indicatorCategory": "CPU",
            "indicatorType": "float",
            "collectionType": "cumulative",
            "indicatorUnit": "GHz",
            "@baseType": "PerformanceIndicatorSpecificationRef",
            "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
            "@type": "PerformanceIndicatorSpecificationRef",
            "@referredType": "PerformanceIndicatorSpecification"
        }
    ],
    "@baseType": "PerformanceIndicatorGroupSpecification",
    "@schemaLocation":
"https://myschemas/PerformanceIndicatorGroupSpecification.schema.json",
    "@type": "PerformanceIndicatorGroupSpecification"
}
```

Updates partially a PerformanceIndicatorGroupSpecification

`PATCH /performanceIndicatorGroupSpecification/{id}?fields=…`

Description

This operation allows partial updates of a PerformanceIndicatorGroupSpecification entity. Support of json/merge (https://tools.ietf.org/html/rfc7386) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optiona. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

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Non Patchable Attributes	Rule
href	

Patchable Attributes	Rule
groupCategory	
id	
name	
performanceIndicatorSpecification	

Usage samples

Update a PerformanceIndicatorGroupSpecification using json

Request

```
PATCH /performanceIndicatorGroupSpecification/{id}
Content-Type: application/json

{
    "groupCategory" : "MyGroupCategory",
    "performanceIndicatorSpecification": [
        {
            "id": "4000",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/4000",
            "name": "Gateway Response Time",
            "indicatorCategory": "Latency",
            "indicatorType": "float",
            "collectionType": "cumulative",
            "indicatorUnit": "ms",
            "version": "1.0",
            "@baseType": "PerformanceIndicatorSpecificationRef",
            "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
            "@type": "PerformanceIndicatorSpecificationRef",
            "@referredType": "PerformanceIndicatorSpecification"
        },
        {
            "id": "4001",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/4001",
            "name": "Router Response Time",
            "indicatorCategory": "Latency",
            "indicatorType": "float",
            "collectionType": "cumulative",
            "indicatorUnit": "ms",
            "version": "1.0",
            "@baseType": "PerformanceIndicatorSpecificationRef",
            "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
            "@type": "PerformanceIndicatorSpecificationRef",
            "@referredType": "PerformanceIndicatorSpecification"
        },
        {
            "id": "4003",
```

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```
            "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/4003",
            "name": "RAN Response Time",
            "indicatorCategory": "Latency",
            "indicatorType": "float",
            "collectionType": "cumulative",
            "indicatorUnit": "ms",
            "version": "1.0",
            "@baseType": "PerformanceIndicatorSpecificationRef",
            "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
            "@type": "PerformanceIndicatorSpecificationRef",
            "@referredType": "PerformanceIndicatorSpecification"
        }
    ],
    "@type": "PerformanceIndicatorGroupSpecification"
}
```


Response


```
200

{
    "id": "100",
    "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorGroupSpecification/100",
    "name": "MyPerformanceIndicatorGroupSpecification",
    "groupCategory" : "MyGroupCategory",
    "performanceIndicatorSpecification": [
        {
            "id": "4000",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/4000",
            "name": "Gateway Response Time",
            "indicatorCategory": "Latency",
            "indicatorType": "float",
            "collectionType": "cumulative",
            "indicatorUnit": "ms",
            "version": "1.0",
            "@baseType": "PerformanceIndicatorSpecificationRef",
            "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
            "@type": "PerformanceIndicatorSpecificationRef",
            "@referredType": "PerformanceIndicatorSpecification"
        },
        {
            "id": "4001",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/4001",
            "name": "Router Response Time",
            "indicatorCategory": "Latency",
            "indicatorType": "float",
            "collectionType": "cumulative",
            "indicatorUnit": "ms",
            "version": "1.0",
            "@baseType": "PerformanceIndicatorSpecificationRef",
            "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
            "@type": "PerformanceIndicatorSpecificationRef",
            "@referredType": "PerformanceIndicatorSpecification"
```


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```
        },
        {
            "id": "4003",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/4003",
            "name": "RAN Response Time",
            "indicatorCategory": "Latency",
            "indicatorType": "float",
            "collectionType": "cumulative",
            "indicatorUnit": "ms",
            "version": "1.0",
            "@baseType": "PerformanceIndicatorSpecificationRef",
            "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
            "@type": "PerformanceIndicatorSpecificationRef",
            "@referredType": "PerformanceIndicatorSpecification"
        }
    ],
    "@baseType": "PerformanceIndicatorGroupSpecification",
    "@schemaLocation":
"https://myschemas/PerformanceIndicatorGroupSpecification.schema.json",
    "@type": "PerformanceIndicatorGroupSpecification"
}
```

Update Without query parameters a PerformanceIndicatorGroupSpecification using json-patch+json

Request

```
PATCH /performanceIndicatorGroupSpecification/{id}
Content-Type: application/json-patch+json

[
    {   "op":"replace",
        "path": "/name",
        "value": "MyPerformanceIndicatorGroupSpecification"
    },
    {   "op":"replace",
        "path" : "/performanceIndicatorSpecification/0/indicatorCategory",
        "value" : "Latency"
    },
    {   "op":"replace",
        "path" : "/performanceIndicatorSpecification/2/collectionType",
        "value" : "cumulative"
    }
]
```

Response

```
200

{
    "id": "100",
    "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorGroupSpecification/100",
    "name": "MyPerformanceIndicatorGroupSpecification",
    "groupCategory" : "MyGroupCategory",
```

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```
    "performanceIndicatorSpecification": [
        {
            "id": "4000",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/4000",
            "name": "Gateway Response Time",
            "indicatorCategory": "Latency",
            "indicatorType": "float",
            "collectionType": "cumulative",
            "indicatorUnit": "ms",
            "version": "1.0",
            "@baseType": "PerformanceIndicatorSpecificationRef",
            "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
            "@type": "PerformanceIndicatorSpecificationRef",
            "@referredType": "PerformanceIndicatorSpecification"
        },
        {
            "id": "4001",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/4001",
            "name": "Router Response Time",
            "indicatorCategory": "Latency",
            "indicatorType": "float",
            "collectionType": "cumulative",
            "indicatorUnit": "ms",
            "version": "1.0",
            "@baseType": "PerformanceIndicatorSpecificationRef",
            "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
            "@type": "PerformanceIndicatorSpecificationRef",
            "@referredType": "PerformanceIndicatorSpecification"
        },
        {
            "id": "4003",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/4003",
            "name": "RAN Response Time",
            "indicatorCategory": "Latency",
            "indicatorType": "float",
            "collectionType": "cumulative",
            "indicatorUnit": "ms",
            "version": "1.0",
            "@baseType": "PerformanceIndicatorSpecificationRef",
            "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
            "@type": "PerformanceIndicatorSpecificationRef",
            "@referredType": "PerformanceIndicatorSpecification"
        }
    ],
    "@baseType": "PerformanceIndicatorGroupSpecification",
    "@schemaLocation":
"https://myschemas/PerformanceIndicatorGroupSpecification.schema.json",
    "@type": "PerformanceIndicatorGroupSpecification"
}
```


Update With query parameters a PerformanceIndicatorGroupSpecification using json-patch+json

Request

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```
PATCH
/performanceIndicatorGroupSpecification/{id}?fields=id,name,groupCategory,performanc
eIndicatorSpecification,@type
Content-Type: application/json-patch+json

[
    {   "op":"replace",
        "path": "/name",
        "value": "MyPerformanceIndicatorGroupSpecification"
    },
    {   "op":"replace",
        "path" : "/performanceIndicatorSpecification/0/indicatorCategory",
        "value" : "Latency"
    },
    {   "op":"replace",
        "path" : "/performanceIndicatorSpecification/2/collectionType",
        "value" : "cumulative"
    }
]
```

Response

```
200

{
    "id": "100",
    "name": "MyPerformanceIndicatorGroupSpecification",
    "groupCategory" : "MyGroupCategory",
    "performanceIndicatorSpecification": [
        {
            "id": "4000",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/4000",
            "name": "Gateway Response Time",
            "indicatorCategory": "Latency",
            "indicatorType": "float",
            "collectionType": "cumulative",
            "indicatorUnit": "ms",
            "version": "1.0",
            "@baseType": "PerformanceIndicatorSpecificationRef",
            "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
            "@type": "PerformanceIndicatorSpecificationRef",
            "@referredType": "PerformanceIndicatorSpecification"
        },
        {
            "id": "4001",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/4001",
            "name": "Router Response Time",
            "indicatorCategory": "Latency",
            "indicatorType": "float",
            "collectionType": "cumulative",
            "indicatorUnit": "ms",
            "version": "1.0",
            "@baseType": "PerformanceIndicatorSpecificationRef",
            "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
            "@type": "PerformanceIndicatorSpecificationRef",
```

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```
            "@referredType": "PerformanceIndicatorSpecification"
        },
        {
            "id": "4003",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/4003",
            "name": "RAN Response Time",
            "indicatorCategory": "Latency",
            "indicatorType": "float",
            "collectionType": "cumulative",
            "indicatorUnit": "ms",
            "version": "1.0",
            "@baseType": "PerformanceIndicatorSpecificationRef",
            "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
            "@type": "PerformanceIndicatorSpecificationRef",
            "@referredType": "PerformanceIndicatorSpecification"
        }
    ],
    "@type": "PerformanceIndicatorGroupSpecification"
}
```

Update a PerformanceIndicatorGroupSpecification using json-patch-query+json

Request

```
PATCH /performanceIndicatorGroupSpecification/{id}
Content-Type: application/json-patch-query+json

[
    {   "op":"replace",
        "path": "/performanceIndicatorSpecification[0]?indicatorCategory=Latency",
        "value": "CPU"
    },
    {   "op":"replace",
        "path" : "/performanceIndicatorSpecification[1]?indicatorCategory=Latency",
        "value" : "CPU"
    }
]
```

Response

```
200

{
    "id": "100",
    "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorGroupSpecification/100",
    "name": "MyPerformanceIndicatorGroupSpecification",
    "groupCategory" : "MyGroupCategory",
    "performanceIndicatorSpecification": [
        {
            "id": "4000",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/4000",
            "name": "Gateway Response Time",
            "indicatorCategory": "Latency",
```

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```
            "indicatorType": "float",
            "collectionType": "cumulative",
            "indicatorUnit": "ms",
            "version": "1.0",
            "@baseType": "PerformanceIndicatorSpecificationRef",
            "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
            "@type": "PerformanceIndicatorSpecificationRef",
            "@referredType": "PerformanceIndicatorSpecification"
        },
        {
            "id": "4001",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/4001",
            "name": "Router Response Time",
            "indicatorCategory": "Latency",
            "indicatorType": "float",
            "collectionType": "cumulative",
            "indicatorUnit": "ms",
            "version": "1.0",
            "@baseType": "PerformanceIndicatorSpecificationRef",
            "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
            "@type": "PerformanceIndicatorSpecificationRef",
            "@referredType": "PerformanceIndicatorSpecification"
        },
        {
            "id": "4003",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/4003",
            "name": "RAN Response Time",
            "indicatorCategory": "Latency",
            "indicatorType": "float",
            "collectionType": "cumulative",
            "indicatorUnit": "ms",
            "version": "1.0",
            "@baseType": "PerformanceIndicatorSpecificationRef",
            "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
            "@type": "PerformanceIndicatorSpecificationRef",
            "@referredType": "PerformanceIndicatorSpecification"
        }
    ],
    "@baseType": "PerformanceIndicatorGroupSpecification",
    "@schemaLocation":
"https://myschemas/PerformanceIndicatorGroupSpecification.schema.json",
    "@type": "PerformanceIndicatorGroupSpecification"
}
```


Update a PerformanceIndicatorGroupSpecification using merge-patch+json

Request

```
PATCH /performanceIndicatorGroupSpecification/{id}
Content-Type: application/merge-patch+json

{
    "groupCategory" : "MyGroupCategory",
    "performanceIndicatorSpecification": [
        {
```

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```
            "id": "4000",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/4000",
            "name": "Gateway Response Time",
            "indicatorCategory": "Latency",
            "indicatorType": "float",
            "collectionType": "cumulative",
            "indicatorUnit": "ms",
            "version": "1.0",
            "@baseType": "PerformanceIndicatorSpecificationRef",
            "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
            "@type": "PerformanceIndicatorSpecificationRef",
            "@referredType": "PerformanceIndicatorSpecification"
        },
        {
            "id": "4001",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/4001",
            "name": "Router Response Time",
            "indicatorCategory": "Latency",
            "indicatorType": "float",
            "collectionType": "cumulative",
            "indicatorUnit": "ms",
            "version": "1.0",
            "@baseType": "PerformanceIndicatorSpecificationRef",
            "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
            "@type": "PerformanceIndicatorSpecificationRef",
            "@referredType": "PerformanceIndicatorSpecification"
        },
        {
            "id": "4003",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/4003",
            "name": "RAN Response Time",
            "indicatorCategory": "Latency",
            "indicatorType": "float",
            "collectionType": "cumulative",
            "indicatorUnit": "ms",
            "version": "1.0",
            "@baseType": "PerformanceIndicatorSpecificationRef",
            "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
            "@type": "PerformanceIndicatorSpecificationRef",
            "@referredType": "PerformanceIndicatorSpecification"
        }
    ],
    "@type": "PerformanceIndicatorGroupSpecification"
}
```


Response


```
200

{
    "id": "100",
    "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorGroupSpecification/100",
    "name": "MyPerformanceIndicatorGroupSpecification",
```


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```
    "groupCategory" : "MyGroupCategory",
    "performanceIndicatorSpecification": [
        {
            "id": "4000",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/4000",
            "name": "Gateway Response Time",
            "indicatorCategory": "Latency",
            "indicatorType": "float",
            "collectionType": "cumulative",
            "indicatorUnit": "ms",
            "version": "1.0",
            "@baseType": "PerformanceIndicatorSpecificationRef",
            "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
            "@type": "PerformanceIndicatorSpecificationRef",
            "@referredType": "PerformanceIndicatorSpecification"
        },
        {
            "id": "4001",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/4001",
            "name": "Router Response Time",
            "indicatorCategory": "Latency",
            "indicatorType": "float",
            "collectionType": "cumulative",
            "indicatorUnit": "ms",
            "version": "1.0",
            "@baseType": "PerformanceIndicatorSpecificationRef",
            "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
            "@type": "PerformanceIndicatorSpecificationRef",
            "@referredType": "PerformanceIndicatorSpecification"
        },
        {
            "id": "4003",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/4003",
            "name": "RAN Response Time",
            "indicatorCategory": "Latency",
            "indicatorType": "float",
            "collectionType": "cumulative",
            "indicatorUnit": "ms",
            "version": "1.0",
            "@baseType": "PerformanceIndicatorSpecificationRef",
            "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
            "@type": "PerformanceIndicatorSpecificationRef",
            "@referredType": "PerformanceIndicatorSpecification"
        }
    ],
    "@baseType": "PerformanceIndicatorGroupSpecification",
    "@schemaLocation":
"https://myschemas/PerformanceIndicatorGroupSpecification.schema.json",
    "@type": "PerformanceIndicatorGroupSpecification"
}
```


Deletes a PerformanceIndicatorGroupSpecification

`DELETE /performanceIndicatorGroupSpecification/{id}`

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Description

This operation deletes a PerformanceIndicatorGroupSpecification entity.

Usage samples

Here's an example of a request for deleting a Performance Indicator Group Specification.

Request

```
DELETE /performanceIndicatorGroupSpecification/5000
Content-Type: application/json
```

Response

```
204
```

Operations on PerformanceIndicatorSpecification

List or find PerformanceIndicatorSpecification objects

`GET /performanceIndicatorSpecification?fields=…`

Description

This operation list PerformanceIndicatorSpecification entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving multiple Performance Indicator Specifications.

Request

```
GET
/performanceIndicatorSpecification?collectionType=cumulative&indicatorCategory=CPU
Content-Type: application/json
```

Response

```
200

[
    {
        "id": "5000",
        "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5000",
        "name": "CPU speed",
        "description": "The average CPU speed",
```

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```
        "perspective": "aggregation over a collection of systems",
        "indicatorCategory": "CPU",
        "indicatorType": "float",
        "derivationAlgorithm": "na",
        "derivationMethod": "average",
        "validFor": {
            "startDateTime": "2023-04-01T00:00:00.001Z",
            "endDateTime": "2023-04-30T23:59:59.999Z"
            },
        "collectionType": "cumulative",
        "indicatorUnit": "GHz",
        "@baseType": "PerformanceIndicatorSpecification",
        "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
        "@type": "PerformanceIndicatorSpecification"
    },
    {
        "id": "5001",
        "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5001",
        "name": "CPU speed",
        "description": "The average peak CPU speed",
        "perspective": "aggregation over a collection of systems",
        "indicatorCategory": "CPU",
        "indicatorType": "float",
        "derivationAlgorithm": "na",
        "derivationMethod": "average",
        "validFor": {
            "startDateTime": "2023-04-01T00:00:00.001Z",
            "endDateTime": "2023-04-30T23:59:59.999Z"
            },
        "collectionType": "cumulative",
        "indicatorUnit": "GHz",
        "@baseType": "PerformanceIndicatorSpecification",
        "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
        "@type": "PerformanceIndicatorSpecification"
    }
]
```


Retrieves a PerformanceIndicatorSpecification by ID

`GET /performanceIndicatorSpecification/{id}?fields=…`

Description

This operation retrieves a PerformanceIndicatorSpecification entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving a specific Performance Indicator Specification.

Request

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```
GET /performanceIndicatorSpecification/5000
Content-Type: application/json
```

Response

```
200

{
    "id": "5000",
    "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5000",
    "name": "CPU speed",
    "description": "The average CPU speed",
    "perspective": "aggregation over a collection of systems",
    "indicatorCategory": "CPU",
    "indicatorType": "float",
    "derivationAlgorithm": "na",
    "derivationMethod": "average",
    "validFor": {
        "startDateTime": "2023-04-01T00:00:00.001Z",
        "endDateTime": "2023-04-30T23:59:59.999Z"
        },
    "collectionType": "cumulative",
    "indicatorUnit": "GHz",
    "@baseType": "PerformanceIndicatorSpecification",
    "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
    "@type": "PerformanceIndicatorSpecification"
}
```

Creates a PerformanceIndicatorSpecification

`POST /performanceIndicatorSpecification?fields=…`

Description

This operation creates a PerformanceIndicatorSpecification entity.

Mandatory Attributes

Mandatory Attributes	Rule
collectionType	
indicatorCategory	
indicatorType	
indicatorUnit	
name	

Usage samples

Here's an example of a request and response for creating a new

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PerformanceIndicatorSpecification

Request

```
POST /performanceIndicatorSpecification
Content-Type: application/json

{
    "derivationAlgorithm": "na",
    "derivationMethod": "average",
    "description": "The average CPU speed",
    "indicatorCategory": "CPU",
    "indicatorUnit": "GHz",
    "name": "CPU Speed",
    "perspective": "aggregation over a collection of systems",
    "collectionType": "cumulative",
    "indicatorType": "float",
    "performanceIndicatorSpecRelationship": [
        {
            "perfSpec": {
                "href":"https://host:port/tmf-
api/performanceManagement/v5/PerformanceIndicatorSpecification/300"
            },
            "relationshipType": "aggregation",
            "role": "member",
            "validFor": {
              "endDateTime": "2025-04-12T23:20:50.52Z",
              "startDateTime": "2022-04-12T23:20:50.52Z"
            },
            "@baseType": "PerformanceIndicatorSpecRelationship",
            "@schemaLocation":
"https://myschemas.com/PerformanceIndicatorSpecRelationship.schema.json",
            "@type": "PerformanceIndicatorSpecRelationship"
          }
    ],
    "validFor": {
        "endDateTime": "2022-04-12T23:20:50.52Z",
        "startDateTime": "2022-04-12T23:20:50.52Z"
    },
    "@baseType": "PerformanceIndicatorSpecification",
    "@schemaLocation":
"https://myschemas.com/PerformanceIndicatorSpecification.schema.json",
    "@type": "PerformanceIndicatorSpecification"
}
```

Response

```
201

{
    "id": "200",
    "href": "https://host:port/tmf-
api/performanceManagement/v5/PerformanceIndicatorSpecification/200",
    "derivationAlgorithm": "na",
    "derivationMethod": "average",
    "description": "The average CPU speed",
    "indicatorCategory": "CPU",
    "indicatorUnit": "GHz",
    "name": "CPU Speed",
```

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```
    "perspective": "aggregation over a collection of systems",
    "collectionType": "cumulative",
    "indicatorType": "float",
    "performanceIndicatorSpecRelationship": [
        {
            "perfSpec": {
                "href":"https://host:port/tmf-
api/performanceManagement/v5/PerformanceIndicatorSpecification/300"
            },            "relationshipType": "aggregation",
            "role": "member",
            "validFor": {
              "endDateTime": "2025-04-12T23:20:50.52Z",
              "startDateTime": "2022-04-12T23:20:50.52Z"
            },
            "@baseType": "PerformanceIndicatorSpecRelationship",
            "@schemaLocation":
"https://myschemas.com/PerformanceIndicatorSpecRelationship.schema.json",
            "@type": "PerformanceIndicatorSpecRelationship"
          }
    ],
    "validFor": {
        "endDateTime": "2022-04-12T23:20:50.52Z",
        "startDateTime": "2022-04-12T23:20:50.52Z"
    },
    "@baseType": "PerformanceIndicatorSpecification",
    "@schemaLocation":
"https://myschemas.com/PerformanceIndicatorSpecification.schema.json",
    "@type": "PerformanceIndicatorSpecification"
}
```


Updates partially a PerformanceIndicatorSpecification

`PATCH /performanceIndicatorSpecification/{id}?fields=…`

Description

This operation allows partial updates of a PerformanceIndicatorSpecification entity. Support of json/merge (https://tools.ietf.org/html/rfc7386) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optiona. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
href	

Patchable Attributes	Rule
collectionType	
derivationAlgorithm	
derivationMethod	
description	

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Patchable Attributes	Rule
id	
indicatorCategory	
indicatorType	
indicatorUnit	
name	
performanceIndicatorSpecRelationship	
perspective	
validFor	

Usage samples

Update a PerformanceIndicatorSpecification using json

Request

```
PATCH /performanceIndicatorSpecification/{id}
Content-Type: application/json

{
    "description": "The average CPU speed",
    "indicatorCategory": "CPU",
    "indicatorUnit": "GHz",
    "name": "CPU Speed",
    "perspective": "aggregation over a collection of systems",
    "collectionType": "cumulative",
    "indicatorType": "float",
    "validFor": {
        "endDateTime": "2022-04-12T23:20:50.52Z",
        "startDateTime": "2022-04-12T23:20:50.52Z"
    },
    "@type": "PerformanceIndicatorSpecification"
}
```

Response

```
200

{
    "id": "200",
    "href": "https://host:port/tmf-
api/performanceManagement/v5/PerformanceIndicatorSpecification/200",
    "derivationAlgorithm": "na",
    "derivationMethod": "average",
    "description": "The average CPU speed",
    "indicatorCategory": "CPU",
    "indicatorUnit": "GHz",
    "name": "CPU Speed",
    "perspective": "aggregation over a collection of systems",
    "collectionType": "cumulative",
```

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```
    "indicatorType": "float",
    "performanceIndicatorSpecRelationship": [
        {
            "perfSpec": {
                "href":"https://host:port/tmf-
api/performanceManagement/v5/PerformanceIndicatorSpecification/300"
              },
            "relationshipType": "aggregation",
            "role": "member",
            "validFor": {
              "endDateTime": "2025-04-12T23:20:50.52Z",
              "startDateTime": "2022-04-12T23:20:50.52Z"
            },
            "@baseType": "PerformanceIndicatorSpecRelationship",
            "@schemaLocation":
"https://myschemas.com/PerformanceIndicatorSpecRelationship.schema.json",
            "@type": "PerformanceIndicatorSpecRelationship"
          }
    ],
    "validFor": {
        "endDateTime": "2022-04-12T23:20:50.52Z",
        "startDateTime": "2022-04-12T23:20:50.52Z"
    },
    "@baseType": "PerformanceIndicatorSpecification",
    "@schemaLocation":
"https://myschemas.com/PerformanceIndicatorSpecification.schema.json",
    "@type": "PerformanceIndicatorSpecification"
}
```


Update Without query parameters a PerformanceIndicatorSpecification using json-patch+json

Request

```
PATCH /performanceIndicatorSpecification/{id}
Content-Type: application/json-patch+json

[
    {   "op":"replace",
        "path": "/indicatorCategory",
        "value": "CPU"
    },
    {   "op":"replace",
        "path" : "/validFor/endDateTime",
        "value" : "2022-04-12T23:20:50.52Z"
    }
]
```

Response


```
200

{
    "id": "200",
    "href": "https://host:port/tmf-
api/performanceManagement/v5/PerformanceIndicatorSpecification/200",
    "derivationAlgorithm": "na",
```


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```
    "derivationMethod": "average",
    "description": "The average CPU speed",
    "indicatorCategory": "CPU",
    "indicatorUnit": "GHz",
    "name": "CPU Speed",
    "perspective": "aggregation over a collection of systems",
    "collectionType": "cumulative",
    "indicatorType": "float",
    "performanceIndicatorSpecRelationship": [
        {
            "perfSpec": {
                "href":"https://host:port/tmf-
api/performanceManagement/v5/PerformanceIndicatorSpecification/300"
              },
            "relationshipType": "aggregation",
            "role": "member",
            "validFor": {
              "endDateTime": "2025-04-12T23:20:50.52Z",
              "startDateTime": "2022-04-12T23:20:50.52Z"
            },
            "@baseType": "PerformanceIndicatorSpecRelationship",
            "@schemaLocation":
"https://myschemas.com/PerformanceIndicatorSpecRelationship.schema.json",
            "@type": "PerformanceIndicatorSpecRelationship"
          }
    ],
    "validFor": {
        "endDateTime": "2022-04-12T23:20:50.52Z",
        "startDateTime": "2022-04-12T23:20:50.52Z"
    },
    "@baseType": "PerformanceIndicatorSpecification",
    "@schemaLocation":
"https://myschemas.com/PerformanceIndicatorSpecification.schema.json",
    "@type": "PerformanceIndicatorSpecification"
}
```


Update With query parameters a PerformanceIndicatorSpecification using json-patch+json

Request

```
PATCH
/performanceIndicatorSpecification/{id}?fields=id,href,indicatorCategory,indicatorUn
it,name,collectionType,indicatorType,validFor,@type
Content-Type: application/json-patch+json

[
    {   "op":"replace",
        "path": "/indicatorCategory",
        "value": "CPU"
    },
    {   "op":"replace",
        "path" : "/validFor/endDateTime",
        "value" : "2022-04-12T23:20:50.52Z"
    }
]
```

Response

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```
200

{
    "id": "200",
    "href": "https://host:port/tmf-
api/performanceManagement/v5/PerformanceIndicatorSpecification/200",
    "indicatorCategory": "CPU",
    "indicatorUnit": "GHz",
    "name": "CPU Speed",
    "collectionType": "cumulative",
    "indicatorType": "float",
    "validFor": {
        "endDateTime": "2022-04-12T23:20:50.52Z",
        "startDateTime": "2022-04-12T23:20:50.52Z"
    },
    "@type": "PerformanceIndicatorSpecification"
}
```

Update a PerformanceIndicatorSpecification using json-patch-query+json

Request

```
PATCH /performanceIndicatorSpecification/{id}
Content-Type: application/json-patch-query+json

[
    {   "op":"replace",
        "path": "/indicatorUnit?indicatorCategory=Latency",
        "value": "CPU"
    },
    {   "op":"replace",
        "path" :
"/perspective/performanceIndicatorSpecRelationship[0]?relationshipType=aggregation",
        "value" : "aggregation over a collection of systems"
    }
]
```

Response

```
200

{
    "id": "200",
    "href": "https://host:port/tmf-
api/performanceManagement/v5/PerformanceIndicatorSpecification/200",
    "derivationAlgorithm": "na",
    "derivationMethod": "average",
    "description": "The average CPU speed",
    "indicatorCategory": "CPU",
    "indicatorUnit": "GHz",
    "name": "CPU Speed",
    "perspective": "aggregation over a collection of systems",
    "collectionType": "cumulative",
    "indicatorType": "float",
    "performanceIndicatorSpecRelationship": [
        {
```

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```
            "perfSpec": {
                "href":"https://host:port/tmf-
api/performanceManagement/v5/PerformanceIndicatorSpecification/300"
              },
            "relationshipType": "aggregation",
            "role": "member",
            "validFor": {
              "endDateTime": "2025-04-12T23:20:50.52Z",
              "startDateTime": "2022-04-12T23:20:50.52Z"
            },
            "@baseType": "PerformanceIndicatorSpecRelationship",
            "@schemaLocation":
"https://myschemas.com/PerformanceIndicatorSpecRelationship.schema.json",
            "@type": "PerformanceIndicatorSpecRelationship"
          }
    ],
    "validFor": {
        "endDateTime": "2022-04-12T23:20:50.52Z",
        "startDateTime": "2022-04-12T23:20:50.52Z"
    },
    "@baseType": "PerformanceIndicatorSpecification",
    "@schemaLocation":
"https://myschemas.com/PerformanceIndicatorSpecification.schema.json",
    "@type": "PerformanceIndicatorSpecification"
}
```

Update a PerformanceIndicatorSpecification using merge-patch+json

Request

```
PATCH /performanceIndicatorSpecification/{id}
Content-Type: application/merge-patch+json

{
    "description": "The average CPU speed",
    "indicatorCategory": "CPU",
    "indicatorUnit": "GHz",
    "name": "CPU Speed",
    "perspective": "aggregation over a collection of systems",
    "collectionType": "cumulative",
    "indicatorType": "float",
    "validFor": {
        "endDateTime": "2022-04-12T23:20:50.52Z",
        "startDateTime": "2022-04-12T23:20:50.52Z"
    },
    "@type": "PerformanceIndicatorSpecification"
}
```

Response

```
200

{
    "id": "200",
    "href": "https://host:port/tmf-
api/performanceManagement/v5/PerformanceIndicatorSpecification/200",
    "derivationAlgorithm": "na",
```

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```
    "derivationMethod": "average",
    "description": "The average CPU speed",
    "indicatorCategory": "CPU",
    "indicatorUnit": "GHz",
    "name": "CPU Speed",
    "perspective": "aggregation over a collection of systems",
    "collectionType": "cumulative",
    "indicatorType": "float",
    "performanceIndicatorSpecRelationship": [
        {
            "perfSpec": {
                "href":"https://host:port/tmf-
api/performanceManagement/v5/PerformanceIndicatorSpecification/300"
              },
            "relationshipType": "aggregation",
            "role": "member",
            "validFor": {
              "endDateTime": "2025-04-12T23:20:50.52Z",
              "startDateTime": "2022-04-12T23:20:50.52Z"
            },
            "@baseType": "PerformanceIndicatorSpecRelationship",
            "@schemaLocation":
"https://myschemas.com/PerformanceIndicatorSpecRelationship.schema.json",
            "@type": "PerformanceIndicatorSpecRelationship"
          }
    ],
    "validFor": {
        "endDateTime": "2022-04-12T23:20:50.52Z",
        "startDateTime": "2022-04-12T23:20:50.52Z"
    },
    "@baseType": "PerformanceIndicatorSpecification",
    "@schemaLocation":
"https://myschemas.com/PerformanceIndicatorSpecification.schema.json",
    "@type": "PerformanceIndicatorSpecification"
}
```


Deletes a PerformanceIndicatorSpecification

`DELETE /performanceIndicatorSpecification/{id}`

Description

This operation deletes a PerformanceIndicatorSpecification entity.

Usage samples

Here's an example of a request for deleting a Performance Indicator Specification.

Request

```
DELETE /performanceIndicatorSpecification/5000
Content-Type: application/json
```

Response

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```
204
```

Operations on TrackingRecord

List or find TrackingRecord objects

`GET /trackingRecord?fields=…`

Description

This operation list TrackingRecord entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving multiple TrackingRecord's.

Request

```
GET /trackingRecord?user=myUserName
Content-Type: application/json
```

Response

```
200

[
    {
        "id": "1001",
        "description": "acknowledge ,clear",
        "systemId": "5001",
        "time": "2022-05-15T15:51:04.176Z",
        "user": "myUserName",
        "characteristic": [
          {
            "id": "001",
            "name": "myCharName",
            "valueType": "string",
            "characteristicRelationship": [
              {
                "id": "400",
                "href": "https://host:port/tmf-
api/performanceManagement/v5/characteristicRelationship/400",
                "relationshipType": "sub-part",
                "@baseType": "CharacteristicRelationship",
                "@schemaLocation":
"https://myschemas.com/CharacteristicRelationship.schema.json",
                "@type": "CharacteristicRelationship"
              }
            ],
            "value": "myCharactericticValue",
            "@baseType": "Characteristic",
            "@schemaLocation":
```

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```
"https://myschemas.com/StringCharacteristic.schema.json",
            "@type": "StringCharacteristic"
          }
        ],
        "@baseType": "TrackingRecord",
        "@schemaLocation": "https://myschemas.com/TrackingRecord.schema.json",
        "@type": "TrackingRecord"
      },
      {
        "id": "1002",
        "description": "acknowledge ,clear",
        "systemId": "5002",
        "time": "2022-05-15T16:51:04.176Z",
        "user": "myUserName",
        "characteristic": [
          {
            "id": "002",
            "name": "myCharName2",
            "valueType": "string",
            "characteristicRelationship": [
              {
                "id": "401",
                "href": "https://host:port/tmf-
api/performanceManagement/v5/characteristicRelationship/401",
                "relationshipType": "sub-part",
                "@baseType": "CharacteristicRelationship",
                "@schemaLocation":
"https://myschemas.com/CharacteristicRelationship.schema.json",
                "@type": "CharacteristicRelationship"
              }
            ],
            "value": "myCharactericticValue",
            "@baseType": "Characteristic",
            "@schemaLocation":
"https://myschemas.com/StringCharacteristic.schema.json",
            "@type": "StringCharacteristic"
          }
        ],
        "@baseType": "TrackingRecord",
        "@schemaLocation": "https://myschemas.com/TrackingRecord.schema.json",
        "@type": "TrackingRecord"
      }
]
```


Retrieves a TrackingRecord by ID

`GET /trackingRecord/{id}?fields=…`

Description

This operation retrieves a TrackingRecord entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving a specific TrackingRecord.

Request

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```
GET /trackingRecord/1001
Content-Type: application/json
```

Response

```
200

{
    "id": "1001",
    "description": "acknowledge ,clear",
    "systemId": "5001",
    "time": "2022-05-15T15:51:04.176Z",
    "user": "myUserName",
    "characteristic": [
      {
        "id": "001",
        "name": "myCharName",
        "valueType": "string",
        "characteristicRelationship": [
          {
            "id": "400",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/characteristicRelationship/400",
            "relationshipType": "sub-part",
            "@baseType": "CharacteristicRelationship",
            "@schemaLocation":
"https://myschemas.com/CharacteristicRelationship.schema.json",
            "@type": "CharacteristicRelationship"
          }
        ],
        "value": "myCharactericticValue",
        "@baseType": "Characteristic",
        "@schemaLocation": "https://myschemas.com/Characteristic.schema.json",
        "@type": "StringCharacteristic"
      }
    ],
    "@baseType": "TrackingRecord",
    "@schemaLocation": "https://myschemas.com/TrackingRecord.schema.json",
    "@type": "TrackingRecord"
  }
```

Operations on PerformanceMeasurement

List or find PerformanceMeasurement objects

`GET /performanceMeasurement?fields=…`

Description

This operation list PerformanceMeasurement entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

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Here's an example of a request for retrieving multiple PerformanceMeasurement's.

Request

```
GET /performanceMeasurement?measurementCollectionJob.id=4000
Content-Type: application/json
```

Response

```
200

[
    {
        "id": "1",
        "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceMeasurement/1",
        "description": "The first performance measurement in the database",
        "measurementCollectionJob": {
          "id": "4000",
          "href": "https://host:port/tmf-
api/performanceManagement/v5/measurementCollectionJob/4000",
          "name": "Measurement Collection Job from system A",
          "@baseType": "MeasurementCollectionJobRef",
          "@schemaLocation":
"https://myschemas/MeasurementCollectionJob.schema.json",
          "@type": "MeasurementCollectionJobRef",
          "@referredType": "MeasurementCollectionJob"
        },
        "performanceIndicatorValue": [
            {
              "performanceIndicatorSpecification":{
                "id": "1",
                "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/1",
                "@baseType": "PerformanceIndicatorSpecificationRef",
                "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
                "@type": "PerformanceIndicatorSpecificationRef",
                "@referredType": "PerformanceIndicatorSpecification"
              },
              "observedValue": "2.1"
            },
            {
              "performanceIndicatorSpecification":{
                "id": "2",
                "name": "CPU Speed Indicator Specification",
                "indicatorCategory": "SPEEDCategory",
                "indicatorName": "CPU Speed",
                "indicatorType": "float",
                "indicatorUnit": "GHz",
                "@baseType": "PerformanceIndicatorSpecification",
                "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
                "@type": "PerformanceIndicatorSpecification"
              },
              "observedValue": "3.25"
            }
        ],
        "validFor": {
```

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```
            "endDateTime": "1985-04-12T23:20:50.52Z",
            "startDateTime": "1985-04-12T23:20:50.52Z"
        },
        "@baseType": "PerformanceMeasurement",
        "@schemaLocation":
"https://github.com/repo/PerformanceMeasurementAtomic.schema.json",
        "@type": "PerformanceMeasurementAtomic"
    },
    {
        "id": "2",
        "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceMeasurement/2",
        "description": "The second performance measurement in the database",
        "measurementCollectionJob": {
          "id": "4000",
          "href": "https://host:port/tmf-
api/performanceManagement/v5/measurementCollectionJob/4000",
          "name": "Measurement Collection Job from system A",
          "@baseType": "MeasurementCollectionJobRef",
          "@schemaLocation":
"https://myschemas/MeasurementCollectionJob.schema.json",
          "@type": "MeasurementCollectionJobRef",
          "@referredType": "MeasurementCollectionJob"
        },
        "performanceIndicatorValue": [
            {
              "performanceIndicatorSpecification":{
                "id": "1",
                "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/1",
                "@baseType": "PerformanceIndicatorSpecificationRef",
                "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
                "@type": "PerformanceIndicatorSpecificationRef",
                "@referredType": "PerformanceIndicatorSpecification"
              },
              "observedValue": "2.2",
              "tag":{
                "imsi" : "1111000000",
                "zip" : "93165"
              }
            },
            {
              "performanceIndicatorSpecification":{
                "id": "2",
                "name": "Network Latency Indicator Specification",
                "indicatorCategory": "LatencyCategory",
                "indicatorName": "Latency",
                "indicatorType": "float",
                "indicatorUnit": "ms",
                "@baseType": "PerformanceIndicatorSpecification",
                "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
                "@type": "PerformanceIndicatorSpecification"
              },
              "observedValue": "3.25"
            }
        ],
        "validFor": {
            "endDateTime": "1985-04-12T23:20:50.52Z",
            "startDateTime": "1985-04-12T23:20:50.52Z"
        },
        "tag":{
```


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```
          "country" : "Belgium",
          "region" : "North"
        },
        "@baseType": "PerformanceMeasurement",
        "@schemaLocation":
"https://github.com/repo/PerformanceMeasurementAtomic.schema.json",
        "@type": "PerformanceMeasurementAtomic"
    },
    {
      "id": "3",
      "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceMeasurement/3",
      "description": "The third group performance measurement in the database",
      "measurementCollectionJob": {
        "id": "4000",
        "href": "https://host:port/tmf-
api/performanceManagement/v5/measurementCollectionJob/4000",
        "name": "Measurement Collection Job from system A",
        "@baseType": "MeasurementCollectionJobRef",
        "@schemaLocation": "https://myschemas/MeasurementCollectionJob.schema.json",
        "@type": "MeasurementCollectionJobRef",
        "@referredType": "MeasurementCollectionJob"
      },
      "performanceIndicatorGroupValue": [{
        "performanceIndicatorGroupSpecification" : {
            "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSGrouppecification/600",
            "id": "600",
            "name": "My critical performance indicators",
            "groupCategory": "CPUGroup",
            "performanceIndicatorSpecification": [
              {
                "id": "5000",
                "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5000",
                "name": "CPU speed",
                "version": "1.0",
                "@baseType": "PerformanceIndicatorSpecification",
                "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
                "@type": "PerformanceIndicatorSpecification"
              },
              {
                "id": "5001",
                "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5001",
                "name": "CPU speed2",
                "version": "1.0",
                "@baseType": "PerformanceIndicatorSpecification",
                "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
                "@type": "PerformanceIndicatorSpecification"
              }
            ],
            "@baseType": "PerformanceIndicatorGroupSpecification",
            "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorGroupSpecificati
on.schema.json",
            "@type": "PerformanceIndicatorGroupSpecification"
          },
          "observedGroupValue": [
```


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```
                {
                  "performanceIndicatorSpecification":{
                    "id": "5000",
                    "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5000",
                    "@baseType": "PerformanceIndicatorSpecificationRef",
                    "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecificationRef.schema.json",
                    "@type": "PerformanceIndicatorSpecificationRef",
                    "@referredType": "PerformanceIndicatorSpecification"
                  },
                  "observedValue": "2.1",
                  "tag":{
                    "imsi" : "1111000000",
                    "zip" : "93165"
                  }
                },
                {
                  "performanceIndicatorSpecification":{
                    "id": "5001",
                    "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5001",
                    "@baseType": "PerformanceIndicatorSpecificationRef",
                    "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecificationRef.schema.json",
                    "@type": "PerformanceIndicatorSpecificationRef",
                    "@referredType": "PerformanceIndicatorSpecification"
                  },
                  "observedValue": "2.2",
                  "tag":{
                    "imsi" : "1112000000",
                    "zip" : "93165"
                  }
                }
            ]
      }],
      "validFor": {
          "endDateTime": "1985-04-12T23:20:50.52Z",
          "startDateTime": "1985-04-12T23:20:50.52Z"
      },
      "@baseType": "PerformanceMeasurement",
      "@schemaLocation":
"https://github.com/repo/PerformanceMeasurementGroup.schema.json",
      "@type": "PerformanceMeasurementGroup"
  }
]
```


Retrieves a PerformanceMeasurement by ID

`GET /performanceMeasurement/{id}?fields=…`

Description

This operation retrieves a PerformanceMeasurement entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

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Here's an example of a request for retrieving a specific Simple PerformanceMeasurement.

Request

```
GET /performanceMeasurement/1
Content-Type: application/json
```

Response

```
200

{
    "id": "1",
    "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceMeasurement/1",
    "description": "The first performance measurement in the database",
    "measurementCollectionJob": {
      "id": "4000",
      "href": "https://host:port/tmf-
api/performanceManagement/v5/measurementCollectionJob/4000",
      "name": "Measurement Collection Job from system A",
      "@baseType": "MeasurementCollectionJobRef",
      "@schemaLocation": "https://myschemas/MeasurementCollectionJob.schema.json",
      "@type": "MeasurementCollectionJobRef",
      "@referredType": "MeasurementCollectionJob"
    },
    "performanceIndicatorValue": [
        {
          "performanceIndicatorSpecification":{
            "id": "1",
            "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/1",
            "@baseType": "PerformanceIndicatorSpecificationRef",
            "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
            "@type": "PerformanceIndicatorSpecificationRef",
            "@referredType": "PerformanceIndicatorSpecification"
          },
          "observedValue": "2.1",
          "tag":{
            "imsi" : "1111000000",
            "zip" : "93165"
          }
        },
        {
          "performanceIndicatorSpecification":{
            "id": "2",
            "name": "CPU Speed Indicator Specification",
            "indicatorCategory": "SPEEDCategory",
            "indicatorName": "CPU Speed",
            "indicatorType": "float",
            "indicatorUnit": "GHz",
            "@baseType": "PerformanceIndicatorSpecification",
            "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecification.schema.json",
            "@type": "PerformanceIndicatorSpecification"
          },
          "observedValue": "3.25"
        }
```

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```
    ],
    "validFor": {
        "endDateTime": "1985-04-12T23:20:50.52Z",
        "startDateTime": "1985-04-12T23:20:50.52Z"
    },
    "tag":{
      "country" : "Belgium",
      "region" : "North"
    },
    "@baseType": "PerformanceMeasurement",
    "@schemaLocation":
"https://github.com/repo/PerformanceMeasurementAtomic.schema.json",
    "@type": "PerformanceMeasurementAtomic"
}
```

Here's an example of a request for retrieving a specific Group PerformanceMeasurement.

Request

```
GET /performanceMeasurement/1
Content-Type: application/json
```

Response

```
200

{
    "id": "1",
    "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceMeasurement/1",
    "description": "The first performance measurement in the database",
    "measurementCollectionJob": {
      "id": "4000",
      "href": "https://host:port/tmf-
api/performanceManagement/v5/measurementCollectionJob/4000",
      "name": "Measurement Collection Job from system A",
      "@baseType": "MeasurementCollectionJobRef",
      "@schemaLocation": "https://myschemas/MeasurementCollectionJob.schema.json",
      "@type": "MeasurementCollectionJobRef",
      "@referredType": "MeasurementCollectionJob"
    },
    "performanceIndicatorGroupValue": [{
          "performanceIndicatorGroupSpecification" : {
              "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSGrouppecification/600",
              "id": "600",
              "name": "My critical performance indicators",
              "groupCategory": "CPUGroup",
              "performanceIndicatorSpecification": [
                {
                  "id": "5000",
                  "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5000",
                  "name": "CPU speed",
                  "version": "1.0",
                  "@baseType": "PerformanceIndicatorSpecification",
                  "@schemaLocation": "https://host:port/tmf-
```

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```
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
                  "@type": "PerformanceIndicatorSpecification"
                },
                {
                  "id": "5001",
                  "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5001",
                  "name": "CPU speed2",
                  "version": "1.0",
                  "@baseType": "PerformanceIndicatorSpecification",
                  "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
                  "@type": "PerformanceIndicatorSpecification"
                }
              ],
              "@baseType": "PerformanceIndicatorGroupSpecification",
              "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorGroupSpecificati
on.schema.json",
              "@type": "PerformanceIndicatorGroupSpecification"
            },
            "observedGroupValue": [
                  {
                    "performanceIndicatorSpecification":{
                      "id": "5000",
                      "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5000",
                      "@baseType": "PerformanceIndicatorSpecificationRef",
                      "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecificationRef.schema.json",
                      "@type": "PerformanceIndicatorSpecificationRef",
                      "@referredType": "PerformanceIndicatorSpecification"
                    },
                    "observedValue": "2.1",
                    "tag":{
                      "imsi" : "1111000000",
                      "zip" : "93165"
                    }
                  },
                  {
                    "performanceIndicatorSpecification":{
                      "id": "5001",
                      "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5001",
                      "@baseType": "PerformanceIndicatorSpecificationRef",
                      "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecificationRef.schema.json",
                      "@type": "PerformanceIndicatorSpecificationRef",
                      "@referredType": "PerformanceIndicatorSpecification"
                    },
                    "observedValue": "2.2",
                    "tag":{
                      "imsi" : "1112000000",
                      "zip" : "93165"
                    }
                  }
            ]
      },
      {
        "performanceIndicatorGroupSpecification" : {
            "href": "https://host:port/tmf-
```


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```
api/performanceManagement/v5/performanceIndicatorSGrouppecification/601",
            "id": "601",
            "name": "My critical performance indicators extension",
            "groupCategory": "CPUGroup2",
            "performanceIndicatorSpecification": [
              {
                "id": "5003",
                "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5003",
                "name": "CPU speed3",
                "version": "1.0",
                "@baseType": "PerformanceIndicatorSpecification",
                "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
                "@type": "PerformanceIndicatorSpecification"
              },
              {
                "id": "5004",
                "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5004",
                "name": "CPU speed4",
                "version": "1.0",
                "@baseType": "PerformanceIndicatorSpecification",
                "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorSpecification.sc
hema.json",
                "@type": "PerformanceIndicatorSpecification"
              }
            ],
            "@baseType": "PerformanceIndicatorGroupSpecification",
            "@schemaLocation": "https://host:port/tmf-
api/performanceManagement/v5/schemas/Tmf/common/PerformanceIndicatorGroupSpecificati
on.schema.json",
            "@type": "PerformanceIndicatorGroupSpecification"
          },
          "observedGroupValue": [
                {
                  "performanceIndicatorSpecification":{
                    "id": "5003",
                    "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5003",
                    "@baseType": "PerformanceIndicatorSpecificationRef",
                    "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecificationRef.schema.json",
                    "@type": "PerformanceIndicatorSpecificationRef",
                    "@referredType": "PerformanceIndicatorSpecification"
                  },
                  "observedValue": "2.3",
                  "tag":{
                    "imsi" : "1113000000",
                    "zip" : "93165"
                  }
                },
                {
                  "performanceIndicatorSpecification":{
                    "id": "5004",
                    "href": "https://host:port/tmf-
api/performanceManagement/v5/performanceIndicatorSpecification/5004",
                    "@baseType": "PerformanceIndicatorSpecificationRef",
                    "@schemaLocation":
"https://myschemas/PerformanceIndicatorSpecificationRef.schema.json",
                    "@type": "PerformanceIndicatorSpecificationRef",
```


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```
                    "@referredType": "PerformanceIndicatorSpecification"
                  },
                  "observedValue": "2.4",
                  "tag":{
                    "imsi" : "1114000000",
                    "zip" : "93165"
                  }
                }
            ]
    }],
    "validFor": {
      "endDateTime": "1985-04-12T23:20:50.52Z",
      "startDateTime": "1985-04-12T23:20:50.52Z"
    },
    "tag":{
      "country" : "Belgium",
      "region" : "North"
    },
    "@baseType": "PerformanceMeasurement",
    "@schemaLocation":
"https://github.com/repo/PerformanceMeasurementGroup.schema.json",
    "@type": "PerformanceMeasurementGroup"
  }
```

Operations on DataAccessEndpoint

Retrieves a DataAccessEndpoint by ID

`GET /dataAccessEndpoint/{id}?fields=…`

Description

This operation retrieves a DataAccessEndpoint entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving a specific DataAccessEndpoint.

Request

```
GET /dataAccessEndpoint/123-xxx
Content-Type: application/json
```

Response

```
200


{
  "id": "123-xxx",
  "href": "https://host:port/tmf-
api/performanceManagement/v5/dataAccessEndpoint/123-xxx",
  "apiType": "REST",
```

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```
  "category": "topic",
  "description": "Measurement collection job 123-xxx subscriber topic",
  "endOperatingDate": "2022-05-13T10:13:42.971Z",
  "name": "123-xxx subscriber topic",
  "resourceVersion": "1.0",
  "startOperatingDate": "2022-05-13T10:13:42.971Z",
  "uri": "https://pubsub.googleapis.com/v1/",
  "uriQueryFilter":{
    "mappings": [
        {
            "filterTemplate": {
                "name" : "myRangeFilterSearch0",
                "@type" : "DataFilterTemplateRange",
                "criteria":{
                    "lowerBound" : {
                        "boundValue": -10,
                        "isInclusive": true
                    },
                    "upperBound" : {
                        "boundValue": -10,
                        "isInclusive": true
                    }
                }
            },
            "stringArray": {
                "value": [
                "field01", "field02", "field03"
                 ]
            }
        },
        {
            "filterTemplate": {
                "name": "myRegExSampleSearch0",
                "@type": "DataFilterTemplateRegex",
                "criteria": "dallas|houston|venise|paris|new-york"
            },
            "stringArray": {
                "value":[
                    "field11", "field12", "field13"
                ]
            }
        },
        {
            "filterTemplate": {
                "name" : "myBooleanFilterSearch0",
                "criteria" : true,
                "@type" : "DataFilterTemplateBoolean"
            },
            "stringArray": {
                "value": [
                    "field1", "field2", "field3"
                ],
                "@type": "DataFilterAttributeStringArray"
            }
        }
    ],
    "@type": "DataFilterMap"
  },
  "value": "myLogicalResourceValue",
  "administrativeState": "locked",
  "note": [
    {
      "id": "afa9-3d90aaa8da0f",
```


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```
      "author": "Mr. Franklin",
      "date": "2022-13-05T08:00:00Z",
      "text": "For further details on how to create a subscription to this topic,
visit
https://cloud.google.com/pubsub/docs/reference/rest/v1/projects.subscriptions/create
",
      "@baseType": "Note",
      "@schemaLocation": "https://myschemas.com/Note.schema.json",
      "@type": "Note"
    }
  ],
  "operationalState": "enable",
  "resourceCharacteristic": [
    {
      "id": "2100",
      "name": "myCharName",
      "valueType": "string",
      "characteristicRelationship": [
        {
          "id": "21",
          "href": "https://host:port/tmf-
api/performanceManagement/v5/characteristicRelationship/21",
          "relationshipType": "string",
          "@baseType": "CharacteristicRelationship",
          "@schemaLocation":
"https://myschemas.com/CharacteristicRelationship.schema.json",
          "@type": "CharacteristicRelationship"
        }
      ],
      "value": "myCharValue",
      "@baseType": "Characteristic",
      "@schemaLocation": "https://myschemas.com/StringCharacteristic.schema.json",
      "@type": "StringCharacteristic"
    }
  ],
  "resourceSpecification": {
    "id": "1",
    "href": "https://host:port/tmf-
api/performanceManagement/v5/ResourceSpecification/1",
    "name": "mySpecName",
    "version": "1.0",
    "@baseType": "ResourceSpecification",
    "@schemaLocation": "https://mySchemas.com/ResourceSpecification.schema.json",
    "@type": "ResourceSpecification",
    "@referredType": "ResourceSpecification"
  },
  "resourceStatus": "standby",
  "usageState": "idle",
  "@baseType": "DataAccessEndpoint",
  "@schemaLocation": "https://myschemas.com/dataAccessEndpoint",
  "@type": "DataAccessEndpoint"
}
```


List or find DataAccessEndpoint objects

`GET /dataAccessEndpoint?fields=…`

Description

This operation list DataAccessEndpoint entities. Attribute selection is enabled for all first

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level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving multiple DataAccessEndpoint's.

Request

```
GET /dataAccessEndpoint?apiType=REST
Content-Type: application/json
```

Response

```
200

[
  {
      "id": "123-xxx",
      "href": "https://host:port/tmf-
api/performanceManagement/v5/dataAccessEndpoint/123-xxx",
      "apiType": "REST",
      "category": "topic",
      "description": "Measurement collection job 123-xxx subscriber topic",
      "endOperatingDate": "2022-05-13T10:13:42.971Z",
      "name": "123-xxx subscriber topic",
      "resourceVersion": "1.0",
      "startOperatingDate": "2022-05-13T10:13:42.971Z",
      "uri": "https://pubsub.googleapis.com/v1/",
      "uriQueryFilter":{
        "mappings": [
            {
                "filterTemplate": {
                    "name" : "myRangeFilterSearch0",
                    "@type" : "DataFilterTemplateRange",
                    "criteria":{
                        "lowerBound" : {
                            "boundValue": -10,
                            "isInclusive": true
                        },
                        "upperBound" : {
                            "boundValue": -10,
                            "isInclusive": true
                        }
                    }
                },
                "stringArray": {
                    "value": [
                    "field01", "field02", "field03"
                    ]
                }
            },
            {
                "filterTemplate": {
                    "name": "myRegExSampleSearch0",
                    "@type": "DataFilterTemplateRegex",
                    "criteria": "dallas|houston|venise|paris|new-york"
                },
```

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```
                "stringArray": {
                    "value":[
                        "field11", "field12", "field13"
                    ]
                }
            },
            {
                "filterTemplate": {
                    "name" : "myBooleanFilterSearch0",
                    "criteria" : true,
                    "@type" : "DataFilterTemplateBoolean"
                },
                "stringArray": {
                    "value": [
                        "field1", "field2", "field3"
                    ]
                }
            }
        ],
        "@type": "DataFilterMap"
      },
      "value": "myLogicalResourceValue",
      "administrativeState": "locked",
      "note": [
        {
          "id": "afa9-3d90aaa8da0f",
          "author": "Mr. Franklin",
          "date": "2022-13-05T08:00:00Z",
          "text": "For further details on how to create a subscription to this
topic, visit
https://cloud.google.com/pubsub/docs/reference/rest/v1/projects.subscriptions/create
",
          "@baseType": "Note",
          "@schemaLocation": "https://myschemas.com/Note.schema.json",
          "@type": "Note"
        }
      ],
      "operationalState": "enable",
      "resourceCharacteristic": [
        {
          "id": "2100",
          "name": "myCharName",
          "valueType": "string",
          "characteristicRelationship": [
            {
              "id": "21",
              "href": "https://host:port/tmf-
api/performanceManagement/v5/characteristicRelationship/21",
              "relationshipType": "string",
              "@baseType": "CharacteristicRelationship",
              "@schemaLocation":
"https://myschemas.com/CharacteristicRelationship.schema.json",
              "@type": "CharacteristicRelationship"
            }
          ],
          "value": "myCharValue",
          "@baseType": "Characteristic",
          "@schemaLocation":
"https://myschemas.com/StringCharacteristic.schema.json",
          "@type": "StringCharacteristic"
        }
      ],
      "resourceSpecification": {
```


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```
        "id": "1",
        "href": "https://host:port/tmf-
api/performanceManagement/v5/ResourceSpecification/1",
        "name": "mySpecName",
        "version": "1.0",
        "@baseType": "ResourceSpecification",
        "@schemaLocation":
"https://mySchemas.com/ResourceSpecification.schema.json",
        "@type": "ResourceSpecification",
        "@referredType": "ResourceSpecification"
      },
      "resourceStatus": "standby",
      "usageState": "idle",
      "@baseType": "DataAccessEndpoint",
      "@schemaLocation": "https://myschemas.com/dataAccessEndpoint",
      "@type": "DataAccessEndpoint"
    },
    {
      "id": "123-xx0",
      "href": "https://host:port/tmf-
api/performanceManagement/v5/dataAccessEndpoint/123-xx0",
      "apiType": "REST",
      "category": "topic",
      "description": "Measurement collection job 123-xx0 subscriber topic",
      "endOperatingDate": "2022-05-13T10:13:42.971Z",
      "name": "123-xx0 subscriber topic",
      "resourceVersion": "1.0",
      "startOperatingDate": "2022-05-13T10:13:42.971Z",
      "uri": "https://pubsub.googleapis.com/v1/",
      "uriQueryFilter":{
        "mappings": [
            {
                "filterTemplate": {
                    "name" : "myRangeFilterSearch0",
                    "@type" : "DataFilterTemplateRange",
                    "criteria":{
                        "lowerBound" : {
                            "boundValue": -10,
                            "isInclusive": true
                        },
                        "upperBound" : {
                            "boundValue": -10,
                            "isInclusive": true
                        }
                    }
                },
                "stringArray": {
                    "value": [
                    "field01", "field02", "field03"
                    ]
                }
            },
            {
                "filterTemplate": {
                    "name": "myRegExSampleSearch0",
                    "@type": "DataFilterTemplateRegex",
                    "criteria": "dallas|houston|venise|paris|new-york"
                },
                "stringArray": {
                    "value":[
                        "field11", "field12", "field13"
                    ]
                }
```


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```
            },
            {
                "filterTemplate": {
                    "name" : "myBooleanFilterSearch0",
                    "criteria" : true,
                    "@type" : "DataFilterTemplateBoolean"
                },
                "stringArray": {
                    "value": [
                        "field1", "field2", "field3"
                    ]
                }
            }
        ],
        "@type": "DataFilterMap"
      },
      "value": "myLogicalResourceValue",
      "administrativeState": "locked",
      "note": [
        {
          "id": "afa9-3d90aaa8da0f",
          "author": "Mr. Franklin",
          "date": "2022-13-05T08:00:00Z",
          "text": "For further details on how to create a subscription to this
topic, visit
https://cloud.google.com/pubsub/docs/reference/rest/v1/projects.subscriptions/create
",
          "@baseType": "Note",
          "@schemaLocation": "https://myschemas.com/Note.schema.json",
          "@type": "Note"
        }
      ],
      "operationalState": "enable",
      "resourceCharacteristic": [
        {
          "id": "2100",
          "name": "myCharName",
          "valueType": "string",
          "characteristicRelationship": [
            {
              "id": "21",
              "href": "https://host:port/tmf-
api/performanceManagement/v5/characteristicRelationship/21",
              "relationshipType": "string",
              "@baseType": "CharacteristicRelationship",
              "@schemaLocation":
"https://myschemas.com/CharacteristicRelationship.schema.json",
              "@type": "CharacteristicRelationship"
            }
          ],
          "value": "myCharValue",
          "@baseType": "Characteristic",
          "@schemaLocation":
"https://myschemas.com/StringCharacteristic.schema.json",
          "@type": "StringCharacteristic"
        }
      ],
      "resourceSpecification": {
        "id": "1",
        "href": "https://host:port/tmf-
api/performanceManagement/v5/ResourceSpecification/1",
        "name": "mySpecName",
        "version": "1.0",
```


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```
        "@baseType": "ResourceSpecification",
        "@schemaLocation":
"https://mySchemas.com/ResourceSpecification.schema.json",
        "@type": "ResourceSpecification",
        "@referredType": "ResourceSpecification"
      },
      "resourceStatus": "standby",
      "usageState": "idle",
      "@baseType": "DataAccessEndpoint",
      "@schemaLocation": "https://myschemas.com/dataAccessEndpoint",
      "@type": "DataAccessEndpoint"
    }
]
```


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API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

Register listener

POST /hub

Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

Usage Samples

Here's an example of a request for registering a listener.

Request

```
POST /api/hub

Accept: application/json

{
    "callback": "http://in.listener.com"
}
```

Response

```
201

Content-Type: application/json

Location: /api/hub/42

{
    "id":"42",
    "callback": "http://in.listener.com",
    "query":""
}
```

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Unregister listener

DELETE /hub/{id}

Description

Clears the communication endpoint address that was set by creating the Hub..

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

Usage Samples

Here's an example of a request for un-registering a listener.

Request

```
DELETE /api/hub/42

Accept: application/json
```

Response

```
204
```

Publish Event to listener

POST /client/listener

Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

Usage Samples

Here's an example of a notification received by the listener. In this example "EVENT TYPE" should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

Request

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```
POST /client/listener

Accept: application/json

{
    "event": {
        EVENT BODY
    },
    "eventType": "EVENT_TYPE"
}
```

Response

```
201
```

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

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Acknowledgements

Release History

Release Number	Date	Release led by:	Description
v5.0.0	15/06/2023	Ranjitha Majety	Created v5.0.0. Removed TrackingRecord resource.
v5.0.0	05/09/2023	Gervais-Martial Ngueko	Updated v5.0.0. add more entities and bug fixes.

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